MATHS-IA

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FUNCTIONS

Long answer Questions:

L13

1) If the function f is defined by $f(x) = \begin{cases} x+2 & x>1 \\ 2 & -1 \le x \le 1 \\ x-1 & -3<x<-1 \end{cases}$ then find the Vaule of

a) $f(3)$ b) $f(0)$ c) $f(-1.5)$ d) $f(2)+f(-2)$ e) $f(-5)$

Sol:- To find $f(3)$

$f(x) = x+2$

$f(3) = 3+2$

$= 5$

b) To find $f(0)$

$f(x) = 2$

$f(0) = 2$

c) To find $f(-1.5)$

$f(x) = x-1$

$f(-1.5) = -1.5-1$

$= -2.5$

→ (3M)

d) To find $f(2)+f(-2)$

$f(x) = x+2$

$f(2) = 2+2$

$= 4$

$f(x) = x-1$

$f(-2) = -2-1$

$= -3$

e) To find $f(-5)$

$f(x)$ = Not defined

$f(-5)$ = Not defined

Now $f(2)+f(-2) = 4-3 = 1$ → (4M)

L18

2) If the function f is defined by $f(x) = \begin{cases} 3x-2 & x>3 \\ x^2-2 & -2 \le x \le 2 \\ 2x+1 & x<-3 \end{cases}$ then find the Vaule of $f(4)$, $f(2.5)$, $f(-2)$ $f(-4)$

Sol:- To find $f(4)$

$f(x) = 3x-2$

$f(4) = 3(4)-2$

$= 12-2$

$= 10$

To find $f(2.5)$

$f(x)$ = Not defined

$f(2.5)$ = Not defined

→ (2M)

To find $f(-2)$

$f(x) = x^2-2$

$f(-2) = (-2)^2-2$

$= 4-2$

$= 2$

To find $f(-4)$

$f(x) = 2x+1$

$f(-4) = 2(-4)+1$

$= -8+1$

$= -7$

→ (2M)

To find $f(0)$

$f(x) = x^2 - 2$

$f(0) = 0^2 - 2$

$= -2$

To find $f(-7)$

$f(x) = 2x+1$

$f(-7) = 2(-7)+1$

$= -14+1$

$= -13$ $\longrightarrow (3M)$

L18

3) If the function $f: R \to R$ defined by $f(x) = \frac{3^x + 3^{-x}}{2}$ then show that $f(x+y) + f(x-y) = 2f(x) \cdot f(y)$

Sol:- $f(x) = \frac{3^x + 3^{-x}}{2}$

$f(x+y) = \frac{3^{x+y} + 3^{-(x+y)}}{2} \Rightarrow ①$ $\longrightarrow (1M)$

$f(x-y) = \frac{3^{x-y} + 3^{-(x-y)}}{2} \Rightarrow ②$ $\longrightarrow (1M)$

By adding ① + ②

$= \frac{3^{x+y} + 3^{-(x+y)} + 3^{x-y} + 3^{-(x-y)}}{2}$ $\longrightarrow (2M)$

$= \frac{3^x \cdot 3^y + 3^{-x} \cdot 3^{-y} + 3^x \cdot 3^{-y} + 3^{-x} \cdot 3^{+y}}{2}$

$= \frac{1}{2}\left[3^x\left[3^y + 3^{-y}\right] + 3^{-x}\left[3^{-y} + 3^y\right]\right]$

$= 2\left[\frac{3^x + 3^{-x}}{2}\right]\left[\frac{3^y + 3^{-y}}{2}\right]$

$\Rightarrow f(x+y) + f(x-y) = 2f(x) \cdot f(y)$ $\longrightarrow (3M)$

L18 4) If $f(x) = \cos(\log x)$ then show that $f\left(\frac{1}{x}\right) f\left(\frac{1}{y}\right) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$

Sol:- $f\left(\frac{1}{x}\right) = \cos\left(\log\left(\frac{1}{x}\right)\right)$

$= \cos(\log(1) - \log x)$

$= \cos(0 - \log x)$

$= \cos(-\log x)$

$= \cos(\log x)$ $\longrightarrow (2M)$

$\text{lly}\quad f\left(\frac{1}{y}\right) = \cos(\log y)$

$f\left(\frac{1}{y}\right) = \cos\left(\log\left(\frac{x}{y}\right)\right)$

$= \cos(\log x - \log y) \rightarrow ①$ $\rightarrow$ (1M)

$f(xy) = \cos[\log(xy)]$

$= \cos(\log x + \log y) \rightarrow ②$ $\rightarrow$ (1M)

By adding ① + ②

$= f\left(\frac{x}{y}\right) + f(xy) = \cos(\log x - \log y) + \cos(\log x + \log y)$

$= 2\cos(\log x)\cos(\log y)$

$\boxed{\cos(A-B) + \cos(A+B) = 2\cos A\cos B}$

$\therefore f\left(\frac{1}{x}\right) f\left(\frac{1}{y}\right) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right] = \cos(\log x)\cos(\log y) - \frac{1}{2}[2\cos\log x \cos\log y]$

$= \dfrac{2\cos(\log x)\cos(\log y) - 2\cos\log x\,\cos\log y}{2}$

$= \frac{0}{2}$

$= 0$ $\rightarrow$ (3M)

L18 5) If the function $f: R \rightarrow R$ is defined by $f(x) = \frac{4^x}{4^x+2}$, then show that $f(-x) = 1 - f(x)$ & hence deduce the Vaule of $f\left(\frac{1}{4}\right) + 2f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)$

Sol:- $L.H.S = f(1-x)$

$= \dfrac{4^{1-x}}{4^{1-x}+2}$ $\rightarrow$ (1M)

$= \dfrac{\frac{4}{4^x}}{\frac{4}{4^x}+2}$

$= \dfrac{\frac{4}{4^x}}{\frac{4+2\cdot 4^x}{4^x}}$

$$= \frac{4}{4+2\cdot 4^x}$$

$$= \frac{(4+2\cdot 4^x) - 2\cdot 4^x}{4+2\cdot 4^x}$$

$$= \frac{1-2\cdot 4^x}{4+2\cdot 4^x}$$

$$= \frac{1-4^x\cdot \cancel{2}}{\cancel{2}(4^x+2)}$$

$$= 1 - \frac{4^x}{4^x+2}$$

$$= 1 - F(x)$$

= R.H.S $\rightarrow$ (2M)

If $x = 1/4$ then

$$f\left(\tfrac{3}{4}\right) = 1 - f\left(\tfrac{1}{4}\right)$$

$$f\left(\tfrac{3}{4}\right) = 1 - f\left(\tfrac{1}{4}\right)$$

$$f\left(\tfrac{3}{4}\right) + f\left(\tfrac{1}{4}\right) = 1 \longrightarrow ①$$ $\rightarrow$ (1M)

If $x = 1/2$ then

$$f\left(\tfrac{1}{2}\right) = 1 - f\left(\tfrac{1}{2}\right)$$

$$2f\left(\tfrac{1}{2}\right) = 1 \longrightarrow ②$$ $\rightarrow$ (1M)

By adding ① & ②

① + ②

$$f\left(\tfrac{1}{4}\right) + 2f\left(\tfrac{1}{2}\right) + f\left(\tfrac{3}{4}\right) = 1+1 = 2.$$ $\rightarrow$ (2M)

MATRICES

Long Answer Question's:

Solve the following Simultaneous linear equation by using crammer's rule.

Eg(1):- $3x+4y+5z=18$, $2x-y+8z=13$, $5x-2y+7z=20$

Sol:- Given equation can be written as

$$A=\begin{bmatrix}3&4&5\\2&-1&8\\5&-2&7\end{bmatrix} \quad X=\begin{pmatrix}x\\y\\z\end{pmatrix} \quad D=\begin{bmatrix}18\\13\\20\end{bmatrix} \longrightarrow (1M)$$

This is in the form of $AX=D$.

Now, $$\Delta=\begin{vmatrix}\overset{+}{3}&\overset{-}{4}&\overset{+}{5}\\2&-1&8\\5&-2&7\end{vmatrix}$$

$$=3\begin{vmatrix}-1&8\\-2&7\end{vmatrix}-4\begin{vmatrix}2&8\\5&7\end{vmatrix}+5\begin{vmatrix}2&-1\\5&-2\end{vmatrix}$$

$$=3(-7+16)-4(14+40)+5(-4+5)$$

$$=3(9)-4(-26)+5(1)=27+104+5=136. \longrightarrow (1M)$$

$$\Delta_1=\begin{vmatrix}18&4&5\\13&-1&8\\20&-2&7\end{vmatrix}$$

$$=18\begin{vmatrix}-1&8\\-2&7\end{vmatrix}-4\begin{vmatrix}13&8\\20&7\end{vmatrix}+5\begin{vmatrix}13&-1\\20&-2\end{vmatrix}$$

$$=18(-7+16)-4(91-160)+5(-26+20)$$

$$=18(9)-4(-69)+5(-6)$$

$$=162+276-30=438-30=408 \longrightarrow (1M)$$

$$\Delta_2=\begin{vmatrix}3&18&5\\2&13&8\\5&20&7\end{vmatrix}$$

$$=3\begin{vmatrix}13&8\\20&7\end{vmatrix}-18\begin{vmatrix}2&8\\5&7\end{vmatrix}+5\begin{vmatrix}2&13\\5&20\end{vmatrix}$$

$= 3(91-160) - 18(14-40) + 5(40-65)$

$= 3(-69) - 18(-26) + 5(-25)$

$= -207 + 468 - 125 = 136.$ $\longrightarrow$ (1M)

$$\Delta_3 = \begin{vmatrix} 3 & 4 & 18 \\ 2 & -1 & 13 \\ 5 & -2 & 20 \end{vmatrix}$$

$$= 3\begin{vmatrix} -1 & 13 \\ -2 & 20 \end{vmatrix} - 4\begin{vmatrix} 2 & 13 \\ 5 & 20 \end{vmatrix} + 18\begin{vmatrix} 2 & -1 \\ 5 & -2 \end{vmatrix}$$

$= 3(-20+26) - 4(40-65) + 18(-4+5)$

$= 18 + 100 + 18 = 136.$ $\longrightarrow$ (1M)

By Cramer's rule,

$$x = \frac{\Delta_1}{\Delta} = \frac{408}{136} = 3$$

$$y = \frac{\Delta_2}{\Delta} = \frac{136}{136} = 1$$

$$z = \frac{\Delta_3}{\Delta} = \frac{136}{136} = 1$$

$x=3, y=1, z=1$ is the solution for given system of equation.

$\longrightarrow$ (2M)

(ii) $x+y+z=9,\ 2x+5y+7z=52,\ 2x+y-z=0$

Sol: $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $D = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$

This is in the form of $AX=D$ $\longrightarrow$ (1M)

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{vmatrix}$$

$$= 1\begin{vmatrix} 5 & 7 \\ 1 & -1 \end{vmatrix} - 1\begin{vmatrix} 2 & 7 \\ 2 & -1 \end{vmatrix} + 1\begin{vmatrix} 2 & 5 \\ 2 & 1 \end{vmatrix}$$

$= 1(-5-7) - 1(2-14) + 1(2-10)$

$= -12 + 16 - 8 = -4.$ $\longrightarrow$ (1M)

$$\Delta_1 = \begin{vmatrix} 9 & 1 & 1 \\ 52 & 5 & 7 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= 9\begin{vmatrix} 5 & 7 \\ 1 & -1 \end{vmatrix} - 1\begin{vmatrix} 52 & 7 \\ 0 & -1 \end{vmatrix} + 1\begin{vmatrix} 52 & 5 \\ 0 & 1 \end{vmatrix}$$

$$= 9(-5-7) - 1(-52-0) + 1(52-0)$$

$$= 9(-12) - 1(-52) + 1(52)$$

$$= -108 + 52 + 52 = -4. \longrightarrow (1M)$$

$$\Delta_2 = \begin{vmatrix} 1 & 9 & 1 \\ 2 & 52 & 7 \\ 2 & 0 & -1 \end{vmatrix}$$

$$= 1\begin{vmatrix} 52 & 7 \\ 0 & -1 \end{vmatrix} - 9\begin{vmatrix} 2 & 7 \\ 2 & -1 \end{vmatrix} + 1\begin{vmatrix} 2 & 52 \\ 2 & 0 \end{vmatrix}$$

$$= 1(-52-0) - 9(-2-14) + 1(0-104)$$

$$= -52 + 144 - 104 = -12. \longrightarrow (1M)$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 9 \\ 2 & 5 & 52 \\ 2 & 1 & 0 \end{vmatrix}$$

$$= 1\begin{vmatrix} 5 & 52 \\ 1 & 0 \end{vmatrix} - 1\begin{vmatrix} 2 & 52 \\ 2 & 0 \end{vmatrix} + 9\begin{vmatrix} 2 & 5 \\ 2 & 1 \end{vmatrix}$$

$$= 1(0-52) - 1(0-104) + 9(2-10)$$

$$= 1(-52) - 1(-104) + 9(-8)$$

$$= -52 + 104 - 72 = 104 - 124 = -20. \longrightarrow (1M)$$

By cramer's rule.

$$x = \frac{\Delta_1}{\Delta} = \frac{-4}{-4} = 1 \quad y = \frac{\Delta_2}{\Delta} = \frac{-12}{-4} = 3 \quad z = \frac{\Delta_3}{\Delta} = \frac{-20}{-4} = 5$$

$x = 1$, $y = 3$, $z = 5$. $\longrightarrow (2M)$

4a(iii):- $2x - y + 3z = 9$, $x + y + z = 6$, $x - y + z = 2$

Sol:- given equations can be written as

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad D = \begin{bmatrix} 9 \\ 6 \\ 2 \end{bmatrix}$$

This is in the form of $AX=D$. $\rightarrow$ (1M)

$$\Delta = \begin{vmatrix} 2 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= 2\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} + 1\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + 3\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= 2(1+1) + 1(1-1) + 3(-1-1) = 4+0-6 = -2.$$ $\rightarrow$ (1M)

$$\Delta_1 = \begin{vmatrix} 9 & -1 & 3 \\ 6 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = 9\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} + 1\begin{vmatrix} 6 & 1 \\ 2 & 1 \end{vmatrix} + 3\begin{vmatrix} 6 & 1 \\ 2 & -1 \end{vmatrix}$$

$$= 9(1+1) + 1(6-2) + 3(-6-2) = 18+4-24 = -2.$$ $\rightarrow$ (1M)

$$\Delta_2 = \begin{vmatrix} 2 & 9 & 3 \\ 1 & 6 & 1 \\ 1 & 2 & 1 \end{vmatrix}$$

$$= 2\begin{vmatrix} 6 & 1 \\ 2 & 1 \end{vmatrix} - 9\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + 3\begin{vmatrix} 1 & 6 \\ 1 & 2 \end{vmatrix}$$

$$= 2(6-2) - 9(1-1) + 3(2-6) = 2(4) - 9(0) + 3(-4)$$

$$= 8-12 = -4.$$ $\rightarrow$ (1M)

$$\Delta_3 = \begin{vmatrix} 2 & -1 & 9 \\ 1 & 1 & 6 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 2\begin{vmatrix} 1 & 6 \\ -1 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 6 \\ 1 & 2 \end{vmatrix} + 9\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= 2(2+6) + 1(2-6) + 9(-1-1)$$

$$= 2(8) + 1(-4) + 9(-2) = 16-4-18 = -6.$$ $\rightarrow$ (1M)

$$x = \frac{\Delta_1}{\Delta} = \frac{-2}{-2} = 1 \quad y = \frac{\Delta_2}{\Delta} = \frac{-4}{-2} = 2, \; z = \frac{\Delta_3}{\Delta} = \frac{-6}{-2} = 3.$$

$x=1$, $y=2$, $z=3$. $\rightarrow$ (2M)

4(a)(4):- $x-y+3z=5,\ 4x+2y-z=0,\ -x+3y+z=5$

<u>Sol</u>:- Given system of eqn can be written as

$$A=\begin{bmatrix}1&-1&3\\4&2&-1\\-1&3&1\end{bmatrix}\quad X=\begin{bmatrix}x\\y\\z\end{bmatrix}\quad D=\begin{bmatrix}5\\0\\5\end{bmatrix}$$

This is in the form of $AX=D$ $\longrightarrow$ (1M)

$$\Delta=\begin{vmatrix}1&-1&3\\4&2&-1\\-1&3&1\end{vmatrix}$$

$$=1\begin{vmatrix}2&-1\\3&1\end{vmatrix}+1\begin{vmatrix}4&-1\\-1&1\end{vmatrix}+3\begin{vmatrix}4&2\\-1&3\end{vmatrix}$$

$$=1(2+3)+1(4-1)+3(12+2)=5+3+42=50$$ $\longrightarrow$ (1M)

$$\Delta_1=\begin{vmatrix}5&-1&3\\0&2&-1\\5&3&1\end{vmatrix}$$

$$=5\begin{vmatrix}2&-1\\3&1\end{vmatrix}+1\begin{vmatrix}0&-1\\5&1\end{vmatrix}+3\begin{vmatrix}0&2\\5&3\end{vmatrix}$$

$$=5(2+3)+1(0+1)+3(0-10)$$

$$=25+5-30=0$$ $\longrightarrow$ (1M)

$$\Delta_2=\begin{vmatrix}1&5&3\\4&0&-1\\-1&5&1\end{vmatrix}$$

$$=1\begin{vmatrix}0&-1\\5&1\end{vmatrix}-5\begin{vmatrix}4&-1\\-1&1\end{vmatrix}+3\begin{vmatrix}4&0\\-1&5\end{vmatrix}$$

$$=1(0+5)-5(4-1)+3(20-0)$$

$$=5-15+60=50$$ $\longrightarrow$ (1M)

$$\Delta_3=\begin{vmatrix}1&-1&5\\4&2&0\\-1&3&5\end{vmatrix}$$

$$=1\begin{vmatrix}2&0\\3&5\end{vmatrix}+1\begin{vmatrix}4&0\\-1&5\end{vmatrix}+5\begin{vmatrix}4&2\\-1&3\end{vmatrix}$$

$$=1(10)+1(20)+5(12+2)=10+20+70=100$$ $\longrightarrow$ (1M)

$x=\frac{\Delta_1}{\Delta}=\frac{0}{50}=0$, $y=\frac{\Delta_2}{\Delta}=\frac{50}{50}=1$, $z=\frac{\Delta_3}{\Delta}=\frac{100}{50}=2$.

$x=0, y=1, z=2$. $\longrightarrow$ (2M)

19(v) $x+y+z=1$, $2x+2y+3z=6$, $x+4y+9z=3$

Sol:- given system of eqn can be written as.

$$A=\begin{bmatrix}1&1&1\\2&2&3\\1&4&9\end{bmatrix} \quad X=\begin{bmatrix}x\\y\\z\end{bmatrix} \quad D=\begin{bmatrix}1\\6\\3\end{bmatrix}$$

It can be written as $AX=D$ $\longrightarrow$ (1M)

$$\Delta=\begin{vmatrix}1&1&1\\2&2&3\\1&4&9\end{vmatrix}$$

$$=1\begin{vmatrix}2&3\\4&9\end{vmatrix}-1\begin{vmatrix}2&3\\1&9\end{vmatrix}+1\begin{vmatrix}2&2\\1&4\end{vmatrix}$$

$$=1(18-12)-1(18-3)+1(8-2)=6-15+6=-3$$ $\longrightarrow$ (1M)

$$\Delta_1=\begin{vmatrix}1&1&1\\6&2&3\\3&4&9\end{vmatrix}$$

$$=1\begin{vmatrix}2&3\\4&9\end{vmatrix}-1\begin{vmatrix}6&3\\3&9\end{vmatrix}+1\begin{vmatrix}6&2\\3&4\end{vmatrix}$$

$$=(18-12)-(54-9)+(24-6)$$

$$=6-45+18=-21$$ $\longrightarrow$ (1M)

$$\Delta_2=\begin{vmatrix}1&1&1\\2&6&3\\1&3&9\end{vmatrix}$$

$$=1\begin{vmatrix}6&3\\3&9\end{vmatrix}-1\begin{vmatrix}2&3\\1&9\end{vmatrix}+1\begin{vmatrix}2&6\\1&3\end{vmatrix}$$

$$=(54-9)-18(-3)+(6-6)$$

$$=45-15=30.$$ $\longrightarrow$ (1M)

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 6 \\ 1 & 4 & 3 \end{vmatrix}$$

$= 1\begin{vmatrix} 2 & 6 \\ 4 & 3 \end{vmatrix} - 1\begin{vmatrix} 2 & 6 \\ 1 & 3 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix}$

$= (6-24) - (6-6) + \cancel{8}(8-2)$

$= -18+6 = -12.$ $\longrightarrow$ (1M)

$x = \frac{\Delta_1}{\Delta} = \frac{-21}{-3} = 7$, $y = \frac{\Delta_2}{\Delta} = \frac{30}{-3} = -10$, $z = \frac{\Delta_3}{\Delta} = \frac{-12}{-3} = 4$

$x = 7,\ y = -10,\ z = 4.$ $\longrightarrow$ (2M)

$\angle_{19}$(vi) $2x - y + 3z = 8,\ -x + 2y + z = 4,\ 3x + y - 4z = 0$

Sol:- given system of eqn can be written as

$A = \begin{pmatrix} 2 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & -4 \end{pmatrix}$ $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $D = \begin{pmatrix} 8 \\ 4 \\ 0 \end{pmatrix}$

This is in the form of $AX = D$. $\longrightarrow$ (1M)

$$\Delta = \begin{vmatrix} 2 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & -4 \end{vmatrix}$$

$= 2\begin{vmatrix} 2 & 1 \\ 1 & -4 \end{vmatrix} + 1\begin{vmatrix} -1 & 1 \\ 3 & -4 \end{vmatrix} + 3\begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix}$

$= 2(-8-1) + 1(4-3) + 3(-1-6)$

$= 2(-9) + 1(1) + 3(-7)$

$= -18 + 1 - 21 = -38$ $\longrightarrow$ (1M)

$$\Delta_1 = \begin{vmatrix} 8 & -1 & 3 \\ 4 & 2 & 1 \\ 0 & 1 & -4 \end{vmatrix}$$

$= 8\begin{vmatrix} 2 & 1 \\ 1 & -4 \end{vmatrix} + 1\begin{vmatrix} 4 & 1 \\ 0 & -4 \end{vmatrix} + 3\begin{vmatrix} 4 & 2 \\ 0 & 1 \end{vmatrix}$

$= 8(-8-1) + 1(\cancel{16}(-16-0) + 3(4-0)$

$\cancel{= -18+1-21 = -38}$

$= 8(-9) - 16 + 12$

$= -72 - 4 = -76.$ $\longrightarrow$ (1M)

$$\Delta_2 = \begin{vmatrix} 2 & 8 & 3 \\ -1 & 4 & 1 \\ 3 & 0 & -4 \end{vmatrix}$$

$$= 2\begin{vmatrix} 4 & 1 \\ 0 & -4 \end{vmatrix} - 8\begin{vmatrix} -1 & 1 \\ 3 & -4 \end{vmatrix} + 3\begin{vmatrix} -1 & 4 \\ 3 & 0 \end{vmatrix}$$

$= 2(-16) - 8(4-3) + 3(0-12)$

$= -32 - 8 - 36 = -76.$ $\longrightarrow$ (1M)

$$\Delta_3 = \begin{vmatrix} 2 & -1 & 8 \\ -1 & 2 & 4 \\ 3 & 1 & 0 \end{vmatrix}$$

$$= 2\begin{vmatrix} 2 & 4 \\ 1 & 0 \end{vmatrix} + 1\begin{vmatrix} -1 & 4 \\ 3 & 0 \end{vmatrix} + 8\begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix}$$

$= 2(0-4) + 1(0-12) + 8(-1-6)$

$= -8 - 12 - 56 = -76$ $\longrightarrow$ (1M)

$x = \dfrac{\Delta_1}{\Delta} = \dfrac{-76}{-38} = 2$, $y = \dfrac{\Delta_2}{\Delta} = \dfrac{-76}{-38} = 2$, $z = \dfrac{\Delta_3}{\Delta} = \dfrac{-76}{-38} = 2.$

$x = 2,\ y = 2,\ z = 2.$ $\longrightarrow$ (2M)

Solve the following simultaneous linear equation by using "matrix inversion".

L_{20}(i) $3x + 4y + 5z = 18$, $2x - y + 8z = 13$, $5x - 2y + 7z = 20$

Sol: given eqn can be written as

$$A = \begin{bmatrix} 3 & 4 & 5 \\ 2 & -1 & 8 \\ 5 & -2 & 7 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad D = \begin{bmatrix} 18 \\ 13 \\ 20 \end{bmatrix}$$

This is in the form of $AX = D$. $\longrightarrow$ (1M)

$$|A| = \begin{vmatrix} 3 & 4 & 5 \\ 2 & -1 & 8 \\ 5 & -2 & 7 \end{vmatrix}$$

$$= 3\begin{vmatrix} -1 & 8 \\ -2 & 7 \end{vmatrix} - 4\begin{vmatrix} 2 & 8 \\ 5 & 7 \end{vmatrix} + 5\begin{vmatrix} 2 & -1 \\ 5 & -2 \end{vmatrix}$$

$$= 3(-7+16) - 4(14-40) + 5(-4+5)$$

$$= 3(9) - 4(-26) + 5(1)$$

$$= 27 + 104 + 15 = 136 \neq 0$$

$\det A \neq 0$, A^{-1} is exit $\rightarrow$ (1M)

$$\begin{array}{cccc} -1 & 8 & 2 & -1 \\ -2 & 7 & 5 & -2 \\ 4 & 5 & 3 & 4 \\ -1 & 8 & 2 & -1 \end{array}$$

$$\text{adj} A = \begin{bmatrix} -7+16 & 40-14 & -4+5 \\ -10-28 & 21-25 & 20+6 \\ 32+5 & 10-24 & -3-8 \end{bmatrix}^T$$

$$= \begin{bmatrix} 9 & 26 & 1 \\ -38 & -4 & 26 \\ 37 & -14 & -11 \end{bmatrix}^T = \begin{bmatrix} 9 & -38 & 37 \\ 26 & -4 & -14 \\ 1 & 26 & -11 \end{bmatrix}$$

Hence by matrix inversion method, $\rightarrow$ (2M)

$$X = A^{-1}D \Rightarrow X = \frac{\text{adj}A}{\det A} \cdot D.$$

$$A^{-1} = \frac{\text{adj}A}{\det A}.$$

$$X = \frac{1}{136}\begin{bmatrix} 9 & -38 & 37 \\ 26 & -4 & -14 \\ 1 & 26 & -11 \end{bmatrix}\begin{bmatrix} 18 \\ 13 \\ 20 \end{bmatrix}$$

$$= \frac{1}{136}\begin{bmatrix} 162-494+740 \\ 468-52-280 \\ 18+338-220 \end{bmatrix}$$

$$= \frac{1}{136}\begin{bmatrix} 408 \\ 136 \\ 136 \end{bmatrix} = \frac{136}{136}\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore x = 3$ $y = 1, z = 1$ are solution of given equation. $\rightarrow$ (3M)

$L_{20}(b)$: $x+y+z=9$, $2x+5y+7z=52$, $2x+y-z=0$

Sol: given eqn are

$$A=\begin{bmatrix}1&1&1\\2&5&7\\2&1&-1\end{bmatrix} \quad x=\begin{bmatrix}x\\y\\z\end{bmatrix} \quad D=\begin{bmatrix}9\\52\\0\end{bmatrix}$$

This is in the form of $Ax=D$ $\longrightarrow$ (1M)

$$|A|=\begin{vmatrix}1&1&1\\2&5&7\\2&1&-1\end{vmatrix}$$

$$=1\begin{vmatrix}5&7\\1&-1\end{vmatrix}-1\begin{vmatrix}2&7\\2&-1\end{vmatrix}+1\begin{vmatrix}2&5\\2&1\end{vmatrix}$$

$$=1(-5-7)-1(2-14)+1(2-10)$$

$$=-12+16-8=-4\neq 0$$

$|A|=0$ $\longrightarrow$ (1M)

A^{-1} exists.

$$\text{adj}A=\begin{bmatrix}-5-7&14+2&2-10\\1+1&-1-2&2-1\\7-5&2-7&5-2\end{bmatrix}^T$$

$$=\begin{bmatrix}-12&16&-8\\2&-3&1\\2&-5&3\end{bmatrix}^T=\begin{bmatrix}-12&2&2\\16&-3&-5\\-8&1&3\end{bmatrix}$$

$$\begin{matrix}5&7&2&5\\1&-1&2&1\\1&1&1&1\\5&7&2&5\end{matrix}$$

$\longrightarrow$ (2M)

Hence by matrix inversion method, $x=A^{-1}D=\left(\frac{\text{adj}A}{\det A}\right)D$

$$x=\frac{1}{-4}\begin{bmatrix}-12&2&2\\16&-3&-5\\-8&1&3\end{bmatrix}\begin{bmatrix}9\\52\\0\end{bmatrix}$$

$$x=\frac{1}{-4}\begin{bmatrix}-108+104+0\\144-156+0\\-72+52+0\end{bmatrix}=\frac{-1}{4}\begin{bmatrix}-4\\-12\\-20\end{bmatrix}$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{-4}{-4}\begin{bmatrix}1\\3\\5\end{bmatrix} \Rightarrow \begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\3\\5\end{bmatrix}$$ $\longrightarrow$ (3M)

$x=1$, $y=3$, $z=5$ are solutions of given equation

$L_{20}(3)$ $2x-y+3z=9$, $x+y+z=6$, $x-y+z=2$.

Sol: given equations can be written as

$$A=\begin{bmatrix}2&-1&3\\1&1&1\\1&-1&1\end{bmatrix}\quad X=\begin{bmatrix}x\\y\\z\end{bmatrix}\quad D=\begin{bmatrix}9\\6\\2\end{bmatrix}$$

This is in the form of $AX=D$ $\longrightarrow$ (1M)

$$|A|=\begin{vmatrix}2&-1&3\\1&1&1\\1&-1&1\end{vmatrix}=2\begin{vmatrix}1&1\\-1&1\end{vmatrix}+1\begin{vmatrix}1&1\\1&1\end{vmatrix}+3\begin{vmatrix}1&1\\1&-1\end{vmatrix}$$

$|A|=2(1+1)+1(1-1)+3(-1-1)=4+0-6$.

$|A|=-2\neq 0$.

A^{-1} exists $\longrightarrow$ (1M)

$$\text{adj}\,A=\begin{bmatrix}1+1&1-1&-1-1\\-3+1&2-3&-1+2\\-1-3&3-2&2+1\end{bmatrix}^T$$

$$\begin{array}{cccc}1&1&1&1\\-1&1&1&-1\\-1&3&2&-1\\1&1&1&1\end{array}$$

$$=\begin{bmatrix}2&0&-2\\-2&-1&1\\-4&1&3\end{bmatrix}^T=\begin{bmatrix}2&-2&-4\\0&-1&1\\-2&1&3\end{bmatrix}$$

$\longrightarrow$ (2M)

Hence by matrix inversion method, $X=A^{-1}D=\dfrac{\text{adj}\,A}{\det A}$

$$X=\frac{1}{-2}\begin{bmatrix}2&-2&-4\\0&-1&1\\-2&1&3\end{bmatrix}\begin{bmatrix}9\\6\\2\end{bmatrix}$$

$$X=\frac{1}{-2}\begin{bmatrix}18-12-8\\0-6+2\\-18+6+6\end{bmatrix}=\frac{1}{-2}\begin{bmatrix}-2\\-4\\-6\end{bmatrix}$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{-2}{-2}\begin{bmatrix}1\\2\\3\end{bmatrix}\Rightarrow\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\2\\3\end{bmatrix}.$$

$x=1$, $y=2$, $z=3$. are solutions of given equation.

$\longrightarrow$ (3M)

L_{20}(iv) $x-y+3z=5,\ 4x+2y-z=0,\ -x+3y+z=5$

Sol:- given eqn can be written as

$$A=\begin{bmatrix}1&-1&3\\4&2&-1\\-1&3&1\end{bmatrix}\quad X=\begin{bmatrix}x\\y\\z\end{bmatrix}\quad D=\begin{bmatrix}5\\0\\5\end{bmatrix}$$

This is in the form of $AX=D$ $\longrightarrow$ (1M)

$$|A|=\begin{vmatrix}1&-1&3\\4&2&-1\\-1&3&1\end{vmatrix}=1\begin{vmatrix}2&-1\\3&1\end{vmatrix}+1\begin{vmatrix}4&-1\\-1&1\end{vmatrix}+3\begin{vmatrix}4&2\\-1&3\end{vmatrix}$$

$|A|=1(2+3)+1(4-1)+3(12+2)$

$|A|=5+3+42=50\neq A$

$\det A\neq 0$ A^{-1} exists $\longrightarrow$ (1M)

$$\operatorname{adj}A=\begin{bmatrix}2+3&1-4&12+2\\9+1&1+3&1-3\\1-6&12+1&2+4\end{bmatrix}^T$$

$$\begin{array}{cccc}2&-1&4&2\\3&1&-1&3\\-1&3&1&-1\\2&-1&4&2\end{array}$$

$$=\begin{bmatrix}5&-3&14\\10&4&-2\\-5&13&6\end{bmatrix}^T=\begin{bmatrix}5&10&-5\\-3&4&13\\14&-2&6\end{bmatrix}$$

Hence by matrix inversion method $\longrightarrow$ (2M)

$$X=A^{-1}D=\frac{\operatorname{adj}A}{\det A}\cdot D$$

$$X=\frac{1}{50}\begin{bmatrix}5&10&-5\\-3&4&13\\14&-2&6\end{bmatrix}\begin{bmatrix}5\\0\\5\end{bmatrix}$$

$$X=\frac{1}{50}\begin{bmatrix}25+0-25\\-15+0+65\\70-0+30\end{bmatrix}=\frac{1}{50}\begin{bmatrix}0\\50\\100\end{bmatrix}$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{50}{50}\begin{bmatrix}0\\1\\2\end{bmatrix}\Rightarrow\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\1\\2\end{bmatrix}$$

$\therefore$ $x=0,\ y=1,\ z=2$ are solutions of given equation. $\longrightarrow$ (3M)

L_{20} (v) $x+y+z=1$, $2x+2y+3z=6$, $x+4y+9z=3$

Sol:- given eqn can be written as

$$A=\begin{bmatrix}1&1&1\\2&2&3\\1&4&9\end{bmatrix} \quad X=\begin{bmatrix}x\\y\\z\end{bmatrix} \quad D=\begin{bmatrix}1\\6\\3\end{bmatrix}$$

This is in the form of $AX=D$ $\longrightarrow$ (1M)

$$|A|=\begin{vmatrix}1&1&1\\2&2&3\\1&4&9\end{vmatrix}=1\begin{vmatrix}2&3\\4&9\end{vmatrix}-1\begin{vmatrix}2&3\\1&9\end{vmatrix}+1\begin{vmatrix}2&2\\1&4\end{vmatrix}$$

$=1(18-12)-1(18-3)+1(8-2)$

$=6-15+6$

$=-3\neq 0$

A^{-1} exists. $\longrightarrow$ (1M)

$$\text{adj } A=\begin{bmatrix}18-12&3-18&8-2\\4-9&9-1&1-4\\3-2&2-3&2-2\end{bmatrix}^T$$

$$=\begin{bmatrix}6&-15&6\\-5&8&-3\\1&-1&0\end{bmatrix}^T=\begin{bmatrix}6&-5&1\\-15&8&-1\\6&-3&0\end{bmatrix}$$

$$\begin{matrix}2&3&2&2\\4&9&1&4\\1&1&1&1\\2&3&2&2.\end{matrix}$$

$\longrightarrow$ (2M)

Hence by matrix inversion method, $X=A^{-1}D=\frac{\text{adj}A}{\det A}D$.

$$X=\frac{1}{-3}\begin{bmatrix}6&-5&1\\-15&8&-1\\6&-3&0\end{bmatrix}\begin{bmatrix}1\\6\\3\end{bmatrix}$$

$$X=\frac{1}{-3}\begin{bmatrix}6-30+3\\-15+48-3\\6-18+0\end{bmatrix}=\frac{1}{-3}\begin{bmatrix}-21\\30\\-12\end{bmatrix}$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{-3}{-3}\begin{bmatrix}7\\-10\\4\end{bmatrix}$$

$x=7$, $y=-10$, $z=4$. $\longrightarrow$ (3M)

(20)(vi) $2x-y+3z=8,\ -x+2y+z=4,\ 3x+y-4z=0$

Sol:- given eqn can be written as $AX=D$

$$A=\begin{bmatrix}2&-1&3\\-1&2&1\\3&1&-4\end{bmatrix}\quad X=\begin{bmatrix}x\\y\\z\end{bmatrix}\quad D=\begin{bmatrix}8\\4\\0\end{bmatrix}$$ $\longrightarrow$ (1M)

$$|A|=\begin{vmatrix}2&-1&3\\-1&2&1\\3&1&-4\end{vmatrix}=2\begin{vmatrix}2&1\\1&-4\end{vmatrix}+1\begin{vmatrix}-1&1\\3&-4\end{vmatrix}+3\begin{vmatrix}-1&2\\3&1\end{vmatrix}$$

$|A|=2(-8-1)+1(4-3)+3(-1-6)$

$|A|=-18+1-21=-38\neq 0$

A^{-1} exists $\longrightarrow$ (1M)

$$\text{adj}\,A=\begin{bmatrix}-8-1&3-4&-1-6\\3-4&-8-9&-3-2\\-1-6&-3-2&4-1\end{bmatrix}^T$$

$$\begin{array}{cccc}2&1&-1&2\\1&-4&3&1\\-1&3&2&-1\\2&1&-1&2\end{array}$$

$$\text{adj}\,A=\begin{bmatrix}-9&-1&-7\\-1&-17&-5\\-7&-5&3\end{bmatrix}^T=\begin{bmatrix}-9&-1&-7\\-1&-17&-5\\-7&-5&3\end{bmatrix}$$

Hence by matrix inversion method. $\longrightarrow$ (2M)

$$X=A^{-1}D=\frac{\text{adj}\,A}{\det A}\cdot D.$$

$$X=\frac{1}{-38}\begin{bmatrix}-9&-1&-7\\-1&-17&-5\\-7&-5&3\end{bmatrix}\begin{bmatrix}8\\4\\0\end{bmatrix}$$

$$X=\frac{1}{-38}\begin{bmatrix}-72-4+0\\-8-68+0\\-56-20+0\end{bmatrix}=\frac{1}{-38}\begin{bmatrix}-76\\-76\\-76\end{bmatrix}$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix}=\frac{-38}{-38}\begin{bmatrix}2\\2\\2\end{bmatrix}\Rightarrow\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}2\\2\\2\end{bmatrix}$$

$x=2,\ y=2,\ z=2$

$\longrightarrow$ (3M)

9) If $A=\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix}$ then find $A^3-3A^2-A-3\underline{I}$

Sol:- given $A=\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix}$

$$A^2=A\cdot A=\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix}\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix} \longrightarrow (1M)$$

$$=\begin{bmatrix}1+0+3 & -2-2-1 & 1+2+1\\ 0+0-3 & 0+1+1 & 0-1-1\\ 3+0+3 & -6-1-1 & 3+1+1\end{bmatrix}$$

$$=\begin{bmatrix}4 & -5 & 4\\ -3 & 2 & -2\\ 6 & -8 & 5\end{bmatrix}$$

$$A^3=A^2\cdot A=\begin{bmatrix}4 & -5 & 4\\ -3 & 2 & -2\\ 6 & -8 & 5\end{bmatrix}\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix} \longrightarrow (1M)$$

$$=\begin{bmatrix}4+0+12 & -8-5-4 & 4+5+4\\ -3+0-6 & 6+2+2 & -3-2-2\\ 6+0+15 & -12-8-5 & 6+8+5\end{bmatrix}=\begin{bmatrix}16 & -17 & 3\\ -9 & 10 & -7\\ 21 & -25 & 19\end{bmatrix} \longrightarrow (1M)$$

$$A^3-3A^2-A-3\underline{I}=\begin{bmatrix}16 & -17 & 13\\ -9 & 10 & -7\\ 21 & -25 & 19\end{bmatrix}-3\begin{bmatrix}4 & -5 & 4\\ -3 & 2 & -2\\ 6 & -8 & 5\end{bmatrix}-\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix}-3\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}16 & -17 & 13\\ -9 & 10 & -7\\ 21 & -25 & 19\end{bmatrix}-\begin{bmatrix}12 & -15 & 12\\ -9 & 6 & -6\\ 18 & -24 & 15\end{bmatrix}-\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -1\\ 3 & -1 & 1\end{bmatrix}-\begin{bmatrix}3 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}16-12-1-3 & -17+15+2-0 & 13-12-1-0\\ -9+9-0-0 & 10-6-1-3 & -7+6+1-0\\ 21-18-3-0 & -25+24+1-0 & 19-15-1-3\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}=0$$

$\therefore A^3-3A^2-A-3\underline{I}=0$. $\longrightarrow (4M)$

L3

(8) If $A=\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}$ then show that $A^2-4A-5\underline{I}=0$

Sol:- given $A=\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}$

$$A^2=A\cdot A=\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix} \longrightarrow (1M)$$

$$A^2=\begin{bmatrix}1+4+4&2+2+4&2+4+2\\2+2+4&4+1+4&4+2+2\\2+4+2&4+2+2&4+4+1\end{bmatrix}$$

$$A^2=\begin{bmatrix}9&8&8\\8&9&8\\8&8&9\end{bmatrix} \longrightarrow (1M)$$

$$4A=4\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}=\begin{bmatrix}4&8&8\\8&4&8\\8&8&4\end{bmatrix}$$

$$5\underline{I}=5\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}5&0&0\\0&5&0\\0&0&5\end{bmatrix} \longrightarrow (2M)$$

Now, $A^2-4A-5\underline{I}=\begin{bmatrix}9&8&8\\8&9&8\\8&8&9\end{bmatrix}-\begin{bmatrix}4&8&8\\8&4&8\\8&8&4\end{bmatrix}-\begin{bmatrix}5&0&0\\0&5&0\\0&0&5\end{bmatrix}$

$$=\begin{bmatrix}9-4-5&8-8&8-8\\8-8&9-4-5&8-8\\8-8&8-8&9-4-5\end{bmatrix}$$

$$=\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}=0$$

$\therefore A^2-4A-5\underline{I}=0.$

$\longrightarrow (3M)$

ADDITION OF VECTORS

LAQ'S: $\boxed{L_{21}}$

1) If Points whose position Vectors are $3\bar{i}-2\bar{j}-\bar{k}$, $2\bar{i}+3\bar{j}-4\bar{k}$, $-\bar{i}+\bar{j}+2\bar{k}$, $4\bar{i}+5\bar{j}+\lambda\bar{k}$ are Coplanar then show that $\lambda = \frac{-146}{7}$

Sol, Let 'O' be the Origin

$\boxed{\text{Given Vectors } \overline{AB}, \overline{AC}, \overline{AD} \text{ are Coplanar } [\overline{AB}\ \overline{AC}\ \overline{AD}]=0}$

$\overline{OA} = 3\bar{i}-2\bar{j}-\bar{k}$

$\overline{OB} = 2\bar{i}+3\bar{j}-4\bar{k}$

$\overline{OC} = -\bar{i}+\bar{j}+2\bar{k}$

$\overline{OD} = 4\bar{i}+5\bar{j}+\lambda\bar{k}$ $\longrightarrow$ (2M)

$\overline{AB} = \overline{OB}-\overline{OA}$

$= 2\bar{i}+3\bar{j}-4\bar{k}-(3\bar{i}-2\bar{j}-\bar{k})$

$= 2\bar{i}+3\bar{j}-4\bar{k}-3\bar{i}+2\bar{j}+\bar{k}$ $\longrightarrow$ (1M)

$\overline{AB} = -\bar{i}+5\bar{j}-3\bar{k}$

$\overline{AC} = \overline{OC}-\overline{OA}$

$= -\bar{i}+\bar{j}+2\bar{k}-(3\bar{i}-2\bar{j}-\bar{k})$

$= -\bar{i}+\bar{j}+2\bar{k}-3\bar{i}+2\bar{j}+\bar{k}$

$\overline{AC} = -4\bar{i}+3\bar{j}+3\bar{k}$ $\longrightarrow$ (1M)

$\overline{AD} = \overline{OA}+\overline{OD}$

$= 3\bar{i}-2\bar{j}-\bar{k}-(4\bar{i}+5\bar{j}+\lambda\bar{k})$ $\longrightarrow$ (1M)

$= -i-7j+(\lambda+1)\bar{k}$

Given Vectors are Coplanar then $[\overline{AB}\ \overline{AC}\ \overline{AD}]=0$

$$\begin{vmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & \lambda+1 \end{vmatrix} [\bar{a}\ \bar{b}\ \bar{c}]=0$$

$$-1\begin{vmatrix} 3 & 3 \\ 7 & \lambda+1 \end{vmatrix} - 5\begin{vmatrix} -4 & 3 \\ 1 & \lambda+1 \end{vmatrix} + (-3)\begin{vmatrix} -4 & 3 \\ 1 & 7 \end{vmatrix} = 0$$

$-1(3(\lambda+1)-21)-5(-4(\lambda+1)-3)-3(-28-3)=0$

$-1(3\lambda+3-21)-5((-4\lambda+4)-3)-3(-31)=0$

$-1(3\lambda-18)+20\lambda+35+93=0$

$-3\lambda+18+20\lambda+35+93=0$

$17\lambda+146=0$

$17\lambda=-146$

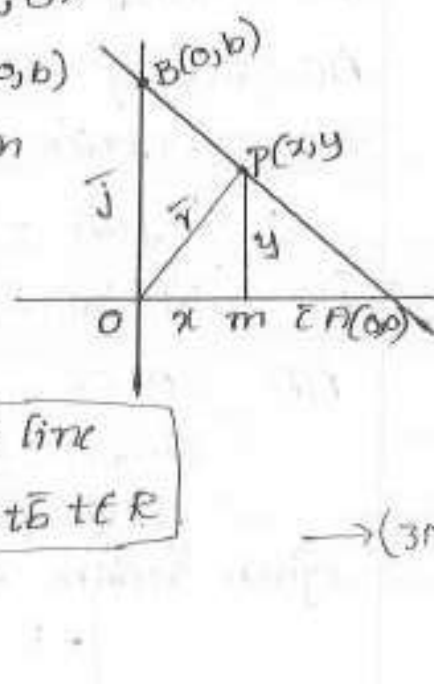
$\lambda=\frac{-146}{17}$ $\longrightarrow$ (2M)

L22 2) In two dimensional plane prove by using Vector Method equation of the line whose intercept on the axes are 'a' and 'b'. $\frac{x}{a}+\frac{y}{b}=1$

Sol) Let $\bar{i}, \bar{j}$ be unit vectors along $\overleftrightarrow{OX}, \overleftrightarrow{OY}$

Let $\overline{OA}=a\bar{i}, \overline{OB}=b\bar{j}$, and $A(a,0)$ $B(0,b)$

Let $\bar{r}=\overline{OP}$, Let m be the projection of x-axis



$\bar{r}=\overline{om}+\overline{mp}$

$\bar{r}=x\bar{i}+y\bar{j}$

Vector equation of the form of line Passing through A,B is $\bar{r}=(1-t)\bar{a}+t\bar{b}$ $t\in R$ $\longrightarrow$ (3M)

$\bar{r}=(1-t)\overline{OA}+t(\overline{OB}), t\in R$

$\bar{r}=(1-t)a\bar{i}+tb\bar{j} \rightarrow$ ②

Comparing i and j Coefficient

$x=(1-t)a \quad \Big| \quad y=tb$

$\frac{x}{a}=1-t \rightarrow (3) \quad \Big| \quad \frac{y}{b}=t \rightarrow (4) \qquad \longrightarrow (2M)$

Add 3 & 4

$\frac{x}{a}+\frac{y}{b}=1-\cancel{t}+\cancel{t}$

$\frac{x}{a}+\frac{y}{b}=1 \qquad \longrightarrow (2M)$

3) Find the vector equation of the Plane Passing through Point $4\bar{i}+3\bar{j}-\bar{k}$, $3\bar{i}+7\bar{j}-10\bar{k}$ and $2\bar{i}+5\bar{j}-7\bar{k}$ and show that the Point $\bar{i}+2\bar{j}-3\bar{k}$ lies in the plane

Sol) Given Points are

$\overline{OA}=4\bar{i}-3\bar{j}-\bar{k} \quad \overline{OB}=3\bar{i}+7\bar{j}-10\bar{k} \quad \overline{OC}=2\bar{i}+5\bar{j}-7\bar{k} \quad \overline{OD}=\bar{i}+2\bar{j}+3\bar{k}$

$\boxed{\text{Vector equation of the Plane Passing the Points } A, B, C \text{ is } \bar{r}=(1-t-s)(\overline{OA})+t(\overline{OB})+s(\overline{OC}),\ s,t \in R} \qquad \longrightarrow (2M)$

$\bar{r}=(1-t-s)(4\bar{i}-3\bar{j}-\bar{k})+t(3\bar{i}+7\bar{j}-10\bar{k})+s(2\bar{i}+5\bar{j}-7\bar{k})$

$\overline{AB}=\overline{OB}-\overline{OA}=3\bar{i}+7\bar{j}-10\bar{k}-(4\bar{i}-3\bar{j}-\bar{k})$

$=3\bar{i}+7\bar{j}-10\bar{k}-4\bar{i}+3\bar{j}+\bar{k}$

$\overline{AB}=-\bar{i}+10\bar{j}-9\bar{k} \qquad \longrightarrow (1M)$

$\overline{AC}=\overline{OC}-\overline{OA}$

$=2\bar{i}+5\bar{j}-7\bar{k}-(4\bar{i}-3\bar{j}-\bar{k})$

$=2\bar{i}+5\bar{j}-7\bar{k}+4\bar{i}+3\bar{j}+\bar{k} \qquad \longrightarrow (1M)$

$\overline{AC}=-2\bar{i}+8\bar{j}-6\bar{k}$

$\overline{AD}=\overline{OD}-\overline{OA}=\bar{i}+2\bar{j}-3\bar{k}-(4\bar{i}-3\bar{j}-\bar{k})$

$=\bar{i}+2\bar{j}-3\bar{k}-4\bar{i}+3\bar{j}+\bar{k}$

$=(-3\bar{i}+5\bar{j}-2\bar{k})$

$[\overline{AB}\ \overline{AC}\ \overline{AD}]=0 \qquad \longrightarrow (1M)$

$$= \begin{vmatrix} -1 & 10 & -9 \\ -2 & 8 & -6 \\ 3 & 5 & 2 \end{vmatrix} [\bar{a}\ \bar{b}\ \bar{c}]$$

$$= -1(-16+30) - 10(4-18) - 9(10+24)$$

$$= -1(14) - 10(-14) - 9(14)$$

$$= -14 + 140 - 126 = 140 - 140$$

$$= 0$$

$\therefore$ The Point $\bar{i}+2\bar{j}-3\bar{k}$ lies in the plane $\longrightarrow$ (2M)

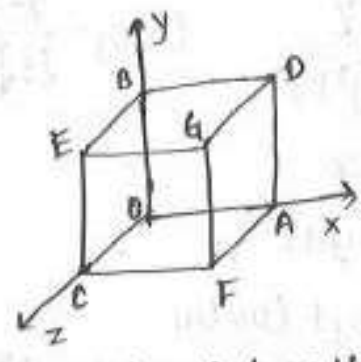
———— x ————

L32 (1)

A line makes angles $\theta_1, \theta_2, \theta_3$ and θ_4 with diagonals of a cube. Show that,

$\cos^2\theta_1 + \cos^2\theta_2 + \cos^2\theta_3 + \cos^2\theta_4 = 4/3$

Sol:



$\rightarrow$ (1M)

Let OABCDEFG be a cube of length 'a' units. Let $\bar{i}, \bar{j}, \bar{k}$ be the unit vectors in the directions of OA, OB, OC respectively.

then $\overline{OA} = a\bar{i}$, $\overline{OB} = a\bar{j}$, $\overline{OC} = a\bar{k}$

Let $\overline{OG}, \overline{AE}, \overline{BF}, \overline{CD}$ be the 4 diagonals of cube

$\overline{OG} = \overline{OA} + \overline{AD} + \overline{DG}$
$= \overline{OA} + \overline{OB} + \overline{OC}$
$= a\bar{i} + a\bar{j} + a\bar{k}$

$\overline{AE} = \overline{AO} + \overline{OB} + \overline{BE}$
$= -\overline{OA} + \overline{OB} + \overline{OC}$
$= -a\bar{i} + a\bar{j} + a\bar{k}$

$\overline{BF} = \overline{BO} + \overline{OA} + \overline{AF}$
$= \overline{OA} - \overline{OB} + \overline{OC}$
$= a\bar{i} - a\bar{j} + a\bar{k}$

$\overline{CD} = \overline{CF} + \overline{FG} + \overline{GD}$
$= \overline{OA} + \overline{OB} - \overline{OC}$
$= a\bar{i} + a\bar{j} - a\bar{k}$ $\rightarrow$ (2M)

Let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$ be the line making angles $\theta_1, \theta_2, \theta_3, \theta_4$ with diagonals of a cube.

$$\boxed{\cos\theta = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}}$$ $\rightarrow$ (1M)

Then, $\cos\theta_1 = \dfrac{\bar{r}\cdot\overline{OG}}{|\bar{r}||\overline{OG}|} = \dfrac{ax + ay + az}{\sqrt{x^2+y^2+z^2}\sqrt{a^2+a^2+a^2}} = \dfrac{a(x+y+z)}{\sqrt{x^2+y^2+z^2}\sqrt{3a^2}}$

$$\cos\theta_1 = \frac{a(x+y+z)}{\sqrt{3}(a)\sqrt{x^2+y^2+z^2}}$$

$$\cos\theta_1 = \frac{x+y+z}{\sqrt{3}\sqrt{x^2+y^2+z^2}} \longrightarrow (1M)$$

similarly,

$$\cos\theta_2 = \frac{-x+y+z}{\sqrt{3}\sqrt{x^2+y^2+z^2}} \qquad \cos\theta_3 = \frac{x-y+z}{\sqrt{3}\sqrt{x^2+y^2+z^2}}$$

$$\cos\theta_4 = \frac{x+y-z}{\sqrt{3}\sqrt{x^2+y^2+z^2}}$$

$$LHS = \cos^2\theta_1 + \cos^2\theta_2 + \cos^2\theta_3 + \cos^2\theta_4$$

$$= \frac{(x+y+z)^2}{3(x^2+y^2+z^2)} + \frac{(-x+y+z)^2}{3(x^2+y^2+z^2)} + \frac{(x-y+z)^2}{3(x^2+y^2+z^2)} + \frac{(x+y-z)^2}{3(x^2+y^2+z^2)}$$

$$= \frac{4(x^2+y^2+z^2)}{3(x^2+y^2+z^2)}$$

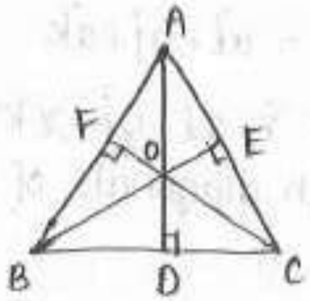
$$= \frac{4}{3} = RHS \longrightarrow (2M)$$

$$\therefore \cos^2\theta_1 + \cos^2\theta_2 + \cos^2\theta_3 + \cos^2\theta_4 = \frac{4}{3}$$

L22 (2)

Show that in any triangle, the altitudes are concurrent

Sol:- In the given ΔABC,



the altitudes AD and BE meet at 'O'

Take 'O' as origin.

$\overline{BC} = \overline{BO} + \overline{OC} = -\overline{b} + \overline{c}$

$\overline{CA} = \overline{CO} + \overline{OA} = -\overline{c} + \overline{a}$

$\overline{AB} = \overline{AO} + \overline{OB} = -\overline{a} + \overline{b} \longrightarrow (2M)$

since $\overline{AD} \perp \overline{BC}$, $\overline{a}\cdot(\overline{c} - \overline{b}) = 0$

Hence, $\overline{a}\cdot\overline{c} - \overline{a}\cdot\overline{b} = 0 \longrightarrow (1M)$

$$\bar{a}\cdot\bar{c} = \bar{a}\cdot\bar{b} \longrightarrow ①$$

Also, since $\overline{BE} \perp \overline{AC}$, $\bar{b}\cdot(\bar{a}-\bar{c}) = 0$

$$\bar{b}\cdot\bar{a} - \bar{b}\cdot\bar{c} = 0$$

$$\bar{b}\cdot\bar{a} = \bar{b}\cdot\bar{c} \longrightarrow ② \qquad \rightarrow (2M)$$

From equations ① & ②, we have,

$$\bar{a}\cdot\bar{c} = \bar{b}\cdot\bar{c}$$

Hence $\bar{c}\cdot(\bar{b}-\bar{a}) = 0$

$$\therefore \overline{CF} \perp \overline{AB}$$

Hence, the altitudes of a triangle are concurrent. $\longrightarrow$ (2M)

<u>L22(3)</u>

Let $\bar{a} = 4\bar{i}+5\bar{j}-\bar{k}$, $\bar{b} = \bar{i}-4\bar{j}+5\bar{k}$ and $\bar{c} = 3\bar{i}+\bar{j}-\bar{k}$. Find the vector which is perpendicular to both $\bar{a}$ and $\bar{b}$ whose magnitude is twenty one times the magnitude of $\bar{c}$.

<u>Sol:</u> $\bar{a} = 4\bar{i}+5\bar{j}-\bar{k}$, $\bar{b} = \bar{i}-4\bar{j}+5\bar{k}$ and $\bar{c} = 3\bar{i}+\bar{j}-\bar{k}$

$$\bar{a}\times\bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 4 & 5 & -1 \\ 1 & -4 & 5 \end{vmatrix} \longrightarrow (1M)$$

$$= \bar{i}(25-4) - \bar{j}(20+1) + \bar{k}(-16-5)$$

$$= 21\bar{i} - 21\bar{j} - 21\bar{k}$$

$$\bar{a}\times\bar{b} = 21(\bar{i}-\bar{j}-\bar{k}) \longrightarrow (1M)$$

$$|\bar{a}\times\bar{b}| = 21\sqrt{1+1+1} = 21\sqrt{3}$$

$$|\bar{c}| = \sqrt{9+1+1} = \sqrt{11} \longrightarrow (1M)$$

The vector which is perpendicular to both $\bar{a}$ and $\bar{b}$, whose magnitude is '21' times the magnitude of $\bar{c}$ is

$$\pm 21|\bar{c}|\frac{\bar{a}\times\bar{b}}{|\bar{a}\times\bar{b}|}$$

$\longrightarrow$ (1M)

Required vector is $\pm 21|\bar{c}|\dfrac{\bar{a}\times\bar{b}}{|\bar{a}\times\bar{b}|}$

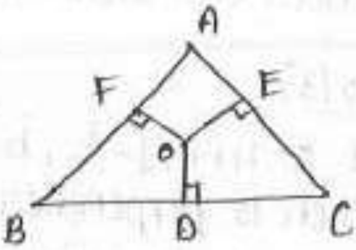
$$= \pm 21\sqrt{11}\cdot\frac{21(\bar{i}-\bar{j}-\bar{k})}{21\sqrt{3}}\cdot\frac{\sqrt{3}}{\sqrt{3}}$$

$$= \pm\frac{21\sqrt{33}}{3}(\bar{i}-\bar{j}-\bar{k})$$

$$= \pm 7\sqrt{33}(\bar{i}-\bar{j}-\bar{k}) \longrightarrow (3M)$$

L22 (4)

Show that in a triangle, the perpendicular bisectors of the sides are concurrent.



Sol: In the given ΔABC, let the midpoints of BC, CA, AB be D, F and F respectively. Let the perpendicular bisectors drawn to BC and CA at D and E meet at 'O'. Join OE with respect to O. Let the position vectors of A, B and C be $\bar{a}, \bar{b}$ and $\bar{c}$

from the figure. $\bar{OA}=\bar{a}$, $\bar{OB}=\bar{b}$, $\bar{OC}=\bar{c}$ $\longrightarrow (2M)$

$\bar{BC}=-\bar{b}+\bar{c}$ $\quad \bar{CA}=-\bar{c}+\bar{a}$, $\bar{AB}=-\bar{a}+\bar{b}$

$\bar{OD}=\frac{1}{2}(\bar{b}+\bar{c})$, $\bar{OE}=\frac{1}{2}(\bar{c}+\bar{a})$, $\bar{OF}=\frac{1}{2}(\bar{a}+\bar{b})$

since $\bar{OD}\perp\bar{BC}$, $\frac{1}{2}(\bar{b}+\bar{c})\cdot(-\bar{b}+\bar{c})=0$

Hence $|\bar{c}|^2-|\bar{b}|^2=0$ ——— ① $\longrightarrow (2M)$

Since $\bar{OE}\perp\bar{CA}$, $\frac{1}{2}(\bar{c}+\bar{a})\cdot(-\bar{c}+\bar{a})=0$

Hence, $|\bar{a}|^2-|\bar{c}|^2=0$ ——— ② $\longrightarrow (1M)$

On adding equations ① & ②

$$|\bar{a}|^2-|\bar{b}|^2=0 \Longrightarrow |\bar{b}|^2-|\bar{a}|^2=0$$

$$(\bar{b}+\bar{a})\cdot(\bar{b}-\bar{a})=0$$

$$\frac{1}{2}(\bar{b}+\bar{a})\cdot(\bar{b}-\bar{a})=0 \text{ i.e } \bar{OF}\perp\bar{AB}$$

$\therefore$ The perpendicular bisectors of the sides of a triangle are concurrent $\longrightarrow (2M)$

TRANSFORMATIONS

LAQ's

L_{23}(1):- In a $\triangle ABC$, prove that $\cos\frac{A}{2}+\cos\frac{B}{2}+\cos\frac{C}{2} = 4\cos\frac{\pi-A}{4}\cos\frac{\pi-B}{4}\cdot\cos\frac{\pi-C}{4}$

$\boxed{A+B+C=180}$

$\boxed{2\cos A\cos B = \cos(A+B)+\cos(A-B)}$

$\boxed{2\sin A\cos A = \sin A}$

Sol:- Given $A+B+C=\pi$

$$A+B=\pi-C$$
$$\frac{A+B}{2}=90-\frac{C}{2}$$

$$R.H.S = 4\cos\left(\frac{\pi-A}{4}\right)\cos\left(\frac{\pi-B}{4}\right)\cos\left(\frac{\pi-C}{4}\right)$$
$$= 2\left(2\cos\left(\frac{\pi-A}{4}\right)\cos\left(\frac{\pi-B}{4}\right)\right)\cos\left(\frac{\pi-C}{4}\right)$$
$$= 2\left(\cos\left(\frac{\pi-A}{4}+\frac{\pi-B}{4}\right)+\cos\left(\frac{\pi-A}{4}-\frac{\pi-B}{4}\right)\right)\cdot\cos\left(\frac{\pi-C}{4}\right)$$

$\boxed{\cos(A+B)+\cos(A-B) = 2\cos A\cos B}$ $\longrightarrow$ (2M)

$$= 2\left(\cos\left(\frac{\pi-A+\pi-B}{4}\right)+\cos\left(\frac{\pi-A-\pi+B}{4}\right)\right)\cdot\cos\left(\frac{\pi-C}{4}\right)$$
$$= 2\left(\cos\left(\frac{2\pi-(A+B)}{4}\right)+\cos\left(\frac{B-A}{4}\right)\right)\cdot\cos\left(\frac{A+B+C-C}{4}\right)$$
$$= 2\left(\cos\left(\frac{\pi}{2}-\left(\frac{A+B}{4}\right)\right)+\cos\left(\frac{B-A}{4}\right)\right)\cos\left(\frac{A+B}{4}\right)$$
$$= 2\left(\sin\left(\frac{A+B}{4}\right)+\cos\frac{B-A}{4}\right)\cdot\cos\left(\frac{A+B}{4}\right) \longrightarrow (2M)$$
$$= 2\sin\left(\frac{A+B}{4}\right)\cos\frac{A+B}{4} + 2\cos\left(\frac{B+A}{4}\right)\cos\left(\frac{A-B}{4}\right)$$
$$= \sin 2\left(\frac{A+B}{4}\right)+\cos\left(\frac{A+B}{4}+\frac{A-B}{4}\right)+\cos\left(\frac{A+B}{4}-\frac{A-B}{4}\right)$$
$$= \sin\left(\frac{A+B}{2}\right)+\cos\left(\frac{2A}{4}\right)+\cos\left(\frac{2B}{4}\right)$$
$$= \cos\frac{C}{2}+\cos\frac{A}{2}+\cos\frac{C}{2} = \cos\frac{A}{2}+\cos\frac{B}{2}+\cos\frac{C}{2} = L.H.S.$$

$\longrightarrow$ (3M)

L_{23}(2) If A,B,C are angles of a triangle, then prove that $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2} = 1-2\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

Sol:- given that $A+B+C=180$

$$\frac{A+B}{2}=90-\frac{C}{2} \Rightarrow \frac{C}{2}=90-\left(\frac{A+B}{2}\right)$$

$\boxed{A+B+C=180}$

$\boxed{2\cos A\cos B = \cos(A+B)+\cos(A-B)}$

$$L.H.S = \sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}$$
$$= \sin^2\frac{A}{2}+1-\cos^2\frac{B}{2}+\left(-\sin^2\frac{C}{2}\right)$$
$$= 1-\left(\cos^2\frac{B}{2}-\sin^2\frac{A}{2}\right)-\sin^2\frac{C}{2}$$

$\boxed{\cos^2 B-\sin^2 A = \cos(A+B)\cdot\cos(A-B)}$ $\longrightarrow$ (2M)

$= 1 - \left(\cos\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)\right) - \sin^2\frac{C}{2}$

$= 1 - \cos\left(90-\frac{C}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right) - \sin^2\frac{C}{2}$

$= 1 - \sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right) - \sin^2\frac{C}{2}\right.$

$= 1 - \sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \sin\frac{C}{2}\right]$ $\longrightarrow$ (2M)

$= 1 - \sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) + \cos\left(\frac{A+B}{2}\right)\right]$

$= 1 - \sin\frac{C}{2}\left(2\cos\frac{A}{2}\cdot\cos\frac{B}{2}\right)$

$= 1 - \cancel{2\sin}\, 2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

$= R.H.S.$ $\longrightarrow$ (3M)

6(b):- If A, B, C are angles in a triangle, then prove that $\cos A + \cos B - \cos C = -1 + 4\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$.

Sol:- Given A, B, C are angles in a triangle.

$A+B+C = 180$ $\qquad$ $C = 180 - (A+B)$

$A+B = 180 - C$ $\qquad$ $\frac{C}{2} = 90 - \left(\frac{A+B}{2}\right)$

$\frac{A+B}{2} = 90 - \frac{C}{2}$

$\boxed{\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)}$

$\boxed{\cos A = 1 - 2\sin^2\frac{A}{2}}$

$\boxed{\cos(A+B) + \cos(A-B) = 2\cos A\cos B}$

$L.H.S = \cos A + \cos B - \cos C$

$= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) - \cos C$

$= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) - \left(1 - 2\sin^2\frac{C}{2}\right)$

$\boxed{\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)}$

$= 2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) - 1 + 2\sin^2\frac{C}{2}$

$= -1 + 2\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right) + \sin\frac{C}{2}\right)$ $\longrightarrow$ (4M)

$= -1 + 2\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right) + \sin\left(90 - \left(\frac{A+B}{2}\right)\right)\right)$

$= -1 + 2\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right) + \cos\left(\frac{A+B}{2}\right)\right)$

$= -1 + 2\sin\frac{C}{2}\left(2\cos\frac{A}{2}\cdot\cos\frac{B}{2}\right)$

$= -1 + 4\sin\frac{C}{2}\cos\frac{A}{2}\cos\frac{B}{2}$

$= R.H.S$ $\longrightarrow$ (3M)

$L_{(23)}(4)$:- If A,B,C are angles in a triangle. Then prove that

$$\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}=1+4\sin\left(\frac{\pi-A}{4}\right)\cdot\sin\left(\frac{\pi-B}{4}\right)\sin\left(\frac{\pi-C}{4}\right)$$

Sol:- Given $A+B+C=\pi=180$

$$\frac{A+B}{2}=90-\frac{C}{2}$$

$R.H.S = 1+4\sin\left(\frac{\pi-A}{4}\right)\sin\left(\frac{\pi-B}{4}\right)\sin\left(\frac{\pi-C}{4}\right)$

$=1+2\left(2\sin\left(\frac{\pi-A}{4}\right)\sin\left(\frac{\pi-B}{4}\right)\right)\sin\left(\frac{\pi-C}{4}\right)$ → (1M)

$=1+2\left(\cos\left(\frac{\pi-A-\pi+B}{4}\right)-\cos\left(\frac{\pi-A+\pi-B}{4}\right)\right)\cdot\sin\left(\frac{A+B}{4}\right)$

$=1+2\left(\cos\left(\frac{B-A}{4}\right)-\sin\left(\frac{A+B}{4}\right)\right)\cdot\sin\left(\frac{A+B}{4}\right)$

$=1+2\sin\left(\frac{A+B}{4}\right)\cos\left(\frac{B-A}{2}\right)-2\sin^2\left(\frac{A+B}{4}\right)$

$=1-2\sin^2\left(\frac{A+B}{4}\right)+2\sin\left(\frac{A+B}{4}\right)\cdot\cos\left(\frac{A-B}{4}\right)$ → (3M)

$=\cos 2\left(\frac{A+B}{4}\right)+\sin\left(\frac{A+B}{4}+\frac{A-B}{4}\right)+\sin\left(\frac{A+B}{4}-\frac{A-B}{4}\right)$

$=\cos\left(\frac{A+B}{2}\right)+\sin\left(\frac{2A}{4}\right)+\sin\left(\frac{2B}{4}\right)$

$=\sin\frac{C}{2}+\sin\frac{A}{2}+\sin\frac{B}{2}$

$=\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}$

$=L.H.S.$ → (3M)

$L_{(23)}(5)$:- If $A+B+C=180$. then prove that

$$\cos^2\frac{A}{2}+\cos^2\frac{B}{2}+\cos^2\frac{C}{2}=2\left(1+\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$$

Sol:- Given $A+B+C=180$ $\quad C=180-(A+B)$

$$\frac{A+B}{2}=90-\frac{C}{2} \qquad \frac{C}{2}=90-\left(\frac{A+B}{2}\right)$$

$\boxed{\cos^2 A=1-\sin^2 A}$

$\boxed{\cos^2 A-\cos^2 B=\cos(A+B)\cos(A-B)}$

$\boxed{A+B+C=180 \text{ then } \sin\frac{C}{2}=\cos\left(\frac{A+B}{2}\right)}$

$L.H.S=\cos^2\frac{A}{2}+\cos^2\frac{B}{2}+\cos^2\frac{C}{2}$

$=\cos^2\frac{A}{2}+\cos^2\frac{B}{2}+1-\sin^2\frac{C}{2}$

$=1-\sin^2\frac{B}{2}+1-\sin^2\frac{C}{2}+\cos^2\frac{A}{2}$

$=2+\cos^2\frac{A}{2}-\sin^2\frac{B}{2}-\sin^2\frac{C}{2}$

$=2+\cos\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)-\sin^2\frac{C}{2}$

$=2+\cos\left(90-\frac{C}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)-\sin^2\frac{C}{2}$

$=2+\sin\frac{C}{2}\cdot\cos\left(\frac{A-B}{2}\right)-\sin^2\frac{C}{2}$ → (3M)

$= 2+\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right)-\sin\frac{C}{2}\right)$

$= 2+\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right)-\sin\left(90-\frac{(A+B)}{2}\right)\right)$

$\boxed{\cos(A-B)-\cos(A+B) = 2\sin A\sin B}$

$= 2+\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right)$

$= 2+\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right)$

$= 2+\sin\frac{C}{2}\left(2\sin\frac{A}{2}\cdot\sin\frac{B}{2}\right)$

$= 2\left(1+\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$

$=$ R.H.S.

$\longrightarrow$ (4M)

L(23)(6):- In a triangle, if A, B, C are angles then
$\cos A+\cos B+\cos C = 1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

Sol:- Given $A+B+C=180$

$$\frac{A+B}{2} = 90-\frac{C}{2} \Rightarrow \frac{C}{2} = 90-\left(\frac{A+B}{2}\right)$$

L.H.S $= \cos A+\cos B+\cos C$

$= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)+\cos C$

$\boxed{\cos C+\cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)}$

$= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)+1-\sin^2\frac{C}{2}$

$\longrightarrow$ (2M)

$= 1+2\sin\frac{C}{2}\cdot\cos\left(\frac{A-B}{2}\right)-2\sin^2\frac{C}{2}$

$\boxed{\cos A = 1-2\sin^2\frac{A}{2}}$

$= 1+2\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right)-\sin\frac{C}{2}\right)$

$= 1+2\sin\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right)-\sin\left(90-\left(\frac{A+B}{2}\right)\right)\right)$

$\boxed{A+B+C=180^\circ \text{ then } \sin\frac{C}{2} = \cos\left(\frac{A+B}{2}\right)}$

$= 1+2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right]$

$= 1+2\sin\frac{C}{2}\left[2\sin\frac{A}{2}\sin\frac{B}{2}\right]$

$= 1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$=$ R.H.S.

$\longrightarrow$ (5M)

$L_{(23)}(7)$:- If $A+B+C=\pi$, then prove that $\cos^2\frac{A}{2}+\cos^2\frac{B}{2}-\cos^2\frac{C}{2}$
$=2\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\sin\frac{C}{2}$

Sol:- Given that $A+B+C=\pi$

$L\cdot H\cdot S=\cos^2\frac{A}{2}+\cos^2\frac{B}{2}-\cos^2\frac{C}{2}$

$=\cos^2\frac{A}{2}+1-\sin^2\frac{B}{A}-\left[1-\sin^2\frac{C}{2}\right]$ $\quad\boxed{\because \cos^2 A=1-\sin^2 A}$

$=\cos^2\frac{A}{2}-\sin^2\frac{B}{2}+\sin^2\frac{C}{2}$ $\quad\boxed{\because \cos^2 A-\sin^2 B=\cos(A+B)\cos(A-B)}$

$=\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)+\sin^2\frac{C}{2}$ $\quad\boxed{\cos(A+B)+\cos(A-B)=2\cos A\cos B}$

$=\sin\frac{C}{2}\cos\left[\frac{A-B}{2}\right]+\sin^2\frac{C}{2}$

$=\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right)+\sin\frac{C}{2}\right]$ $\longrightarrow$ (4M)

$=\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right)+\cos\left(\frac{A+B}{2}\right)\right]$

$=\sin\frac{C}{2}\left(2\cos\frac{A}{2}\cos\frac{B}{2}\right)$

$=2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

$=R\cdot H\cdot S$ $\longrightarrow$ (3M)

$L_{23}(8)$: If $A+B+C=2S$, then prove that
$\cos(S-A)+\cos(S-B)+\cos(S-C)+\cos S=4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

Sol:- $L\cdot H\cdot S=\cos(S-A)+\cos(S-B)+\cos(S-C)$

$=2\cos\left(\frac{S-A+S-B}{2}\right)\cos\left(\frac{S-A-S+B}{2}\right)+2\cos\left(\frac{S-C+S}{2}\right)\cos\left(\frac{S-C-S}{2}\right)$

$=2\cos\left[\frac{2S-(A+B)}{2}\right]\cdot\cos\left[\frac{-(A-B)}{2}\right]+2\cos\left[\frac{2S-C}{2}\right]\cdot\cos\left(\frac{-C}{2}\right)$ $\longrightarrow$ (2M)

$\boxed{\because \cos(-A)=\cos A}$

$=2\cos\frac{C}{2}\cos\left[\frac{B-A}{2}\right]+2\cos\left[\frac{A+B}{2}\right]\cdot\cos\frac{C}{2}$

$=2\cos\frac{C}{2}\left[\cos\left(\frac{B-A}{2}\right)+\cos\left(\frac{A+B}{2}\right)\right]$

$=2\cos\frac{C}{2}\left[2\cos\left(\frac{B-A+A+B}{4}\right)\cdot\cos\left(\frac{B-A-A-B}{4}\right)\right]$ $\longrightarrow$ (2M)

$=2\cos\frac{C}{2}\left[2\cos\left(\frac{2B}{4}\right)\cdot\cos\left(\frac{-2A}{4}\right)\right]$

$=4\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}$

$=R\cdot H\cdot S\cdot$ $\longrightarrow$ (3M)

4(23)(A)- If $A+B+C=\frac{3\pi}{2}$ prove that $\cos 2A+\cos 2B+\cos 2C=1-4\sin A\sin B\sin C$

Sol:- Given $A+B+C=\frac{3\pi}{2}=270^\circ$

$A+B=270^\circ-C$

$L.H.S=\cos 2A+\cos 2B+\cos 2C$

$=2\cos(A+B)\cos(A-B)+1-2\sin^2 C$ $\quad \boxed{\because \cos 2A=1-2\sin^2 A}$

$=2\cos(270^\circ-C)\cos(A-B)+1-2\sin^2 C$ $\quad \boxed{\because \cos C+\cos D=2\cos\left(\frac{C+D}{2}\right)\cdot\cos\left(\frac{C-D}{2}\right)}$

$=-2\sin C\cos(A-B)+1-2\sin^2 C$

$=1-2\sin C(\cos(A-B)+\sin C)$ → (3m)

$=1-2\sin C(\cos(A-B)+\sin(270^\circ-(A+B)))$

$=1-2\sin C(\cos(A-B)-\cos(A+B))$ $\quad \boxed{\because \cos(A-B)-\cos(A+B)=2\sin A\sin B}$

$=1-2\sin C(2\sin A\sin B)$

$=1-4\sin A\sin B\sin C$

$=R.H.S$ → (4m)

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PROPERTIES OF TRIANGLES

L.A Q's.

L24 (1) If $a=13$, $b=14$, $c=15$ show that $R=\frac{65}{8}$, $r=4$, $r_1=\frac{21}{2}$, $r_2=12$, $r_3=14$

Solution:- Given $a=13$ $b=14$ $c=15$

$$S=\frac{a+b+c}{2}$$

$$S=\frac{13+14+15}{2}$$

$$S=\frac{42}{2}$$

$$S=21$$

$$\boxed{S=\frac{a+b+c}{2}}$$

$$\Delta=\sqrt{S(S-a)(S-b)(S-c)}$$

$$=\sqrt{21(21-13)(21-14)(21-15)}$$

$$=\sqrt{21\times 8\times 7\times 6}$$

$$=\sqrt{7056}=84$$

$$\boxed{\Delta=\sqrt{S(S-a)(S-b)(S-c)}}$$

$\longrightarrow$ (2M)

$$\Delta=84$$

$$R=\frac{abc}{4\Delta}$$

$$R=\frac{13\times 14\times 15}{4\times 84}$$

$$=\frac{65}{8}$$

$$\boxed{R=\frac{abc}{4\Delta}}$$

$\longrightarrow$ (2M)

$$r=\frac{\Delta}{S}$$

$$r=\frac{84}{21}=4$$

$$\boxed{r=\frac{\Delta}{S}}$$

$r_1=\frac{\Delta}{S-a}$	$r_2=\frac{\Delta}{S-b}$	$r_3=\frac{\Delta}{S-c}$
$=\frac{84}{21-13}$	$=\frac{84}{21-14}$	$=\frac{84}{21-15}$
$=\frac{84}{8}$	$=\frac{84}{7}$	$=\frac{84}{6}$
$=\frac{21}{2}$	$=12$	$=14$

$\longrightarrow$ (3M)

L24 (2):- If $r_1=2$, $r_2=3$, $r_3=6$ and $r=1$ prove that $a=3$, $b=4$, $c=5$

Solution:- $r_1=2$ $r_2=3$ $r_3=6$ $r=1$

$$\Delta = \sqrt{r \cdot r_1 \cdot r_2 \cdot r_3} = \sqrt{1 \cdot 2 \cdot 3 \cdot 6} = \sqrt{36} = 6$$

$\boxed{\Delta = \sqrt{r \cdot r_1 \cdot r_2 \cdot r_3}}$

$$r = \frac{\Delta}{s}$$

$\boxed{r = \frac{\Delta}{s}}$

$$1 = \frac{6}{s}$$

$$s = \frac{6}{1}$$

$$s = 6$$

$\longrightarrow$ (4M)

$$r_1 = \frac{\Delta}{s-a}, \quad 2 = \frac{6}{6-a}, \quad 6-a = 3, \quad 6-3 = a, \quad \boxed{a=3}$$

$$r_2 = \frac{\Delta}{s-b}, \quad 3 = \frac{6}{6-b}, \quad 6-b = 2, \quad 6-2 = b, \quad \boxed{b=4}$$

$$r_3 = \frac{\Delta}{s-c}, \quad 6 = \frac{6}{6-c}, \quad 6-c = 1, \quad 6-1 = c, \quad \boxed{c=5}$$

$\therefore a=3,\ b=4,\ c=5$ $\longrightarrow$ (3M)

ii. In ΔABC if $r_1=8$, $r_2=12$, $r_3=24$ find a, b, c

Solution:- Given $r_1=8$, $r_2=12$, $r_3=24$

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$\boxed{\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$

$$\frac{1}{r} = \frac{1}{8} + \frac{1}{12} + \frac{1}{24}$$

$$\frac{1}{r} = \frac{3+2+1}{24} = \frac{6}{24} = \frac{1}{4}$$

$\longrightarrow$ (1M)

$$\Delta = \sqrt{r \cdot r_1 \cdot r_2 \cdot r_3} = \sqrt{4 \cdot 8 \cdot 12 \cdot 24} = \sqrt{9216} = 96$$

$\boxed{\Delta = \sqrt{r \cdot r_1 \cdot r_2 \cdot r_3}}$

$\longrightarrow$ (1M)

$$r = \frac{\Delta}{s}$$

$\boxed{r = \frac{\Delta}{s}}$

$$s = \frac{\Delta}{r} = \frac{96}{4} = 24$$

$$s = 24$$

$\longrightarrow$ (1M)

$r_1 = \frac{\Delta}{s-a}$ $\qquad$ $r_2 = \frac{\Delta}{s-b}$ $\qquad$ $r_3 = \frac{\Delta}{s-c}$

$8 = \frac{96}{24-a}$ $\qquad$ $12 = \frac{96}{24-b}$ $\qquad$ $24 = \frac{96}{24-c}$

$24-a=12$ $\qquad$ $24-b=8$ $\qquad$ $24-c=4$

$24-12=a$ $\qquad$ $24-8=b$ $\qquad$ $24-4=c$

$\boxed{a=12}$ $\qquad$ $\boxed{b=16}$ $\qquad$ $\boxed{c=20}$

$\therefore a=12,\ b=16,\ c=20.$ $\longrightarrow$ (4M)

$L_{24}(3)$:- Show that $\frac{r_1}{bc}+\frac{r_2}{ca}+\frac{r_3}{ab} = \frac{1}{r}-\frac{1}{2R}$

Solution:- $LHS = \frac{r_1}{bc}+\frac{r_2}{ca}+\frac{r_3}{ab}$

$\boxed{a = 2R\sin A}$

$\boxed{r_1 = s\tan\frac{A}{2}}$

$= \frac{ar_1+br_2+cr_3}{abc}$

$= \sum\frac{ar_1}{abc}$

$= \frac{1}{abc}\sum ar_1$

$= \frac{1}{abc}\sum(2R\sin A)s\tan\frac{A}{2}$

$= \frac{2Rs}{abc}\sum 2\sin\frac{A}{2}\cdot\cos\frac{A}{2}\cdot\frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}$

$= \frac{4Rs}{abc}\sum\sin^2 A/2$ $\longrightarrow$ (2M)

$= \frac{4Rs}{abc}(\sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2})$

$= \frac{4Rs}{4R\Delta}(\sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2})$

$= \frac{1}{r}(1-2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2})$

$= \frac{1}{r}(1-\frac{4R\sin A/2\sin B/2\sin C/2}{2R})$ $\longrightarrow$ (3M)

$= \frac{1}{r}(1-\frac{r}{2R})$

$= \frac{1}{r}-\frac{r}{2R}(\frac{1}{r})$

$= \frac{1}{r}-\frac{1}{2R}$

$\therefore LHS = RHS.$ $\longrightarrow$ (2M)

$L_{24}(4)$:- Show that $r+r_1+r_2-r_3 = 4R\cos C$

Solution: $LHS = r+r_1+r_2-r_3$

$$\boxed{\begin{aligned} r &= 4R\sin\tfrac{A}{2}\sin\tfrac{B}{2}\sin\tfrac{C}{2} \\ r_1 &= 4R\sin\tfrac{A}{2}\cos\tfrac{B}{2}\cos\tfrac{C}{2} \\ r_2 &= 4R\cos\tfrac{A}{2}\sin\tfrac{B}{2}\cos\tfrac{C}{2} \\ r_3 &= 4R\cos\tfrac{A}{2}\cos\tfrac{B}{2}\sin\tfrac{C}{2} \end{aligned}}$$

$$\sin A\cos B+\cos A\sin B=\sin(A+B) \longrightarrow (1M)$$

$$r_1+r_2 = 4R\sin\tfrac{A}{2}\cdot\cos\tfrac{B}{2}\cdot\cos\tfrac{C}{2}+4R\cos\tfrac{A}{2}\cdot\sin\tfrac{B}{2}\cdot\cos\tfrac{C}{2}$$
$$=4R\cos\tfrac{C}{2}\left[\sin\tfrac{A}{2}\cos\tfrac{B}{2}+\cos\tfrac{A}{2}\sin\tfrac{B}{2}\right]$$
$$=4R\cos\tfrac{C}{2}\sin\left[\tfrac{A+B}{2}\right]$$
$$=4R\cos\tfrac{C}{2}\sin(90-\tfrac{C}{2})$$
$$=4R\cos\tfrac{C}{2}\cdot\cos\tfrac{C}{2}$$
$$=4R\cos^2\tfrac{C}{2} \text{ --- } (1) \longrightarrow (3M)$$

$$r-r_3 = 4R\sin\tfrac{A}{2}\sin\tfrac{B}{2}\sin\tfrac{C}{2}-4R\cos\tfrac{A}{2}\cos\tfrac{B}{2}\sin\tfrac{C}{2}$$
$$=4R\sin\tfrac{C}{2}\left[\sin\tfrac{A}{2}\sin\tfrac{B}{2}-\cos\tfrac{A}{2}\cos\tfrac{B}{2}\right]$$
$$\boxed{\cos A\cos B-\sin A\sin B=\cos(A+B)}$$
$$=-4R\sin\tfrac{C}{2}\cos\left(\tfrac{A+B}{2}\right)$$
$$=-4R\sin\tfrac{C}{2}\cos(90-\tfrac{C}{2})$$
$$=-4R\sin\tfrac{C}{2}\sin\tfrac{C}{2}=-4R\sin^2\tfrac{C}{2}$$
$$=-4R\sin^2\tfrac{C}{2} \text{ --- } (2)$$

From (1) & (2)

$$r+r_1+r_2-r_3=4R\cos^2\tfrac{C}{2}-4R\sin^2\tfrac{C}{2} \qquad \boxed{\cos 2A=\cos^2 A-\sin^2 A}$$
$$=4R\cos C \longrightarrow (3M)$$

L_{24}(ii):- Show that $r+r_3+r_1-r_2=4R\cos B$

<u>Solution</u>:- $LHS = r+\overline{r_3+r_1}-r_2$

$$r_1+r_3=4R\sin\tfrac{A}{2}\cdot\cos\tfrac{B}{2}\cdot\cos\tfrac{C}{2}+4R\cos\tfrac{A}{2}\cos\tfrac{B}{2}\sin\tfrac{C}{2}$$
$$=4R\cos\tfrac{B}{2}\left[\sin\tfrac{A}{2}\cos\tfrac{C}{2}+\cos\tfrac{A}{2}\sin\tfrac{C}{2}\right]$$
$$\boxed{\sin A\cos B+\cos A\sin B=\sin(A+B)}$$
$$=4R\cos\tfrac{B}{2}\sin\left(\tfrac{A+C}{2}\right)$$
$$=4R\cos\tfrac{B}{2}\sin(90-\tfrac{B}{2})$$

$= 4R\cos\frac{B}{2}\cos\frac{B}{2}$

$r_1+r_3 = 4R\cos^2\frac{B}{2}$ —— ① $\longrightarrow$ (3M)

$r-r_2 = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} - 4R\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$

$= 4R\sin\frac{B}{2}\left[\sin\frac{A}{2}\sin\frac{C}{2} - \cos\frac{A}{2}\cos\frac{C}{2}\right]$

$\boxed{\cos A\cos B - \sin A\sin B = \cos(A+B)}$

$= -4R\sin\frac{B}{2}\left(\cos\left(\frac{A+C}{2}\right)\right)$

$= -4R\sin\frac{B}{2}\cos\left(90-\frac{B}{2}\right)$

$= -4R\sin\frac{B}{2}\sin\frac{B}{2} = -4R\sin^2\frac{B}{2}$

$= -4R\sin^2\frac{B}{2}$ —— ② $\longrightarrow$ (3M)

from ① & ② we have

$r+r_3+r_1-r_2 = 4R\cos^2\frac{B}{2} - 4R\sin^2\frac{B}{2}$ $\qquad \boxed{\cos 2A = \cos^2 A - \sin^2 A}$

$= 4R\cos B.$ $\longrightarrow$ (1M)

L24 (5):- In ΔABC, Prove that $r_1+r_2+r_3-r = 4R$

Solution:- $LHS = r_1+r_2+r_3-r$

$r_2+r_3 = 4R\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2} + 4R\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

$= 4R\cos\frac{A}{2}\left[\sin\frac{B}{2}\cos\frac{C}{2} + \cos\frac{B}{2}\sin\frac{C}{2}\right]$

$= 4R\cos\frac{A}{2}\sin\left(\frac{B+C}{2}\right)$

$= 4R\cos\frac{A}{2}\sin\left(90-\frac{A}{2}\right)$

$= 4R\cos\frac{A}{2}\cos\frac{A}{2}$

$= 4R\cos^2\frac{A}{2}$ —— ① $\longrightarrow$ (3M)

$r_1-r = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} - 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$= 4R\sin\frac{A}{2}\left[\cos\frac{B}{2}\cos\frac{C}{2} - \sin\frac{B}{2}\sin\frac{C}{2}\right]$

$\boxed{\cos A\cos B - \sin A\sin B = \cos(A+B)}$

$= 4R\sin\frac{A}{2}\cos\left(\frac{B+C}{2}\right)$

$= 4R\sin\frac{A}{2}\cos\left(90-\frac{A}{2}\right)$

$= 4R\sin\frac{A}{2}\sin\frac{A}{2}$

$r_1-r = 4R\sin^2\frac{A}{2}$ —— ② $\longrightarrow$ (3M)

from ① & ②

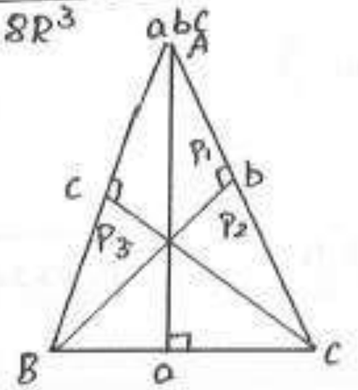
$$r_1+r_2+r_3-r = 4R\cos^2\frac{A}{2}+4R\sin^2\frac{A}{2}$$
$$=4R\left(\cos^2\frac{A}{2}+\sin^2\frac{A}{2}\right)$$
$$=4R$$

$\boxed{\cos^2\frac{A}{2}+\sin^2\frac{A}{2}=1}$

$\therefore$ LHS = RHS. $\longrightarrow$ (1M)

L24(6):- If P_1, P_2, P_3 are the altitudes drawn from vertices A, B, C to the opposite side of a triangle respectively then show that i. $\frac{1}{P_1}+\frac{1}{P_2}+\frac{1}{P_3}=\frac{1}{r}$

ii $P_1 \cdot P_2 \cdot P_3 = \frac{(abc)^2}{8R^3} = \frac{8\Delta^3}{abc}$



Area of the triangle ABC $= \frac{1}{2}\times$ base $\times$ height

$\Delta = \frac{1}{2}aP_1$ $\quad$ $\Delta = \frac{1}{2}bP_2$ $\quad$ $\Delta = \frac{1}{2}cP_3$

$P_1 = \frac{2\Delta}{a}$ $\quad$ $P_2 = \frac{2\Delta}{b}$ $\quad$ $P_3 = \frac{2\Delta}{c}$ $\longrightarrow$ (2M)

i $\frac{1}{P_1}+\frac{1}{P_2}+\frac{1}{P_3} = \frac{1}{\frac{2\Delta}{a}}+\frac{1}{\frac{2\Delta}{b}}+\frac{1}{\frac{2\Delta}{c}}$

$$=\frac{a}{2\Delta}+\frac{b}{2\Delta}+\frac{c}{2\Delta}$$
$$=\frac{a+b+c}{2\Delta} = \frac{2s}{2\Delta}$$
$$=\frac{s}{\Delta}$$
$$=\frac{1}{r}$$ $\longrightarrow$ (2M)

ii $P_1 \cdot P_2 \cdot P_3 = \left(\frac{2\Delta}{a}\right)\left(\frac{2\Delta}{b}\right)\left(\frac{2\Delta}{c}\right) = \frac{8\Delta^3}{abc}$ $\quad$ $\boxed{\Delta = \frac{abc}{4R}}$

$$=\frac{8}{abc}\left(\frac{abc}{4R}\right)^3$$
$$=\frac{8(abc)^3}{abc\cdot 64R^3}$$

$P_1 \cdot P_2 \cdot P_3 = \frac{(abc)^2}{8R^3}$ $\longrightarrow$ (3M)

$$\frac{bc-r_2r_3}{r_1} = \frac{ca-r_3r_1}{r_2} = r.$$

$$\therefore \frac{ab-r_1r_2}{r_3} = \frac{bc-r_2r_3}{r_1} = \frac{ca-r_3r_1}{r_2}. \longrightarrow (1M)$$

L24(8):- Show that $\cos^2\frac{A}{2}+\cos^2\frac{B}{2}+\cos^2\frac{C}{2}=2+\frac{r}{2R}$

Solution:- $LHS=\cos^2\frac{A}{2}+\cos^2\frac{B}{2}+\cos^2\frac{C}{2}$

$=\frac{1+\cos A}{2}+\frac{1+\cos B}{2}+\frac{1+\cos C}{2}$ $\boxed{\cos^2\frac{A}{2}=\frac{1+\cos A}{2}}$

$=\frac{1}{2}[1+\cos A+1+\cos B+1+\cos C]$ $\longrightarrow$ (1M)

$=\frac{1}{2}[3+\cos A+\cos B+\cos C]$

$=\frac{1}{2}[3+1+4\cdot\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}]$ $\longrightarrow$ (2M)

$=\frac{1}{2}[4+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}]$

$=\frac{4}{2}+\frac{4}{2}\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$=2+2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$=2+\frac{4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}{2R}$

$=2+\frac{r}{2R}$ $\longrightarrow$ (4M)

$LHS=RHS$

L24(9):- If $r:R:r_1=2:5:12$ then prove that the triangle is right angled at A

Solution:- Given $r:R:r_1=2:5:12$

$\frac{r}{2}=\frac{R}{5}=\frac{r_1}{12}=K$ (say)

$r=2K$ | $R=5K$ | $r_1=12K$ $\longrightarrow$ (1M)

$r_1-r=12K-2K$

$=10K$

$=2(5K)=2R$

$r_1-r=2R$

$\boxed{\begin{array}{l} r=4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \\ r_1=4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}\end{array}}$ $\longrightarrow$ (2M)

$r_1-r=4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}-4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=2R$

$=4R\sin\frac{A}{2}[\cos\frac{B}{2}\cos\frac{C}{2}-\sin\frac{B}{2}\sin\frac{C}{2}]$

$=4R\sin\frac{A}{2}\cos(\frac{B+C}{2})=2R$

$\boxed{\cos A\cos B-\sin A\sin B=\cos(A+B)}$

$4R\sin\frac{A}{2}\cos(90-\frac{A}{2})=2R$

$4R\sin\frac{A}{2}\sin\frac{A}{2}=2R$ $\longrightarrow$ (2M)

$$4R\sin^2\frac{A}{2} = 2R$$
$$2\sin^2\frac{A}{2} = 1$$
$$\sin^2\frac{A}{2} = \frac{1}{(\sqrt{2})^2}$$
$$\sin\frac{A}{2} = \frac{1}{\sqrt{2}}$$
$$\sin\frac{A}{2} = \sin 45$$
$$\frac{A}{2} = 45$$
$$\boxed{A = 90^\circ} \longrightarrow (2M)$$

L24(10):- Prove that $a^3\cos(B-C) + b^3\cos(C-A) + c^3\cos(A-B) = 3abc$

Solution:- $a^3\cos(B-C) + b^3\cos(C-A) + c^3\cos(A-B)$

$= \Sigma a^3 \cos(B-C)$

$= \Sigma a^2 \cdot a\cos(B-C)$

$= \Sigma a^2(2R\sin A)\cos(B-C)$

$= \Sigma a^2 \cdot 2R\sin(180-(B+C))\cos(B-C)$

$= \Sigma a^2 \cdot R(2R\sin(B+C)\cos(B-C)) \longrightarrow (2M)$

$= \Sigma a^2 \cdot R(\sin(B+C+B-C) + \sin(B+C-B+C))$

$= \Sigma a^2 \cdot R(\sin 2B + \sin 2C)$

$$\boxed{\begin{aligned}\sin 2B &= 2\sin B\cos B\\ \sin 2C &= 2\sin C\cos C\end{aligned}}$$

$= \Sigma a^2 R(2\sin B \cdot \cos B + 2\sin C \cos C)$

$= \Sigma a^2 (2R\sin B\cos B + 2R\sin C \cdot \cos C)$

$$\boxed{\begin{aligned}a &= 2R\sin A\\ b &= 2R\sin B\\ c &= 2R\sin C\end{aligned}}$$

$= \Sigma a^2 (b\cos B + c\cos C)$

$= \Sigma a^2 b\cos B + \Sigma a^2 c\cos C \longrightarrow (3M)$

$= a^2 b\cos B + b^2 c\cos C + c^2 a\cos A + a^2 c\cos C + b^2 a\cos A + c^2 b\cos B$

$= ab(a\cos B + b\cos A) + bc(b\cos C + c\cos B) + ca(c\cos A + a\cos C)$

$= ab(c) + bc(a) + ca(b)$

$$\boxed{\begin{aligned}a &= b\cos B + c\cos C\\ b &= c\cos A + a\cos C\\ c &= b\cos A + a\cos B\end{aligned}}$$

$= 3abc$

$= RHS \longrightarrow (2M)$

L24(11):- $a\cos^2\frac{A}{2} + b\cos^2\frac{B}{2} + c\cos^2\frac{C}{2} = S + \frac{\Delta}{R}$

Solution: $LHS = a\cos^2\frac{A}{2} + b\cos^2\frac{B}{2} + c\cos^2\frac{C}{2}$

$= a\left(\frac{1+\cos A}{2}\right) + b\left(\frac{1+\cos B}{2}\right) + c\left(\frac{1+\cos C}{2}\right) \longrightarrow (1M)$

$= \frac{a}{2} + \frac{a\cos A}{2} + \frac{b}{2} + \frac{b\cos B}{2} + \frac{c}{2} + \frac{c\cos C}{2}$

$= \frac{a+b+c}{2} + \frac{1}{2}[a\cos A + b\cos B + c\cos C] \longrightarrow (2M)$

$$s+\frac{1}{2}\left[2R\sin A\cos A+2R\sin B\cos B+2R\sin C\cos C\right]$$

$$s+\frac{R}{2}\left[2\sin A\cos A+2\sin B\cos B+2\sin C\cos C\right]$$

$$s+\frac{R}{2}\left[\sin 2A+\sin 2B+\sin 2C\right]$$

$$s+\frac{R}{2}\left[4\sin A\sin B\sin C\right]$$

$$s+\frac{2R^2\sin A\sin B\sin C}{R}$$

$$s+\frac{\Delta}{R} \longrightarrow (4M)$$

L24 (12):- Prove that $\dfrac{\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}}{\cot A+\cot B+\cot C}=\dfrac{(a+b+c)^2}{a^2+b^2+c^2}$

Solution:-

$$\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2}=\frac{s(s-a)}{\Delta}+\frac{s(s-b)}{\Delta}+\frac{s(s-c)}{\Delta} \longrightarrow (1M)$$

$$\boxed{\cot\frac{A}{2}=\frac{s(s-a)}{\Delta} \quad \cot\frac{B}{2}=\frac{s(s-b)}{\Delta} \quad \cot\frac{C}{2}=\frac{s(s-c)}{\Delta}}$$

$$=\frac{s}{\Delta}\left[s-a+s-b+s-c\right]$$

$$=\frac{s}{\Delta}\left[3s-(a+b+c)\right] \qquad \boxed{a+b+c=2s}$$

$$=\frac{s}{\Delta}=\left[3s-2s\right]$$

$$\frac{s^2}{\Delta}=\frac{\left(\frac{a+b+c}{2}\right)^2}{\Delta}$$

$$=\frac{(a+b+c)^2}{4\Delta} \text{ —— (1)} \longrightarrow (2M)$$

$$\cot A+\cot B+\cot C=\frac{\cos A}{\sin A}+\frac{\cos B}{\sin B}+\frac{\cos C}{\sin C}$$

$$\boxed{\cos A=\frac{b^2+c^2-a^2}{2bc} \quad \cos B=\frac{a^2+c^2-b^2}{2ac} \quad \cos C=\frac{a^2+b^2-c^2}{2ab}}$$

$$=\frac{b^2+c^2-a^2}{2bc\sin A}+\frac{a^2+c^2-b^2}{2ac\sin B}+\frac{a^2+b^2-c^2}{2ab\sin C}$$

$$=\frac{b^2+c^2-a^2}{4(\frac{1}{2}bc\sin A)}+\frac{a^2+c^2-b^2}{4(\frac{1}{2}ac\sin B)}+\frac{a^2+b^2-c^2}{4(\frac{1}{2}ab\sin C)}$$

$$=\frac{b^2+c^2-a^2}{4\Delta}+\frac{a^2+c^2-b^2}{4\Delta}+\frac{a^2+b^2-c^2}{4\Delta}$$

$$\boxed{\Delta=\frac{1}{2}bc\sin A=\frac{1}{2}ac\sin B=\frac{1}{2}ab\sin C}$$

$$=\frac{b^2+c^2-a^2+a^2+c^2-b^2+a^2+b^2-c^2}{4\Delta}$$

$$=\frac{a^2+b^2+c^2}{4\Delta} \text{ —— (2)} \longrightarrow (3M)$$

$$\frac{(1)}{(2)}=\frac{\frac{(a+b+c)^2}{4\Delta}}{\frac{(a^2+b^2+c^2)}{4\Delta}}$$

$$=\frac{(a+b+c)^2}{a^2+b^2+c^2} \longrightarrow (1M)$$

Matrices SAQ's

$\boxed{S_{11}}$

1. $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos n\theta \end{bmatrix}$ then show that for all positive integers 'A',

$A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$

Sol. Given $A = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

let $s(n)$ be the given statement

$s(n) = A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$

for $n=1$, $L.H.S = A^1 = A$

$R.H.S = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = A$

$L.H.S = R.H.S$

$s(n)$ is true for $n=1$ $\longrightarrow (1M)$

Let us assume that $s(n)$ is true for $n=k$

i.e, $A^k = \begin{bmatrix} \cos k\theta & \sin k\theta \\ -\sin k\theta & \cos k\theta \end{bmatrix}$

$\therefore$ $S(n)$ is true, for $n=k$ $\longrightarrow (1M)$

For $n=k+1$,

$A^{k+1} = A^k \cdot A = \begin{bmatrix} \cos k\theta & \sin k\theta \\ -\sin k\theta & \cos k\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

$= \begin{bmatrix} \cos k\theta\cos\theta - \sin k\theta\sin\theta & \cos k\theta\sin\theta + \sin k\theta\cos\theta \\ -\sin k\theta\cos\theta - \cos k\theta\sin\theta & -\sin k\theta\sin\theta + \cos k\theta\cos\theta \end{bmatrix}$

$= \begin{bmatrix} \cos(k\theta+\theta) & \sin(k\theta+\theta) \\ -\sin(k\theta+\theta) & \cos(k\theta+\theta) \end{bmatrix}$

$= \begin{bmatrix} \cos(k+1)\theta & \sin(k+1)\theta \\ -\sin(k+1)\theta & \cos(k+1)\theta \end{bmatrix}$

$\Rightarrow s(n)$ is true for $n=k+1$ $\longrightarrow (1M)$

∴ By the principle of finite mathematical induction, it is true for all positive integers 'n'. → (1m)

S11 2. If $A = \begin{bmatrix} 3 & 4 \\ 1 & -1 \end{bmatrix}$ then for any integer $n \geq 1$ show that $A^n = \begin{bmatrix} 1+2n & 4n \\ n & 1-2n \end{bmatrix}$

Sol: Given $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ this problem can be solved by using mathematical induction.

Let $s(n)$ be the given statement

$$s(n) = A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$$

For $n=1$, $L.H.S = A^1 = A$

$$R.H.S = \begin{bmatrix} 1+2 & -4 \\ 1 & 1-2 \end{bmatrix} = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = A$$

$L.H.S = R.H.S$

$s(n)$ is true for $n=1$ → (1m)

Let us assume that $s(n)$ is true $n=k$

$$A^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$$ → (1m)

for $n = k+1$, $A^{k+1} = A^k \cdot A$

$$\begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3+6k-4k & -4-8k+4k \\ 3k+1-2k & -4k-1+2k \end{bmatrix}$$

$$= \begin{bmatrix} 1+2(k+1) & -4(k+1) \\ k+1 & 1-2(k+1) \end{bmatrix}$$

⇒ $s(n)$ is true for $n = k+1$

∴ By principle of mathematical induction, the given statement is true for all positive integers. → (2m)

3. If $\theta-\phi = \frac{\pi}{2}$, then show that $\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix} = 0$

Given $\theta-\phi = \frac{\pi}{2} \Rightarrow \boxed{\cos(\theta-\phi)=0}$

sol: L.H.S $= \begin{bmatrix} \cos^2\theta\cos^2\phi + \cos\theta\sin\theta\cos\phi\sin\phi & \cos^2\theta\cos\phi\sin\phi + \cos\theta\sin\theta\sin^2\phi \\ \cos\theta\sin\theta\cos^2\phi + \sin^2\theta\cos\phi\sin\phi & \cos\theta\sin\theta\cos\phi\sin\phi + \sin^2\theta\sin^2\phi \end{bmatrix}$ $\longrightarrow$ (1M)

$= \begin{bmatrix} \cos\theta\cos\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) & \cos\theta\sin\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) \\ \sin\theta\cos\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) & \sin\theta\sin\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) \end{bmatrix}$

$= \begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \sin\theta\cos\phi\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix}$ $\longrightarrow$ (2M)

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$

$\therefore \theta-\phi = \frac{\pi}{2}$ $\longrightarrow$ (1M)

$\cos(\theta-\phi) = \cos\frac{\pi}{2} = 0$

S_{11} 4. If $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ then show that $A^{-1} = A^T$

sol: $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$

$A = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$

Now, $A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$ $\longrightarrow$ (1M)

Consider $AA^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \frac{1}{3}\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$

$= \frac{1}{9}\begin{bmatrix} 1+4+4 & 2+2-4 & -2+4-2 \\ 2+2-4 & 4+1+4 & -4+2+2 \\ -2+4-2 & -4+2+2 & 4+4+1 \end{bmatrix}$

$= \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ $\longrightarrow$ (2M)

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$\therefore AA^T = I \Rightarrow A^{-1} = A^T$ $\longrightarrow$ (1M)

(S11) 5. If $A = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -3 & 0 \\ 5 & 4 \end{bmatrix}$ then verify that $(AB)' = B'A'$

Sol: $A \cdot B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 3 & -4 \end{bmatrix}_{2\times3} \cdot \begin{bmatrix} 1 & -2 \\ -3 & 0 \\ 5 & 4 \end{bmatrix}_{3\times2}$

$$= \begin{bmatrix} 2+3+10 & -4+0+8 \\ 1-9-20 & -2+0-16 \end{bmatrix}_{2\times2}$$

$$= \begin{bmatrix} 15 & 4 \\ -28 & -18 \end{bmatrix}$$

$$(AB)' = \begin{bmatrix} 15 & -28 \\ 4 & -18 \end{bmatrix} \longrightarrow (2M)$$

$B' = \begin{bmatrix} 1 & -3 & 5 \\ -2 & 0 & 4 \end{bmatrix}$ $\quad A' = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 2 & -4 \end{bmatrix}$

$$B'A' = \begin{bmatrix} 1 & -3 & 5 \\ -2 & 0 & 4 \end{bmatrix}_{2\times3} \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 2 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 2+3+10 & 1-9-20 \\ -4+0+8 & -2+0-16 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & -28 \\ 4 & -18 \end{bmatrix} \longrightarrow (2M)$$

(S11) 6. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ then show that $(aI+bE)^3 = a^3I + 3a^2bE$

Sol: Given that $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ & $E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Now, $aI + bE = a\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + b\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$

$$= \begin{bmatrix}a & 0\\ 0 & a\end{bmatrix} + \begin{bmatrix}0 & b\\ 0 & 0\end{bmatrix}$$

$$= \begin{bmatrix}a & b\\ 0 & a\end{bmatrix} \longrightarrow (1m)$$

$(aI+bE)^2 = (aI+bE)(aI+bE) = \begin{bmatrix}a & b\\ 0 & a\end{bmatrix}\begin{bmatrix}a & b\\ 0 & a\end{bmatrix}$

$$= \begin{bmatrix}a^2 & ab+ab\\ 0+0 & 0+a^2\end{bmatrix}$$

$$= \begin{bmatrix}a^2 & 2ab\\ 0 & a^2\end{bmatrix} \longrightarrow (1m)$$

$(aI+bE)^3 = (aI+bE)^2(aI+bE) = \begin{bmatrix}a^2 & 2ab\\ 0 & a^2\end{bmatrix}\begin{bmatrix}a & b\\ 0 & a\end{bmatrix}$

$$= \begin{bmatrix}a^3+0 & a^2b+2a^2b\\ 0+0 & 0+a^3\end{bmatrix}$$

$$= \begin{bmatrix}a^3 & 3a^2b\\ 0 & a^3\end{bmatrix} \rightarrow (1) \longrightarrow (1m)$$

R.H.S =

$a^3I + 3a^2bE = a^3\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + 3a^2b\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$

$$= \begin{bmatrix}a^3 & 0\\ 0 & a^3\end{bmatrix} + \begin{bmatrix}0 & 3a^2b\\ 0 & 0\end{bmatrix}$$

$$= \begin{bmatrix}a^3 & 3a^2b\\ 0 & a^3\end{bmatrix} \longrightarrow (2)$$

From eq (1) & (2) $(aI+bE)^3 = a^3I + 3a^2bE \longrightarrow (1m)$

5(ii)

7. If $A = \begin{bmatrix}2 & 1 & 2\\ 1 & 0 & 1\\ 2 & 2 & 1\end{bmatrix}$ $= 2\begin{vmatrix}0 & 1\\ 2 & 1\end{vmatrix} - 1\begin{vmatrix}1 & 1\\ 2 & 1\end{vmatrix} + 2\begin{vmatrix}1 & 0\\ 2 & 2\end{vmatrix}$

$= 2(0-2) - 1(1-2) + 2(2-0)$

$= -4 + 1 + 4$

$= 1 \neq 0$ $\longrightarrow$ (1M)

$$\text{Adj } A = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 2 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0-2 & 2-1 & 2-0 \\ 4-1 & 2-4 & 2-4 \\ 1-0 & 2-2 & 0-1 \end{bmatrix}^T$$

$$= \begin{bmatrix} -2 & 1 & 2 \\ 3 & -2 & -2 \\ 1 & 0 & -1 \end{bmatrix}^T = \begin{bmatrix} -2 & 3 & 1 \\ 1 & -2 & 0 \\ 2 & -2 & -1 \end{bmatrix}$$ $\longrightarrow$ (2M)

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{1}\begin{bmatrix} -2 & 3 & 1 \\ 1 & -2 & 0 \\ 2 & -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 3 & 1 \\ 1 & -2 & 0 \\ 2 & -2 & -1 \end{bmatrix}$$ $\longrightarrow$ (1M)

5.11

8. If $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ then find $A^{-1} = ?$

Sol: $|A| = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{vmatrix}$

$= 1\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} - 2\begin{vmatrix} 3 & 3 \\ 1 & 2 \end{vmatrix} + 1\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix}$

$= 1(1) - 2(3) + 1(1)$

$= 1 - 6 + 1$

$= -4$

$|A| \neq 0$, A^{-1} is exist $\longrightarrow$ (1M)

$$\text{Adj}A = \begin{bmatrix} 2 & 3 & 3 & 2 \\ 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 2 & 3 & 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & 3-6 & -3-2 \\ 1-4 & 2-1 & 2-1 \\ 6-2 & 3-3 & 2-6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -3 & 1 \\ -3 & 1 & 1 \\ 4 & 0 & -4 \end{bmatrix}^T$$

$$\boxed{A^{-1} = \frac{1}{|A|} \text{adj}A}$$

$$\text{Adj}A = \begin{bmatrix} 1 & -3 & 4 \\ -3 & 1 & 0 \\ 1 & 1 & 4 \end{bmatrix}$$ $\longrightarrow$ (2M)

$$\therefore A^{-1} = \frac{1}{|A|} \begin{bmatrix} 1 & -3 & 4 \\ -3 & 1 & 0 \\ 1 & 1 & 4 \end{bmatrix}$$

$$= \frac{1}{-4} \begin{bmatrix} 1 & -3 & 4 \\ -3 & 1 & 0 \\ 1 & 1 & 4 \end{bmatrix}$$ $\longrightarrow$ (1M)

(S11)

9. If $A = \begin{bmatrix} 7 & -2 \\ -1 & 2 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & -1 \\ 4 & 2 \\ -1 & 0 \end{bmatrix}$ then find AB' & BA'

Sol: $AB' = \begin{bmatrix} 7 & -2 \\ -1 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -2 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix}$

$$= \begin{bmatrix} -14+2 & 28-4 & 7 \\ 2-2 & -4+4 & 1 \\ -10-3 & 20+6 & -5 \end{bmatrix} = \begin{bmatrix} -12 & 24 & -7 \\ 0 & 0 & 1 \\ -13 & 26 & -5 \end{bmatrix}$$ $\longrightarrow$ (2M)

$$BA' = \begin{bmatrix} -2 & -1 \\ 4 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 7 & -1 & 5 \\ -1 & 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -14+2 & 2-2 & -10-3 \\ 28-4 & -4+4 & 20+6 \\ -7-0 & 1+0 & -5+0 \end{bmatrix} = \begin{bmatrix} -12 & 0 & -13 \\ 0 & 0 & 26 \\ -13 & 1 & -5 \end{bmatrix}$$ $\longrightarrow$ (2M)

(S_{11}) 10 If $A=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & -1 & 1\end{bmatrix}$ then show that $A^3=A^{-1}$

Sol: Given,

$$A=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & -1 & 1\end{bmatrix}$$

$$A^2=A\cdot A$$

$$=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & +1 & 1\end{bmatrix}\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & 1 & 1\end{bmatrix}=\begin{bmatrix}9-6+10 & -9+9-4 & 12-12-4\\ 6-6+0 & -6+9-4 & 8-12+4\\ 0-2+0 & 0+3-1 & 0-4+1\end{bmatrix}=\begin{bmatrix}3 & -4 & 4\\ 0 & -1 & 0\\ -2 & 2 & -3\end{bmatrix}$$

$\longrightarrow$ (2M)

Now,

$$A^4=A^2\cdot A^2$$

$$A^4=\begin{bmatrix}3 & -4 & 4\\ 0 & -1 & 0\\ -2 & 2 & -3\end{bmatrix}\begin{bmatrix}3 & -4 & 4\\ 0 & -1 & 0\\ -2 & 2 & -3\end{bmatrix}=\begin{bmatrix}9+0-8 & -12+4+8 & 12+0-12\\ 0+0+0 & 0+1+0 & 0+0+0\\ -6+0+6 & 8+(-2)-6 & -8+0+9\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\underline{I}$$

$\longrightarrow$ (1M)

$$A^4=\underline{I}$$

$$A\cdot A^3=\underline{I}$$

$$A^3=(A^{-1})\underline{I}$$

$$A^{-1}=A^3.$$

$\longrightarrow$ (1M)

(S_{11}) 11 If $A=\begin{bmatrix}2 & -4\\ -5 & 3\end{bmatrix}$ find $A+A'$, $A\cdot A'$

Sol:

$$A+A'=\begin{bmatrix}2 & -4\\ -5 & 3\end{bmatrix}+\begin{bmatrix}2 & -5\\ -4 & 3\end{bmatrix}$$

$$=\begin{bmatrix}2+2 & -4+(-5)\\ -5-4 & 3+3\end{bmatrix}$$

$$=\begin{bmatrix}4 & -9\\ -9 & 6\end{bmatrix}$$

$\longrightarrow$ (2M)

$$AA' = \begin{bmatrix} 2 & -4 \\ -5 & 3 \end{bmatrix}\begin{bmatrix} 2 & -5 \\ -4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4+16 & -10-12 \\ -10-12 & 25+9 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & -22 \\ -22 & 34 \end{bmatrix} \longrightarrow 2M$$

(S11) 2. If $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ then show that $adj\,A = 3A'$ and find A^{-1}

Sol:- Given $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} -1 & 2 & 2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}$$

$$|A| = -1\begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} - (-2)\begin{vmatrix} 2 & -2 \\ 2 & 1 \end{vmatrix} + (-2)\begin{vmatrix} 2 & 1 \\ 2 & -2 \end{vmatrix}$$

$$|A| = -1(1-4) + 2(2+4) + (-2)(-4-2)$$

$$|A| = 3+12+12$$

$$|A| = 27 \neq 0$$

$\therefore A^{-1}$ is exist $\longrightarrow$ (1M)

$$\text{Minor of } A = \begin{bmatrix} \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} & \begin{vmatrix} 2 & -2 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 2 & -2 \end{vmatrix} \\ \begin{vmatrix} -2 & -2 \\ -2 & 1 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 2 & -2 \end{vmatrix} \\ \begin{vmatrix} -2 & -2 \\ 1 & -2 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 2 & -2 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} \end{bmatrix}$$

$$=\begin{bmatrix} 1-4 & 2+4 & -4-2 \\ -2-4 & -1+4 & 2+4 \\ 4+2 & 2+4 & -1+4 \end{bmatrix}$$

$$=\begin{bmatrix} -3 & 6 & -6 \\ -6 & 3 & 6 \\ 6 & 6 & 3 \end{bmatrix} \longrightarrow (2M)$$

Cofactor of $A = \begin{bmatrix} 3 & -6 & -6 \\ 6 & 3 & -6 \\ 6 & -6 & 3 \end{bmatrix}$

$$\text{adj } A = \begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix} = 3\begin{bmatrix} -1 & 2 & 2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}$$

$\therefore$ adj $A = 3A'$

$$A^{-1} = \frac{1}{|A|}\text{adj } A$$

$$A^{-1} = \frac{1}{27}\begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix} = \frac{\cancel{3}}{\underset{9}{\cancel{27}}}\begin{bmatrix} -1 & 2 & 2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}$$

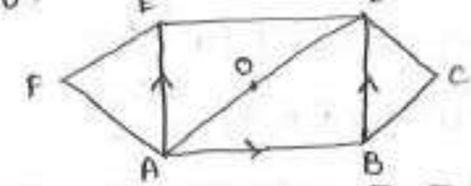
$$A^{-1} = \frac{1}{9}\begin{bmatrix} -1 & 2 & 2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix} \longrightarrow (1M)$$

S12

ADDITION OF VECTORS SAQ

(1) Let ABCDEF be a regular hexagon with centre 'O'. Show that $AB+AC+AD+AE+AF = 3AD = 6AO$

Sol: Given A, B, C, D, E, F be a regular hexagon with centre 'O'.



$\overline{AE} = \overline{BD}$

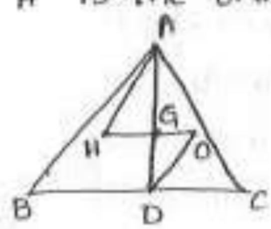
$\overline{AF} = \overline{CD}$

$\overline{AD} = 2\overline{AO}$ → (1M)

$$\text{LHS} = \overline{AB}+\overline{AC}+\overline{AD}+\overline{AE}+\overline{AF}$$
$$= \overline{AB}+\overline{AC}+\overline{AD}+\overline{BD}+\overline{CD}$$
$$= (\overline{AB}+\overline{BD})+\overline{AD}+(\overline{AC}+\overline{CD})$$
$$= \overline{AD}+\overline{AD}+\overline{AD}$$
$$= 3\overline{AD}$$
$$= 3(2\overline{AO})$$
$$= 6\overline{AO}$$ → (3M)

S12 (2) In $\triangle ABC$ if 'O' is circumcentre and H is the orthocentre, then shows that (i) $\overline{OA}+\overline{OB}+\overline{OC} = \overline{OH}$

(ii) $\overline{HA}+\overline{HB}+\overline{HC} = 2\overline{HO}$



Sol: 'O' is circumcentre

'H' is orthocentre

Let 'D' is midpoint of $\overline{BC}$ → (1M)

(i) $\overline{OD} = \dfrac{\overline{OB}+\overline{OC}}{2} \Rightarrow 2\overline{OD} = \overline{OB}+\overline{OC}$ $[\because 2\overline{OD} = \overline{AH}]$

$$= \overline{OA}+\overline{OB}+\overline{OC}$$
$$= \overline{OA}+2(\overline{OD})$$
$$= \overline{OA}+\overline{AH}$$
$$= \overline{OH}$$ → (2M)

(ii) $\overline{HA}+\overline{HB}+\overline{HC} = \overline{OA}-\overline{OH}+\overline{OB}-\overline{OH}+\overline{OC}-\overline{OH} = (\overline{OA}+\overline{OB}+\overline{OC})-3\overline{OH}$

$$= \overline{OH}-3\overline{OH}$$
$$= -2\overline{OH}$$
$$= 2\overline{HO}$$ → (1M)

S12

(3) a, b, c are coplanar prove that following four points are coplanar (i) $-\bar{a}+4\bar{b}-3\bar{c}$, $3\bar{a}+2\bar{b}-5\bar{c}$, $-3\bar{a}+8\bar{b}-5\bar{c}$, $-3\bar{a}+2\bar{b}+\bar{c}$

Sol: Given position vectors are

$\overline{OA} = -\bar{a}+4\bar{b}-3\bar{c}$ $\quad \overline{OB} = 3\bar{a}+2\bar{b}-5\bar{c}$ $\quad \overline{OC} = -3\bar{a}+8\bar{b}-5\bar{c}$ $\quad \overline{OD} = -3\bar{a}+2\bar{b}+\bar{c}$

$\overline{AB} = \overline{OB}-\overline{OA} = 3\bar{a}+2\bar{b}-5\bar{c}-(-\bar{a}+4\bar{b}-3\bar{c}) = 4\bar{a}-2\bar{b}-2\bar{c}$

$\overline{AC} = \overline{OC}-\overline{OA} = -3\bar{a}+8\bar{b}-5\bar{c}-(-\bar{a}+4\bar{b}-3\bar{c}) = -2\bar{a}+4\bar{b}-2\bar{c}$

$\overline{AD} = \overline{OD}-\overline{OA} = -3\bar{a}+2\bar{b}+\bar{c}-(-\bar{a}+4\bar{b}-3\bar{c}) = -2\bar{a}-2\bar{b}+4\bar{c}$ → (1M)

$$[\overline{AB}\ \overline{AC}\ \overline{AD}] = \begin{vmatrix} 4 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{vmatrix} [\bar{a}\ \bar{b}\ \bar{c}] \longrightarrow (1M)$$

$$= \left[4 \begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} - (-2)\begin{vmatrix} -2 & -2 \\ -2 & 4 \end{vmatrix} + (-2)\begin{vmatrix} -2 & 4 \\ -2 & -2 \end{vmatrix}\right] [\bar{a}\ \bar{b}\ \bar{c}]$$

$$= [4\ (16-4) + 2(-8-4) - 2(4+8)]\ [\bar{a}\ \bar{b}\ \bar{c}]$$

$$\Rightarrow [4(12) + 2(-12) - 2(12)]\ [\bar{a}\ \bar{b}\ \bar{c}]$$

$$= [48 - 24 - 24][\bar{a}\ \bar{b}\ \bar{c}]$$

$$= [0]\ [\bar{a}\ \bar{b}\ \bar{c}]$$

$$[\overline{AB}\ \overline{AC}\ \overline{AD}] = 0$$

$\therefore$ Given points are coplanar $\longrightarrow (2M)$

(ii) $6\bar{a}+2\bar{b}-\bar{c}$, $2\bar{a}$, $-\bar{b}+3\bar{c}$, $-\bar{a}+2\bar{b}-4\bar{c}$, $-12\bar{a}-\bar{b}-3\bar{c}$

Sol: Given points are A, B, C, D position vectors are

$\overline{OA} = 6\bar{a}+2\bar{b}-\bar{c}$

$\overline{OB} = 2\bar{a}-\bar{b}+3\bar{c}$

$\overline{OC} = -\bar{a}+2\bar{b}-4\bar{c}$

$\overline{OD} = -12\bar{a}-\bar{b}-3\bar{c}$

$\overline{AB} = \overline{OB} - \overline{OA} = 2\bar{a}-\bar{b}+3\bar{c}-(6\bar{a}+2\bar{b}-\bar{c}) = -4\bar{a}-3\bar{b}+4\bar{c}$

$\overline{AC} = \overline{OC} - \overline{OA} = -\bar{a}+2\bar{b}-4\bar{c}-(6\bar{a}+2\bar{b}-\bar{c}) = -7\bar{a}-3\bar{c}$

$\overline{AD} = \overline{OD} - \overline{OA} = -12\bar{a}-\bar{b}-3\bar{c}-(6\bar{a}+2\bar{b}-\bar{c}) = -18\bar{a}-3\bar{b}-2\bar{c}$ $\longrightarrow (1M)$

$$[\overline{AB}\ \overline{AC}\ \overline{AD}] = \begin{vmatrix} -4 & -3 & 4 \\ -7 & 0 & -3 \\ -18 & -3 & -2 \end{vmatrix} [\bar{a}\ \bar{b}\ \bar{c}]$$

$$= [-4(0-9)+3(14-54)+4(21+0)]\ [\bar{a}\ \bar{b}\ \bar{c}]$$

$$\Rightarrow [4(-9)+3(-40)+4(21)]\ [\bar{a}\ \bar{b}\ \bar{c}]$$

$$\Rightarrow [36-126+84][\bar{a}\ \bar{b}\ \bar{c}]$$

$$\Rightarrow 120-120\ [\bar{a}\ \bar{b}\ \bar{c}]$$

$$\Rightarrow 0$$

$\therefore$ The given points A, B, C, D are coplanar.

$\longrightarrow (3M)$

S12

(4) If i, j, k are unit vectors along the positive directions of the co-ordinates axes then show that four points $4\bar{i}+5\bar{j}+\bar{k}$, $-\bar{j}-\bar{k}$, $3\bar{i}+9\bar{j}+4\bar{k}$, $-4\bar{i}+4\bar{j}+4\bar{k}$ are coplanar.

Sol:- Given points are A, B, C, D be position vectors.

$\overline{OA} = 4\bar{i}+5\bar{j}+\bar{k}$; $\overline{OB} = -\bar{j}-\bar{k}$; $\overline{OC} = 3\bar{i}+9\bar{j}+4\bar{k}$; $\overline{OD} = -4\bar{i}+4\bar{j}+4\bar{k}$

$\overline{AB} = \overline{OB}-\overline{OA} = (-\bar{j}-\bar{k}) - (4\bar{i}+5\bar{j}+\bar{k}) = -4\bar{i}-6\bar{j}-2\bar{k}$

$\overline{AC} = \overline{OC}-\overline{OA} = 3\bar{i}+9\bar{j}+4\bar{k} - (4\bar{i}+5\bar{j}+\bar{k}) = -\bar{i}+4\bar{j}+3\bar{k}$

$\overline{AD} = \overline{OD}-\overline{OA} = -4\bar{i}+4\bar{j}+4\bar{k} - (4\bar{i}+5\bar{j}+\bar{k}) = -8\bar{i}-\bar{j}+3\bar{k}$

$$[\overline{AB}\ \overline{AC}\ \overline{AD}] = \begin{vmatrix} -4 & -6 & -2 \\ -1 & 4 & 3 \\ 8 & -1 & 3 \end{vmatrix} [\bar{a}\ \bar{b}\ \bar{c}] \longrightarrow (2M)$$

$= [-4(12+3)+6(-3+24)-2(1+32)]\ [\bar{a}\ \bar{b}\ \bar{c}]$

$= [-4(15)+6(21)-2(33)]\ [\bar{a}\ \bar{b}\ \bar{c}]$

$= [-60+126-66][\bar{a}\ \bar{b}\ \bar{c}]$

$= 0$

∴ Given points are coplanar. $\longrightarrow (2M)$

S12

(5) Show that the line joining the pair of points $6a-4b+4c$, $-4c$ and the line joining the pair of points $-a-2b-3c$, $a+2b-5c$ intersect at the point $-4c$ and when a, b, c are non-coplanar vectors.

Sol:- Let $\overline{OA} = \bar{a} = 6\bar{a}-4\bar{b}+4\bar{c}$; $\overline{OB} = \bar{b} = -4\bar{c}$; $\overline{OC} = \bar{c} = -\bar{a}-2\bar{b}-3\bar{c}$

$\overline{OD} = \bar{d} = \bar{a}+2\bar{b}-5\bar{c}$

∴ The vector equation of the line joining points whose position vectors are $\overline{OA}$ and $\overline{OB}$ is $\bar{r} = (1-t)\bar{a}+t\bar{b}$ where $t \in R$

$\bar{r} = (1-t)(6\bar{a}-4\bar{b}+4\bar{c})+t(-4\bar{c}) \rightarrow (1)$ $\longrightarrow (1M)$

The vector eq of the line joining points whose position vectors are $\overline{OC}$ and $\overline{OD}$ is, $\bar{r} = (1-s)\overline{OC}+s(\overline{OD})$ where $s \in R$

$\bar{r} = (1-s)(-\bar{a}-2\bar{b}-3\bar{c})+s(\bar{a}+2\bar{b}-5\bar{c}) \rightarrow (2)$ $\longrightarrow (1M)$

from (1) & (2)

$(1-t)(6\bar{a}-4\bar{b}+4\bar{c})+t(-4\bar{c}) = (1-s)(-\bar{a}-2\bar{b}-3\bar{c})+s(\bar{a}+2\bar{b}-5\bar{c})$

Equating the components of $\bar{a}$ & $\bar{b}$

$6-6t = -1+s+s$ $\qquad$ $-4+4t = -2+2s+2s$

$6t+2s-7 = 0 \rightarrow (3)$ $\qquad$ $4t-4s-2 = 0$

$\qquad$ $2t-2s-1 = 0 \rightarrow (4)$

P.O.I of (3) & (4) $\qquad$ $6t+2s-7 = 0$

$2t-2s-1 = 0$

$\longrightarrow (1M)$

$$8t - 8 = 0$$
$$t = 1$$

$\therefore$ Point of intersection of (1) and (2)

$$(1-1)(6\bar{a} - 4\bar{b} + 4\bar{c}) + 1(-4\bar{c}) = -4\bar{c} \longrightarrow (1M)$$

S_{12}

(6) If $\bar{a}, \bar{b}, \bar{c}$ are noncoplanar. find the point of intersection of the line passing through the points $2\bar{a}+3\bar{b}-\bar{c}$, $3\bar{a}+4\bar{b}-2\bar{c}$ with the line joining the points $\bar{a}-2\bar{b}+3\bar{c}$, $\bar{a}-6\bar{b}+6\bar{c}$.

Sol: Let $\overline{OA} = \bar{a} = 2\bar{a}+3\bar{b}-\bar{b}$ $\quad \overline{OB} = 3\bar{a}+4\bar{b}-2\bar{c}$

$\overline{OC} = C = \bar{a}+2\bar{b}+3\bar{c}$ $\quad \overline{OD} = \bar{d} = \bar{a}-6\bar{b}-6\bar{c}$

The vector equation the line joining the points whose position vectors are $\overline{OA}$ and $\overline{OB}$ is $\bar{r} = (1-t)\bar{a}+t\bar{b}$; $t \in R$

$$\bar{r} = (1-t)(2\bar{a}+3\bar{b}-\bar{c}) + t(3\bar{a}+4\bar{b}-2\bar{c}) \rightarrow (1) \longrightarrow (1M)$$

The vector of the line joining points whose position vectors are $\overline{OC}$ and $\overline{OD}$ is $\bar{r} = (1-s)\overline{OC} + s(\overline{OD})$, $s \in R$

$$\bar{r} = (1-s)(\bar{a}-2\bar{b}+3\bar{c}) + s(\bar{a}-6\bar{b}+6\bar{c}) \rightarrow (2) \longrightarrow (1M)$$

from (1) & (2) comparing coefficients $\bar{a}, \bar{b}, \bar{c}$,

$$2-2t+3t = 1-s+s$$
$$2+t = 1$$
$$t = -1$$

from (1)

$$\bar{r} = (1+1)(2\bar{a}+3\bar{b}-\bar{c}) - 1(3\bar{a}+4\bar{b}-2\bar{c})$$
$$\bar{r} = 4\bar{a}+6\bar{b}-2\bar{c}-3\bar{a}-4\bar{b}+2\bar{c}$$
$$= \bar{a}+2\bar{b} \longrightarrow (2M)$$

S_{12}

(7) Find the vector eq of the line parallel to the vector $2\bar{i}-\bar{j}+2\bar{k}$ and passing through point A whose position vector is $3\bar{i}+\bar{j}-\bar{k}$. If P is a point on this line such that $AP = 15$ then find the position vector of P.

Sol: Let $\bar{a} = 3\bar{i}+\bar{j}-\bar{k}$ $\quad \bar{b} = 2\bar{i}-\bar{j}+2\bar{k}$

The vector eq of the line passing through the point whose position vector is $\bar{a}$ and parallel to vector $\bar{b}$ is $\bar{r} = \bar{a}+t\bar{b} \rightarrow (1)$

$$\bar{r} = (3\bar{i}+\bar{j}-\bar{k}) + t(2\bar{i}-\bar{j}+2\bar{k}) \rightarrow (2)$$

from (1) $\quad \overline{OP} = \overline{OA} + t\bar{b} \longrightarrow (2M)$

$$\overline{OP} - \overline{OA} = t\bar{b}$$
$$\overline{AP} = t\bar{b}$$
$$|\overline{AP}| = t|\bar{b}|$$
$$15 = t\sqrt{4+1+4} \Rightarrow 15 = t(\pm 3)$$
$$t = \pm 5$$

$t = 5$ from (2),

$\overline{OP} = (3i+j-k) + 5(2\bar{i}-\bar{j}+2\bar{k})$

$\overline{OP} = 13\bar{i} - 4\bar{j} + 9\bar{k}$

$t = -5$ from (2)

$\overline{OP} = (3\bar{i}+\bar{j}-\bar{k}) - 5(2\bar{i}-\bar{j}+2\bar{k})$

$\overline{OP} = -7\bar{i} + 6\bar{j} - 11\bar{k}$ $\longrightarrow$ (2M)

S_{18} (8) Let $\bar{a}, \bar{b}$ be non-collinear vectors. If $\bar{\alpha} = (x+4y)\bar{a} + (2x+y+1)\bar{b}$ and $\bar{\beta} = (y-2x+2)\bar{a} + (2x-3y-1)\bar{b}$ such that $3\bar{\alpha} = 2\bar{\beta}$ then find x & y.

Sol: Given $3\bar{\alpha} = 2\bar{\beta}$

$3[(x+4y)\bar{a} + (2x+y+1)\bar{b}] = 2[(y-2x+2)\bar{a} + (2x-3y-1)\bar{b}]$

$3(x+4y)\bar{a} + 3(2x+y+1)\bar{b} = 2(y-2x+2)\bar{a} + 2(2x-3y-1)\bar{b}$

comparing coefficients of $\bar{a}$ & $\bar{b}$ $\longrightarrow$ (1M)

$3x+12y = 2y-4x+4 \Rightarrow 7x+10y = 4 \rightarrow (1)$

$6x+3y+3 = 4x-6y-2 \Rightarrow 2x+9y = -5 \rightarrow (2)$

Solving (1) & (2)

$(1)\times 2 \quad 14x+20y = 8 \rightarrow (3)$ $\longrightarrow$ (1M)

$(2)\times 7 \quad 14x+63y = -35 \rightarrow (4)$

$(4)-(3)$

$\Rightarrow -43y = 43$

$y = -1$

$y=-1$ in eq $\rightarrow$ (1)

$7x - 10 = 4$

$7x = 14$

$x = 2$

$\therefore x = 2, \; y = -1$ $\longrightarrow$ (2M)

S_{19} (9) If $\bar{a}+\bar{b}+\bar{c} = \alpha\bar{d}$, $\bar{b}+\bar{c}+\bar{d} = \beta\bar{a}$ & $\bar{a}, \bar{b}, \bar{c}$ are non-coplanar vectors, then show that $\bar{a}+\bar{b}+\bar{c}+\bar{d} = 0$

Sol: Given $\bar{a}+\bar{b}+\bar{c} = \alpha\bar{d} \rightarrow (1)$ $\quad \bar{b}+\bar{c}+\bar{d} = \beta\bar{a} \rightarrow (2)$

from (2) $\bar{d} = \beta\bar{a} - \bar{b} - \bar{c}$ $\longrightarrow$ (1M)

Substitute in eq (1)

$\bar{a}+\bar{b}+\bar{c} = \alpha[\beta\bar{a} - \bar{b} - c] \Rightarrow \bar{a}+\bar{b}+\bar{c} = \alpha\beta\bar{a} - \alpha\bar{b} - \alpha\bar{c}$

Given $\bar{a}, \bar{b}, \bar{c}$ are non coplanar vectors

compare 'b' co-efficient on both $\longrightarrow$ (2M)

$1 = -\alpha \Rightarrow \alpha = -1$

from (1)

$\bar{a}+\bar{b}+\bar{c} = (-1)\bar{d}$

$\bar{a}+\bar{b}+\bar{c} = -\bar{d}$

$\therefore$ Hence proved $\longrightarrow$ (1M)

(S13)

(10) If $\bar{a}, \bar{b}, \bar{c}$ are non-coplanar vectors, then test for the collinearity of the following points whose position vectors are

(i) $\bar{a}-2\bar{b}+3\bar{c}$, $2\bar{a}+3\bar{b}-4\bar{c}$, $-7\bar{b}+10\bar{c}$

Given, $\bar{OA} = \bar{a}-2\bar{b}+3\bar{c}$; $\bar{OB} = 2\bar{a}+3\bar{b}-4\bar{c}$; $\bar{OC} = -7\bar{b}+10\bar{c}$

$$\bar{AB} = \bar{OB}-\bar{OA} = (2\bar{a}+3\bar{b}-4\bar{c})-(\bar{a}-2\bar{b}+3\bar{c}) \rightarrow (1M)$$
$$= \bar{a}+5\bar{b}-7\bar{c} \rightarrow (1)$$
$$\bar{AC} = \bar{OC}-\bar{OA} = -7\bar{b}+10\bar{c}-(\bar{a}-2\bar{b}+3\bar{c})$$
$$= -\bar{a}-5\bar{b}+7\bar{c}$$
$$\bar{AB} = \bar{a}+5\bar{b}-7\bar{c} = (-1)(-\bar{a}-5\bar{b}+7\bar{c})$$
$$= (-)\bar{AC}$$
$$= (-1)\bar{AC}$$
$$\bar{AB} = \lambda(\bar{AC})$$
$$\lambda = -1$$

$\therefore$ Given points are collinear $\rightarrow$ (3m)

(ii) Given $3\bar{a}-4\bar{b}+3\bar{c}$, $-4\bar{a}+5\bar{b}-6\bar{c}$, $4\bar{a}-7\bar{b}+6\bar{c}$

Let $\bar{OA} = 3\bar{a}-4\bar{b}+3\bar{c}$; $\bar{OB} = -4\bar{a}+5\bar{b}-6\bar{c}$; $\bar{OC} = 4\bar{a}-7\bar{b}+6\bar{c}$

$$\bar{AB} = \bar{OB}-\bar{OA} = -4\bar{a}+5\bar{b}-6\bar{c}-(3\bar{a}-4\bar{b}+3\bar{c}) = -7\bar{a}+9\bar{b}-9\bar{c}$$
$$\bar{AC} = \bar{OC}-\bar{OA} = 4\bar{a}-7\bar{b}+6\bar{c}-(3\bar{a}-4\bar{b}+3\bar{c}) = \bar{a}-3\bar{b}+3\bar{c}$$
$$\bar{AB} \neq \lambda(\bar{AC})$$

$\therefore$ Given points are non-collinear $\rightarrow$ (4m)

(iii) $\bar{OA} = 2\bar{a}+5\bar{b}-4\bar{c}$; $\bar{OB} = \bar{a}+4\bar{b}-3\bar{c}$; $\bar{OC} = 4\bar{a}+7\bar{b}-6\bar{c}$

$$\bar{AB} = \bar{OB}-\bar{OA} = \bar{a}+4\bar{b}-3\bar{c}-(2\bar{a}+5\bar{b}-4\bar{c}) = -\bar{a}-\bar{b}-\bar{c}$$
$$\bar{AC} = \bar{OC}-\bar{OA} = 4\bar{a}+7\bar{b}-6\bar{c}-(2\bar{a}+5\bar{b}-4\bar{c}) = 2\bar{a}+2\bar{b}-2\bar{c}$$
$$\bar{AC} = 2\bar{a}+2\bar{b}+(-2\bar{c}) = -2(\bar{a}-\bar{b}+\bar{c})$$
$$\bar{AC} = -2(\bar{AB})$$
$$\bar{AC} = \lambda(\bar{AB})$$
$$\lambda = -2$$

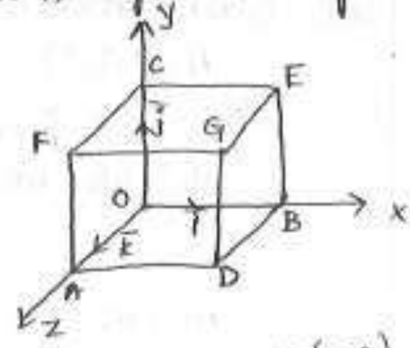
$\therefore$ Given points are collinear (4m)

—— x ——

PRODUCT OF VECTORS SAQ

S13

(1) prove that the smaller angle θ between any two diagonals of a cube given by $\cos\theta = \frac{1}{3}$



Sol: Let OABCDEFG be a cube of Length a units.

Let $\bar{i}, \bar{j}, \bar{k}$ be the unit vectors in the directions of $\overline{OA}, \overline{OB}, \overline{OC}$ respectively.

Let $\overline{OA} = a\bar{i}$; $\overline{OB} = a\bar{j}$; $\overline{OC} = a\bar{k}$

Then diagonals, → (1M)

$$\overline{OG} = \overline{OA} + \overline{AE} + \overline{EG}$$
$$= \overline{OA} + \overline{OB} + \overline{OC}$$
$$= a\bar{i} + a\bar{j} + a\bar{k}$$
$$\overline{AF} = \overline{AD} + \overline{OB} + \overline{BF}$$
$$= -\overline{OA} + \overline{OB} + \overline{OC}$$
$$= -a\bar{i} + a\bar{j} + a\bar{k}$$

→ (1M)

$$\cos\theta = \frac{(\overline{OG} \cdot \overline{AF})}{|\overline{OG}||\overline{AF}|} = \frac{(a)(a) + a(-a) + (a)(a)}{\sqrt{a^2+a^2+a^2}\sqrt{a^2+(-a)^2+a^2}}$$

$$\cos\theta = \frac{a^2 - a^2 + a^2}{\sqrt{3}a^2\sqrt{3}a^2} = \frac{a^2}{3a^2} = \frac{1}{3}$$

$$\theta = \cos^{-1}(1/3)$$ → (2M)

S13

(2) Find the unit vector perpendicular to the plane passing through the points (1,2,3) (2,-1,1) & (1,2,-4)

Sol: Given points are A = (1,2,3) B = (2,-1,1) C = (1,2,-4)

Let 'o' be the origin. The position vectors are $\overline{OA} = \bar{i} + 2\bar{j} + 3\bar{k}$
$\overline{OB} = 2\bar{i} - \bar{j} + \bar{k}$
$\overline{OC} = \bar{i} + 2\bar{j} - 4\bar{k}$

$$\overline{AB} = \overline{OB} - \overline{OA} = 2\bar{i} - \bar{j} + \bar{k} - (\bar{i} + 2\bar{j} + 3\bar{k})$$
$$= 2\bar{i} - \bar{j} + \bar{k} - \bar{i} - 2\bar{j} - 3\bar{k}$$
$$= \bar{i} - 3\bar{j} - 2\bar{k}$$

$$\overline{AC} = \overline{OC} - \overline{OA} = \bar{i} + 2\bar{j} - 4\bar{k} - \bar{i} - 2\bar{j} - 3\bar{k}$$ → (1M)

$$\overline{AC} = -7\bar{k}$$

$$\overline{AB} \times \overline{AC} = \begin{vmatrix} i & j & k \\ 1 & -3 & -2 \\ 0 & 0 & -7 \end{vmatrix} = \bar{i}\begin{vmatrix} -3 & -2 \\ 0 & -7 \end{vmatrix} - \bar{j}\begin{vmatrix} 1 & -2 \\ 0 & -7 \end{vmatrix} + \bar{k}\begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix}$$

$$= \bar{i}(21-0) - \bar{j}(-7+0) + \bar{k}(0-0)$$
$$= 21\bar{i} + 7\bar{j}$$ → (1M)

$$|\overline{AB} \times \overline{AC}| = \sqrt{(21)^2 + (7)^2} = \sqrt{441+49} = \sqrt{490} = 7\sqrt{10}$$

∴ unit vector perpendicular to plane passing through A, B, C

$$\pm\frac{(\overline{AB} \times \overline{AC})}{|\overline{AB} \times \overline{AC}|} = \frac{\pm|(21\bar{i} + 7\bar{j})|}{7\sqrt{10}} = \frac{\pm 7(3\bar{i} + \bar{j})}{7\sqrt{10}} = \frac{\pm(3\bar{i} + \bar{j})}{\sqrt{10}}$$

→ (2M)

S13

(3) Find the area of triangle whose vertices are $A(1,2,3)$ $B=(2,3,1)$ and $C=(3,1,2)$

Sol:- Given vertices of triangles

$A=(1,2,3)\quad B=(2,3,1)\quad C(3,1,2)$

the $\overline{OA}=\bar{i}+2\bar{j}+3\bar{k}\quad \overline{OB}=2\bar{i}+3\bar{j}+\bar{k}\quad \overline{OC}=3\bar{i}+\bar{j}+2\bar{k}$

$$\overline{AB}=\overline{OB}-\overline{OA}=2\bar{i}+3\bar{j}+\bar{k}-(\bar{i}+2\bar{j}+3\bar{k})$$
$$=2\bar{i}+3\bar{j}+\bar{k}-\bar{i}-2\bar{j}-3\bar{k}$$
$$=\bar{i}+\bar{j}-2\bar{k}$$
$$\overline{AC}=\overline{OC}-\overline{OA}=3\bar{i}+\bar{j}+2\bar{k}-(\bar{i}+2\bar{j}+3\bar{k})$$
$$=3\bar{i}+\bar{j}+2\bar{k}-\bar{i}-2\bar{j}-3\bar{k}$$
$$=2\bar{i}-\bar{j}-\bar{k} \longrightarrow (1M)$$

$$(\overline{AB}\times\overline{AC})=\begin{vmatrix}\bar{i} & \bar{j} & \bar{k}\\ 1 & 1 & -2\\ 2 & -1 & -1\end{vmatrix}=\bar{i}\begin{vmatrix}1 & -2\\ -1 & -1\end{vmatrix}-\bar{j}\begin{vmatrix}1 & -2\\ 2 & -1\end{vmatrix}+\bar{k}\begin{vmatrix}1 & 1\\ 2 & -1\end{vmatrix}$$
$$=\bar{i}(-1-2)-\bar{j}(-1+4)+k(-1-2)$$
$$=-3\bar{i}-3\bar{j}-3\bar{k} \longrightarrow (1M)$$

$$|\overline{AB}\times\overline{AC}|=\sqrt{(-3)^2+(-3)^2+(-3)^2}$$
$$=\sqrt{9+9+9}$$
$$=\sqrt{27}$$

Area of the triangle $=\frac{1}{2}|\overline{AB}\times\overline{AC}|$
$$=\frac{1}{2}|\sqrt{27}|$$
$$=\frac{3\sqrt{2}}{2}\text{ sq.units} \longrightarrow (2M)$$

S13 (4) Find the unit vector perpendicular to the plane determined by the points $P(1,-1,2)$ $Q(2,0,-1)$ & $R(0,2,1)$

Sol:- Let 'o' be the origin $\overline{OP}=\bar{i}-\bar{j}+2\bar{k}$; $\overline{OQ}=2\bar{i}-\bar{k}$; $\overline{OR}=2\bar{j}+\bar{k}$

$$\overline{PQ}=\overline{OQ}-\overline{OP}=(2\bar{i}-\bar{k})-(\bar{i}-\bar{j}+2\bar{k})$$
$$=\bar{i}+\bar{j}-3\bar{k}$$
$$\overline{PR}=\overline{OR}-\overline{OP}=2\bar{j}+\bar{k}-(\bar{i}-\bar{j}+2\bar{k})=(-\bar{i}+3\bar{j}-\bar{k}) \longrightarrow (1M)$$

Now $\overline{PQ}\times\overline{PR}=\begin{vmatrix}\bar{i} & \bar{j} & \bar{k}\\ 1 & 1 & -3\\ -1 & 3 & -1\end{vmatrix}=\bar{i}(-1+9)-\bar{j}(-1-3)+\bar{k}(3+1)$
$$=8\bar{i}+4\bar{j}+4\bar{k}$$
$$=4(2\bar{i}+\bar{j}+\bar{k})$$
$$|\overline{PQ}\times\overline{PR}|=4|2\bar{i}+\bar{j}+\bar{k}|=4\sqrt{4+1+1}=4\sqrt{6} \longrightarrow (1M)$$

$\therefore$ Unit vector perpendicular to the plane determined by the points P, Q and R is $\pm\frac{\overline{PQ}\times\overline{PR}}{|\overline{PQ}\times\overline{PR}|}=\pm\frac{4(2\bar{i}+\bar{j}+\bar{k})}{4\sqrt{6}}$
$$=\frac{2\bar{i}+\bar{j}+\bar{k}}{\sqrt{6}} \longrightarrow (2M)$$

S13

(5) If $a = 2\bar{i}+3\bar{j}+4\bar{k}$ $b=\bar{i}+\bar{j}-\bar{k}$ $c=+\bar{i}-\bar{j}+\bar{k}$, then compute $\bar{a}\times(\bar{b}\times\bar{c})$ and verify that it is perpendicular to $\bar{a}$

Sol:- $\bar{b}\times\bar{c} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix} = \bar{i}(1-1)-\bar{j}(1+1)+\bar{k}(-1-1) = -2\bar{j}-2\bar{k}$ $\longrightarrow$ (1M)

$\bar{a}\times(\bar{b}\times\bar{c}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 3 & 4 \\ 0 & -2 & -2 \end{vmatrix} = \bar{i}(-6+8)-\bar{j}(-4-0) = 2\bar{i}+4\bar{j}-4\bar{k}$ $\longrightarrow$ (1M)

Now $[(\bar{a}\times\bar{b})\times\bar{c}]\bar{a} = (2\bar{i}+4\bar{j}-4k)(2\bar{i}+3\bar{j}+4k) = 4+12-16=0$

$\therefore$ $\bar{a}\times(\bar{b}\times\bar{c})$ is perpendicular to $\bar{a}$ $\longrightarrow$ (2M)

S13 (6) If $\bar{a}=\bar{i}-2\bar{j}-3\bar{k}$, $\bar{b}=2\bar{i}+\bar{j}-\bar{k}$, $\bar{c}=\bar{i}+3\bar{j}-2\bar{k}$ verify that $\bar{a}\times(\bar{b}\times\bar{c}) \neq (\bar{a}\times\bar{b})\times\bar{c}$

Sol:- $\bar{a}\times\bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & -2 & -3 \\ 2 & 1 & -1 \end{vmatrix} = \bar{i}(2+3)-\bar{j}(-1+6)+\bar{k}(1-14) = 5\bar{i}-5\bar{j}+5\bar{k}$

$\bar{b}\times\bar{c} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 1 & -1 \\ 1 & 3 & -2 \end{vmatrix} = \bar{i}(-2+3)-\bar{j}(-4+1)+\bar{k}(6-1) = +\bar{i}+3\bar{j}+5\bar{k}$

$\bar{a}\times(\bar{b}\times\bar{c}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & -2 & -3 \\ 1 & 3 & 5 \end{vmatrix} = i(-10+9)-\bar{j}(5+3)+\bar{k}(3+2) = -\bar{i}-8\bar{j}+5\bar{k}$

$(\bar{a}\times\bar{b})\times\bar{c} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 5 & -5 & 5 \\ 1 & 3 & -2 \end{vmatrix} = \bar{i}(10-15)-\bar{j}(-10-5)+\bar{k}(15+5) = -5\bar{i}+15\bar{j}+20\bar{k}$ $\longrightarrow$ (4M)

S13 (7) $a=3\bar{i}-\bar{j}+2\bar{k}$, $b=-\bar{i}+3\bar{j}+2\bar{k}$, $c=4\bar{i}+5\bar{j}-2\bar{k}$ and $d=\bar{i}+3\bar{j}+5\bar{k}$ then compute the following (i) $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})$ & (ii) $(\bar{a}\times\bar{b})\bar{c}-(\bar{a}\times\bar{d}).\bar{b}$

Sol:-

(i) $(\bar{a}\times\bar{b})\times(\bar{c}\times\bar{d})$

$\bar{a}=3\bar{i}-\bar{j}+2\bar{k}$, $\bar{b}=-\bar{i}+3\bar{j}+2\bar{k}$

$\bar{a}\times\bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 3 & -1 & 2 \\ -1 & 3 & 2 \end{vmatrix} = \bar{i}(-2-6)-\bar{j}(6+2)+\bar{k}(9-1)$

$= -8\bar{i}-8\bar{j}+8\bar{k}$

$= 8(-\bar{i}-\bar{j}+\bar{k})$

$\longrightarrow$ (1M)

$\bar{c} = 4\bar{i} + 5\bar{j} - 2\bar{k} \qquad \bar{d} = \bar{i} + 3\bar{j} + 5\bar{k}$

$$\bar{c} \times \bar{d} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 4 & 5 & -2 \\ 1 & 3 & 5 \end{vmatrix} = \bar{i}(25+6) - \bar{j}(20+2) + \bar{k}(12-5)$$
$$= 31\bar{i} - 22\bar{j} + 7\bar{k}$$

$\bar{a} \times \bar{b} = 8(-\bar{i} - \bar{j} + \bar{k}) \quad , \quad \bar{c} \times \bar{d} = 31\bar{i} - 22\bar{j} + 7\bar{k}$ $\longrightarrow$ (2M)

$$(\bar{a} \times \bar{b}) \times (\bar{c} \times \bar{d}) = 8 \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -1 & -1 & 1 \\ 31 & -22 & 7 \end{vmatrix} = 8[i(-7+22) - j(-7-31) + k(22+31)]$$
$$= 8(15\bar{i} + 38\bar{j} + 53\bar{k})$$ $\longrightarrow$ (1M)

(ii) $(\bar{a} \times \bar{b})\bar{c} - (\bar{a} \times \bar{d})\bar{b}$

$$\bar{a} \times \bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 3 & -1 & 2 \\ -1 & 3 & 2 \end{vmatrix} = \bar{i}(-2-6) - \bar{j}(6+2) + \bar{k}(9-1)$$
$$= -8\bar{i} - 8\bar{j} + 8\bar{k}$$
$$= 8(-\bar{i} - \bar{j} + \bar{k})$$

$\therefore (\bar{a} \times \bar{b}) \cdot \bar{c} = 8(-\bar{i} - \bar{j} + \bar{k})(4\bar{i} + 5\bar{j} - 2\bar{k}) = 8[-4-5-2] = -88 \rightarrow ①$ $\longrightarrow$ (1M)

$$\bar{a} \times \bar{d} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 3 & -1 & 2 \\ 1 & 3 & 5 \end{vmatrix} = \bar{i}(-5-6) - \bar{j}(15-2) + \bar{k}(9+1) = -11\bar{i} - 13\bar{j} + 10\bar{k}$$

$\therefore (\bar{a} \times \bar{d}) \cdot \bar{b} = (-11\bar{i} - 13\bar{j} + 10\bar{k})(-\bar{i} + 3\bar{j} + 2\bar{k}) = 11 - 39 + 20 = -8 \rightarrow ②$

Now $(\bar{a} \times \bar{b}) \cdot \bar{c} - (\bar{a} \times \bar{d})\bar{b} = -88 - (-8) = -80$ $\longrightarrow$ (3M)

S13 (8) Find 'λ' in order that the four points $A(3,2,1)$ $B(4,\lambda,5)$ $C(4,2,-2)$ and $D(6,5,-1)$ be coplanar.

Sol:- Let $\bar{OA} = 3\bar{i} + 2\bar{j} + \bar{k}$, $\bar{OB} = 4\bar{i} + \lambda\bar{j} + 5\bar{k}$, $\bar{OC} = 4\bar{i} + 2\bar{j} - 2k$

$\bar{OD} = 6\bar{i} + 5\bar{j} - \bar{k}$

$\bar{AB} = \bar{OB} - \bar{OA} = 4\bar{i} + \lambda\bar{j} + 5\bar{k} - 3\bar{i} - 2\bar{j} - \bar{k} = \bar{i} + (\lambda - 2)\bar{j} + 4\bar{k}$

$\bar{AC} = \bar{OC} - \bar{OA} = 4\bar{i} + 2\bar{j} - 2\bar{k} - 3\bar{i} - 2\bar{j} - \bar{k} = \bar{i} + 0\bar{j} - 3\bar{k}$

$\bar{AD} = \bar{OD} - \bar{OA} = 6\bar{i} + 5\bar{j} - \bar{k} - 3\bar{i} - 2\bar{j} - \bar{k} = 3\bar{i} + 3\bar{j} - 2\bar{k}$ $\longrightarrow$ (1M)

The given points are coplanar $[\bar{AB} \ \bar{AC} \ \bar{AD}] = 0$

$$\begin{vmatrix} 1 & \lambda-2 & 4 \\ 1 & 0 & -3 \\ 3 & 3 & -2 \end{vmatrix} = 0$$

$$1(0+9) - (\lambda-2)(-2+9) + 4(3-0) = 0$$
$$9 - (\lambda-2)(7) + 4(3) = 0$$
$$-7\lambda = -35$$
$$\lambda = \frac{35}{7} = 5$$ $\longrightarrow$ (3M)

S_{13}

(9) If $\bar{a} = 2\bar{i}+\bar{j}-\bar{k}$, $\bar{b} = -\bar{i}+2\bar{j}-4\bar{k}$, $\bar{c} = \bar{i}+\bar{j}+\bar{k}$ then find $(\bar{a}\times\bar{b})\cdot(\bar{b}\times\bar{c})$

Sol:- $\bar{a}\times\bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 1 & -1 \\ -1 & 2 & -4 \end{vmatrix} = \bar{i}(-4+2)-\bar{j}(-8-1)+\bar{k}(4+1)$

$= -2\bar{i}+9\bar{j}+5\bar{k}$

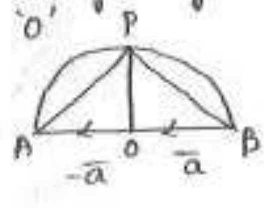
$\bar{b}\times\bar{c} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -1 & 2 & -4 \\ 1 & 1 & 1 \end{vmatrix} = \bar{i}(2+4)-\bar{j}(-1+4)+\bar{k}(-1-2)$

$= +6\bar{i}-3\bar{j}-3\bar{k}$ → (3M)

Required $\boxed{(\bar{a}\times\bar{b})\cdot(\bar{b}\times\bar{c}) = -2(6)+9(-3)+5(-3) = -12-27-15 = -54}$ → (1M)

S_{13} (10) Show that angle in a semicircle is a rightangle

Sol:- Let APB be a semicircle with centre 'O'



$\bar{OA} = \bar{OB} = \bar{OP}$ also $\bar{OB} = -\bar{OA}$

$(\bar{AP})(\bar{BP}) = (\bar{OP}-\bar{OA})(\bar{OP}-\bar{OB})$

$= (\bar{OP}-\bar{OA})(\bar{OP}+\bar{OA})$

$= (OP)^2-(OA)^2$

$= (OA)^2-(OA)^2$

$= 0$ → (2M)

$\bar{AP}\cdot\bar{BP} = 0$

$\angle APB = 90^\circ$

∴ Angle in a semicircle is right angle triangle. → (2M)

S_{13} (11) If $a+b+c=0$, $|a|=3$, $|b|=5$ and $|c|=7$, then find the angle between a and b

Sol:- Given, $|\bar{a}|=3$ $|\bar{b}|=5$ $|\bar{c}|=7$

$\bar{a}+\bar{b}+\bar{c}=0$

$\bar{a}+\bar{b}=-\bar{c}$ → (1M)

(S.O.B.S)

$|\bar{a}|^2+|\bar{b}|^2+2\bar{a}\cdot\bar{b} = |\bar{c}|^2$ $\boxed{\because \bar{a}\cdot\bar{b} = |\bar{a}||\bar{b}|\cos(\bar{a},\bar{b})}$

$|\bar{a}|^2+|\bar{b}|^2+2|\bar{a}||\bar{b}|\cos\theta = |c|^2$

$|a|^2+|b|^2+2|\bar{a}||\bar{b}|\cos(\bar{a},\bar{b}) = |c|^2$

$(3)^2+(5)^2+2(3)(5)\cos(\bar{a},\bar{b}) = (7)^2$

$9+25+30\cos(\bar{a},\bar{b}) = 49$

$30\cos(\bar{a},\bar{b}) = 49-34$

$30\cos(\bar{a},\bar{b}) = 15$

$\cos(\bar{a},\bar{b}) = \frac{1}{2} = \cos 60^\circ$

$(\bar{a},\bar{b}) = 60^\circ = \frac{\pi}{3}$ → (3M)

S3

12 (12) show that the points $(5,-1,1)(7,-4,7)(1,-6,10)$ and $(-1,-3,4)$ are the vertices of Rhombus by vectors.

Sol:- Let 'o' be the origin

A, B, C, D are the vertices of Rhombus

Now $\overline{OA} = 5\bar{i} - \bar{j} + \bar{k}$ $\quad \overline{OB} = 7\bar{i} - 4\bar{j} + 7\bar{k}$ $\quad \overline{OC} = \bar{i} - 6\bar{j} + 10\bar{k}$

$\overline{OD} = -\bar{i} - 3\bar{j} + 4\bar{k}$

$\overline{AB} = \overline{OB} - \overline{OA} = 2\bar{i} - 3\bar{j} + 6\bar{k}$ $\longrightarrow$ (1M)

$|\overline{AB}| = \sqrt{4+9+36} = \sqrt{49} = 7$

$\overline{BC} = \overline{OC} - \overline{OB} = -6\bar{i} - 2\bar{j} + 3\bar{k}$

$|\overline{BC}| = \sqrt{36+4+9} = \sqrt{49} = 7$

$\overline{CD} = \overline{OD} - \overline{OC} = -2\bar{i} + 3\bar{j} - 6\bar{k}$

$|\overline{CD}| = \sqrt{4+9+36} = \sqrt{49} = 7$

$\overline{DA} = \overline{OA} - \overline{OD} = 6\bar{i} + 2\bar{j} - 3\bar{k}$

$|\overline{DA}| = \sqrt{36+4+9} = \sqrt{49} = 7$ $\longrightarrow$ (1M)

$\therefore$ All sides are equal.

$|\overline{AC}| = \sqrt{16+25+81} = \sqrt{122}$

$\overline{BD} = \overline{OD} - \overline{OB} = -8\bar{i} + \bar{j} - 3\bar{k}$

$|\overline{BD}| = \sqrt{64+1+9} = \sqrt{74}$

$\overline{AC} \neq \overline{BD}$

Now $\overline{AC} \cdot \overline{BD} = (-4\bar{i} - 5\bar{j} + 9\bar{k})(-8\bar{i} + \bar{j} - 3\bar{k}) = 0$

$\therefore \overline{AC} \perp \overline{BD}$

$\therefore$ It is a Rhombus. $\longrightarrow$ (2M)

S3 (13) Let $\bar{a}$ & $\bar{b}$ be vectors satisfying $|\bar{a}| = |\bar{b}| = 5$ and $(\bar{a}, \bar{b}) = 45$. Find the area of the triangle having $\bar{a} - 2\bar{b}$ & $3\bar{a} + 2\bar{b}$ as two of its sides

Sol:- Area of triangle $= \frac{1}{2}|[\bar{a} - 2\bar{b}) \times (3\bar{a} + 2\bar{b})|$ $\rightarrow$ ① $\longrightarrow$ (1M)

$= |3(\bar{a}\times\bar{a}) + 2(\bar{a}\times\bar{b}) - 6(\bar{b}\times\bar{a}) - 4(\bar{b}\times\bar{b})|$

~~$\Rightarrow |0 + 2(\bar{a}\times\bar{b})$~~

$\Rightarrow |0 + 2(\bar{a}\times\bar{b}) + 6(\bar{a}\times\bar{b}) - 0|$

$\Rightarrow |8(\bar{a}\times\bar{b})|$ $\longrightarrow$ (1M)

$\Rightarrow 8|a||b| \sin 45^\circ$

$\Rightarrow 8.5.5 \times \frac{1}{\sqrt{2}}$

$\Rightarrow 100\sqrt{2}$

$\longrightarrow$ (2M)

TRIGONOMETRY UPTO TRANSFORMATIONS

S_{14} SAQ's

(1) If $A+B=45^\circ$, then prove that

(i) $(1+\tan A)(1+\tan B)=2$

Sol:

$A+B=\frac{\pi}{4} \Rightarrow \tan(A+B)=\frac{\pi}{4}$

$$\frac{\tan A+\tan B}{1-\tan A\tan B}=1 \qquad \boxed{\because \tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}}$$

$\Rightarrow \tan A+\tan B = 1-\tan A\tan B$ → (1M)

$\tan A+\tan B+\tan A\tan B=1$

Adding '1' on both sides → (1M)

$1+\tan A+\tan B+\tan A\tan B=1+1$

$1(1+\tan A)+\tan B(1+\tan A)=2$

$(1+\tan A)(1+\tan B)=2$ → (2M)

(ii) If $A+B=\frac{\pi}{4}$, prove that $(\cot A-1)(\cot B-1)=2$

Sol: $A+B=\frac{\pi}{4} \Rightarrow \cot(A+B)=\cot\frac{\pi}{4}$

$$\frac{\cot A\cot B-1}{\cot A+\cot B}=1 \qquad \boxed{\because \cot(A+B)=\frac{\cot A\cot B-1}{\cot A+\cot B}}$$

$\cot A\cot B-1=\cot A+\cot B$ → (1M)

$\cot A\cot B-\cot A-\cot B=1$

Adding '1' on both sides → (1M)

$\cot A\cot B-\cot B-\cot A+1=1+1$

$\cot B(\cot A-1)-1(\cot A-1)=2$

$(\cot B-1)(\cot A-1)=2$ → (2M)

(iii) If $A-B=\frac{3\pi}{4}$ then show that

$(1+\tan A)(1+\tan B)=2$

Sol: Given $A-B=\frac{3\pi}{4}$

Applying "tan" on Both sides

$$\tan(A-B) = \tan\left(\frac{3\pi}{4}\right) \qquad \boxed{\tan(A-B) = \frac{\tan A-\tan B}{1+\tan A\tan B}}$$

$$\frac{\tan A-\tan B}{1+\tan A\tan B} = -1 \longrightarrow (1m)$$

$$\tan A-\tan B = -1-\tan A\tan B$$

$$\tan A-\tan B+\tan A\tan B = -1$$

$$-(\tan B-\tan A\tan B-\tan A) = -1$$

$$\tan B-\tan A\tan B-\tan A = 1 \longrightarrow (1M)$$

Adding '1' on both sides

$$\tan B-\tan A\tan B-\tan A+1 = 1+1$$

$$\tan B(1-\tan A)+1(1-\tan A) = 2$$

$$(1-\tan A)(1+\tan B) = 2$$

Hence proved $\longrightarrow (2m)$

S_{15} ② Prove that $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1} = \frac{1+\sin\theta}{\cos\theta}$

Sol: Given, $LHS = \frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}$

$$= \frac{\tan\theta+\sec\theta-(\sec^2\theta-\tan^2\theta)}{\tan\theta-\sec\theta+1} \longrightarrow (1m)$$

$$= \frac{(\tan\theta+\sec\theta)-(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)}{\tan\theta-\sec\theta+1}$$

$\rightarrow (1m)$

$$= \frac{(\tan\theta + \sec\theta)(1 - \cancel{\sec\theta} + \tan\theta)}{(\tan\theta - \cancel{\sec\theta} + 1)}$$

$$= \tan\theta + \sec\theta$$

$$= \frac{\sin\theta}{\cos\theta} + \frac{1}{\cos\theta}$$

$$= \frac{1+\sin\theta}{\cos\theta} \longrightarrow (2M)$$

$= RHS$

S_{16} ③ Prove that $\left(1+\cos\frac{\pi}{10}\right)\left(1+\cos\frac{3\pi}{10}\right)\left(1+\cos\frac{7\pi}{10}\right)\left(1+\cos\frac{9\pi}{10}\right) = \frac{1}{16}$

Sol : $LHS = \left(1+\cos\frac{\pi}{10}\right)\left(1+\cos\frac{3\pi}{10}\right)\left(1+\cos\frac{7\pi}{10}\right)\left(1+\cos\frac{9\pi}{10}\right)$

$= (1+\cos 18)(1+\cos 3(18))(1+\cos 7(18))(1+\cos 9(18))$

$= (1+\cos 18)(1+\cos 54)(1+\cos(180-54))(1+\cos(180-18)) \longrightarrow (1M)$

$= (1+\cos 18)(1+\cos 54)(1-\cos 54)(1-\cos 18)$

$= (1-\cos^2 18)(1-\cos^2 54)$

$= (\sin^2 18)(\sin^2 54)$

$$\boxed{\begin{aligned}\sin 18 &= \frac{\sqrt{5}-1}{4}\\ \sin 54 &= \frac{\sqrt{5}+1}{4}\end{aligned}}$$

$= \left(\frac{\sqrt{5}-1}{4}\right)^2\left(\frac{\sqrt{5}+1}{4}\right)^2$

$= \left[\frac{(\sqrt{5})^2-1^2}{16}\right]^2 \longrightarrow (2M)$

$= \left(\frac{5-1}{16}\right)^2$

$= \left(\frac{4}{16}\right)^2$

$= \left(\frac{1}{4}\right)^2$

$= \frac{1}{16}$

$= RHS \longrightarrow (1M)$

S16 ④ If A is not an integral multiple of 'π' then prove that $\cos A \cos 2A \cos 4A \cos 8A = \frac{\sin 16A}{16 \sin A}$ and hence deduce that $\cos \frac{2\pi}{15} \cdot \cos \frac{4\pi}{15} \cdot \cos \frac{8\pi}{15} \cdot \cos \frac{16\pi}{15} = \frac{1}{16}$

Sol: LHS $= \cos A \cos 2A \cos 4A \cos 8A$

Multiply and divide with $2\sin A$

$$= \frac{2\sin A}{2\sin A} \cos A \cdot \cos 2A \cdot \cos 4A \cdot \cos 8A \longrightarrow (1M)$$

$\boxed{\because \sin 2A = 2\sin A \cos A}$

$$= \frac{2\sin 2A}{4\sin A} \cos 2A \cdot \cos 4A \cdot \cos 8A$$

$$= \frac{2\sin 4A}{8\sin A} \cos 4A \cdot \cos 8A$$

$$= \frac{2\sin 8A}{16\sin A} \cos 8A$$

$$= \frac{\sin 16A}{16\sin A}$$

$= \text{RHS}$

$$\cos A \cdot \cos 2A \cdot \cos 4A \cdot \cos 8A = \frac{\sin 16A}{16\sin A} \longrightarrow (1) \qquad \longrightarrow (2m)$$

Put $A = \frac{2\pi}{15}$ in (1)

$$\cos\frac{2\pi}{15} \cdot \cos\frac{4\pi}{15} \cdot \cos\frac{8\pi}{15} \cdot \cos\frac{16\pi}{15} = \frac{\sin 16\left(\frac{2\pi}{15}\right)}{16\sin\left(\frac{2\pi}{15}\right)}$$

$$= \frac{\sin\left(\frac{32\pi}{15}\right)}{16\sin\left(\frac{2\pi}{15}\right)}$$

$$= \frac{\sin\left(\frac{2\pi}{15}\right)}{16\sin\left(\frac{2\pi}{15}\right)}$$

$$= \frac{1}{16} \qquad \longrightarrow (1M)$$

S_{15} 5 let ABC be a triangle Such that $\cot A + \cot B + \cot C = \sqrt{3}$ then prove that ABC is an equilateral triangle.

Sol: In a Δ^{le} ABC $\Rightarrow A+B+C = 180^\circ$

$\cot A + \cot B + \cot C = \sqrt{3}$

let $\cot A = x$; $\cot B = y$; $\cot C = z$ $\longrightarrow$ (1M)

$\therefore x+y+z = \sqrt{3}$

Then, $\Sigma \cot A \cdot \cot B = 1$ $\boxed{a^2+b^2 = (a+b)^2 - 2ab}$

$$\begin{aligned}\Sigma (x-y)^2 &= \Sigma (x^2+y^2-2xy) \\ &= \Sigma x^2 + \Sigma y^2 - 2\Sigma xy \\ &= x^2+y^2+z^2+y^2+z^2+x^2 - 2[xy+yz+zx] \\ &= 2x^2+2y^2+2z^2 - 2[xy+yz+zx] \\ &= 2\left[(x+y+z)^2 - 2(xy+yz+zx)\right] - 2\left[(xy+yz+zx)\right] \\ &= 2\left[(\sqrt{3})^2 - 2(1)\right] - 2[1] \\ &= 2[3-2] - 2 \\ &= 2-2 \\ &= 0\end{aligned}$$

$\Sigma (x-y)^2 = 0$

$x-y=0$; $y-z=0$; $z-x=0$ $\longrightarrow$ (2M)

$\Rightarrow x=y=z$

$\cot A = \cot B = \cot C$

$A = B = C$

$\therefore \Delta ABC$ is an Equilateral triangle. $\longrightarrow$ (1M)

S_{16} ⑥ Prove that $\tan 70^\circ - \tan 20^\circ = 2\tan 50^\circ$

Sol: $70^\circ - 20^\circ = 50^\circ$

Applying "tan" on both sides

$\tan(70^\circ - 20^\circ) = \tan 50^\circ$ $\boxed{\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A \tan B}}$

$\dfrac{\tan 70 - \tan 20}{1+\tan 70 \tan 20} = \tan 50$ $\longrightarrow$ (2m)

$\dfrac{\tan 70 - \tan 20}{1+\tan(90-20)\tan 20} = \tan 50$

$\dfrac{\tan 70 - \tan 20}{1+\cancel{\cot 20}\cdot\frac{1}{\cancel{\cot 20}}} = \tan 50$

$\tan 70 - \tan 20 = 2\tan 50$ $\longrightarrow$ (2m)

S_{16} ⑦ For $A \in R$, Prove that -

(i) $\sin A \cdot \sin(60+A)\sin(60-A) = \frac{1}{4}\sin 3A$

Sol: $LHS = \sin A\left(\sin(60+A)\cdot\sin(60-A)\right)$

$= \sin A\left(\sin^2 60 - \sin^2 A\right)$ $\boxed{\because \sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B}$ $\longrightarrow$ (1m)

$= \sin A\left[\frac{3}{4} - \sin^2 A\right]$

$= \frac{\sin A}{4}\left[3-4\sin^2 A\right]$

$= \frac{1}{4}\left[3\sin A - 4\sin^3 A\right]$ $\boxed{\because \sin 3A = 3\sin A - 4\sin^3 A}$

$= \frac{1}{4}\sin 3A$ $\longrightarrow$ (2m)

$= RHS$

(ii) Hence deduce that $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ = \frac{3}{16}$

$\boxed{\sin A \sin(60^\circ - A)\sin(60^\circ + A) = \frac{1}{4}\sin 3A}$

$= \frac{1}{4} \sin(3 \times 20^\circ) \cdot \sin 60^\circ$

$= \frac{1}{4} \sin^2 60^\circ$

$= \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$ $\longrightarrow$ (1M)

(iii) Prove that $\cos A \cdot \cos(60+A) \cos(60-A) = \frac{1}{4} \cos 3A$ and hence deduce that $\cos\frac{\pi}{9} \cdot \cos\frac{2\pi}{9} \cdot \cos\frac{3\pi}{9} \cdot \cos\frac{4\pi}{9} = \frac{1}{16}$

Sol: $\cos A \cdot \cos(60+A) \cos(60-A) = \cos A \left(\cos^2 60^\circ - \sin^2 A\right)$

$\boxed{\because \cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B}$

$= \cos A \left(\frac{1}{4} - \sin^2 A\right)$ $\longrightarrow$ (1M)

$= \cos A \left(\frac{1}{4} - (1 - \cos^2 A)\right)$

$= \frac{\cos A \left(1 - 4 + 4\cos^2 A\right)}{4}$

$= \cos A \frac{\left(4\cos^2 A - 3\right)}{4}$

$= \frac{1}{4} \left(4\cos^3 A - 3\cos A\right)$

$= \frac{1}{4} \cos 3A$ $\longrightarrow$ (2M)

$\cos\frac{\pi}{9} \cdot \cos\frac{2\pi}{9} \cdot \cos\frac{3\pi}{9} \cdot \cos\frac{4\pi}{9} = \cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ$

$= \frac{1}{4} \cos(60^\circ) \cdot \cos(60^\circ)$

$\boxed{\cos A \cos(60+A) \cos(60-A) = \frac{1}{4} \cos 3A}$

$= \frac{1}{4} \cos^2 60$

$= \frac{1}{4} \times \frac{1}{4}$

$= \frac{1}{16}$

$\longrightarrow$ (1M)

S16 (8)(i) Prove that $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8} = \frac{3}{2}$

<u>Sol</u>: LHS $= \sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8}$

$$= \sin^4\left(\frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{2}-\frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{2}+\frac{\pi}{8}\right) + \sin^4\left(\pi-\frac{\pi}{8}\right)$$

$$= \sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8} + \cos^4\frac{\pi}{8} + \sin^4\frac{\pi}{8} \longrightarrow (1M)$$

$$= 2\left[\sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8}\right] \qquad \boxed{\sin^2\theta + \cos^2\theta = 1}$$

$$= 2\left[\left(\sin^2\frac{\pi}{8}\right)^2 + \left(\cos^2\frac{\pi}{8}\right)^2\right]$$

$$= 2\left[\left[\sin^2\frac{\pi}{8} + \cos^2\frac{\pi}{8}\right] - 2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right]$$

$$= 2\left[1 - 2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right]$$

$$= 2 - 2^2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}$$

$$= 2 - \left(2\sin\frac{\pi}{8}\cos\frac{\pi}{8}\right)^2 \longrightarrow (2M)$$

$$= 2 - \left[\sin 2\left(\frac{\pi}{8}\right)\right]^2$$

$$= 2 - \sin^2\frac{\pi}{4}$$

$$= 2 - \frac{1}{2}$$

$$= \frac{4-1}{2}$$

$$= \frac{3}{2}$$

$= \text{RHS} \longrightarrow (1M)$

(ii) Prove that $\cos^4\frac{\pi}{8} + \cos^4\frac{3\pi}{8} + \cos^4\frac{5\pi}{8} + \cos^4\frac{7\pi}{8} = \frac{3}{2}$

<u>Sol</u>: LHS $= \cos^4\frac{\pi}{8} + \cos^4\frac{3\pi}{8} + \cos^4\frac{5\pi}{8} + \cos^4\frac{7\pi}{8}$

$$= \cos^4\frac{\pi}{8} + \cos^4\left(\frac{\pi}{2}-\frac{\pi}{8}\right) + \cos^4\left(\frac{\pi}{2}+\frac{\pi}{8}\right) + \cos^4\left(\pi-\frac{\pi}{8}\right) \longrightarrow (1M)$$

$= \cos^4\frac{\pi}{8} + \sin^4\frac{\pi}{8} + \sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8}$

$= 2\left[\sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8}\right]$ $\boxed{a^2+b^2 = (a+b)^2 - 2ab}$

$= 2\left[\left(\sin^2\frac{\pi}{8} + \cos^2\frac{\pi}{8}\right)^2 - 2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right]$

$= 2\left[1 - 2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right]$

$= 2 - 2^2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}$

$= 2 - \left[2\sin\frac{\pi}{8}\cos\frac{\pi}{8}\right]^2$ $\longrightarrow$ (2M)

$= 2 - \left(\sin\frac{\pi}{4}\right)^2$

$= 2 - \frac{1}{2}$

$= \frac{3}{2}$

$=$ RHS $\longrightarrow$ (1M)

5_{15} 9 (i) Prove that $\cos\frac{2\pi}{7}\cdot\cos\frac{4\pi}{7}\cdot\cos\frac{8\pi}{7} = \frac{1}{8}$

Sol: Let $C = \cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{8\pi}{7}$;

$S = \sin\frac{2\pi}{7}\sin\frac{4\pi}{7}\sin\frac{8\pi}{7}$

$CS = \left(\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{8\pi}{7}\right)\left(\sin\frac{2\pi}{7}\sin\frac{4\pi}{7}\sin\frac{8\pi}{7}\right)$ $\longrightarrow$ (1M)

$= \frac{1}{2}\left(\sin\frac{4\pi}{7}\right)\cdot\frac{1}{2}\left(\sin\frac{8\pi}{7}\right)\cdot\frac{1}{2}\left(\sin\frac{16\pi}{7}\right)$ $\boxed{\because \sin 2A = 2\sin A\cos A}$

$= \frac{1}{8}\left(\sin\frac{4\pi}{7}\right)\left(\sin\frac{8\pi}{7}\right)\left(\sin\left(2\pi + \frac{2\pi}{7}\right)\right)$

$= \frac{1}{8}\left(\sin\frac{2\pi}{7}\right)\left(\sin\frac{4\pi}{7}\right)\left(\sin\frac{8\pi}{7}\right)$

$C\cancel{S} = \frac{1}{8}\cancel{S}$

$C = \frac{1}{8}$ $\longrightarrow$ (3M)

(ii) Prove that $\cos\frac{\pi}{11}\cdot\cos\frac{2\pi}{11}\cdot\cos\frac{3\pi}{11}\cdot\cos\frac{4\pi}{11}\cdot\cos\frac{5\pi}{11}=\frac{1}{32}$.

Sol: Let $C=\cos\frac{\pi}{11}\cdot\cos\frac{2\pi}{11}\cdot\cos\frac{3\pi}{11}\cdot\cos\frac{4\pi}{11}\cdot\cos\frac{5\pi}{11}$;

$S=\sin\frac{\pi}{11}\sin\frac{2\pi}{11}\sin\frac{3\pi}{11}\sin\frac{4\pi}{11}\sin\frac{5\pi}{11}$ $\longrightarrow$ (1m)

$C\cdot S=\left(\cos\frac{\pi}{11}\cos\frac{2\pi}{11}\cos\frac{3\pi}{11}\cos\frac{4\pi}{11}\cos\frac{5\pi}{11}\right)\left(\sin\frac{\pi}{11}\sin\frac{2\pi}{11}\sin\frac{3\pi}{11}\sin\frac{4\pi}{11}\sin\frac{5\pi}{11}\right)$

$=\frac{1}{32}\left(2\sin\frac{\pi}{11}\cos\frac{\pi}{11}\right)\left(2\sin\frac{2\pi}{11}\cos\frac{2\pi}{11}\right)\left(2\sin\frac{3\pi}{11}\cos\frac{3\pi}{11}\right)\left(2\cos\frac{4\pi}{11}\sin\frac{4\pi}{11}\right)\left(2\sin\frac{5\pi}{11}\cos\frac{5\pi}{11}\right)$

$=\frac{1}{32}\sin\frac{2\pi}{11}\cdot\sin\frac{4\pi}{11}\sin\frac{6\pi}{11}\sin\frac{8\pi}{11}\cdot\sin\frac{10\pi}{11}$

$=\frac{1}{32}\sin\frac{\pi}{11}\sin\frac{2\pi}{11}\sin\frac{3\pi}{11}\sin\frac{4\pi}{11}\sin\frac{5\pi}{11}$

$C\cancel{S}=\frac{1}{32}\cancel{S}$

$C=\frac{1}{32}$

$\longrightarrow$ (3m)

(S14) (10) Prove that $\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}=\frac{5}{16}$

Sol: LHS $=\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$

$=\sin(36)\sin(2\times 36)\sin(3\times 36)\sin(4\times 36)$

$=\sin 36\ \sin 72\ \sin 108\ \sin 144$ $\longrightarrow$ (1m)

$=\sin 36\ \sin(90-18)\sin(90+18)\sin(180-36)$

$=\sin 36(\cos 18)(\cos 18)(\sin 36)$ $\longrightarrow$ (1m)

$=\sin^2 36\cdot\cos^2 18$

$$\boxed{\sin 36=\frac{\sqrt{10-2\sqrt5}}{4}\qquad \cos 18=\frac{\sqrt{10+2\sqrt5}}{4}}$$

$=\left(\frac{\sqrt{10-2\sqrt5}}{4}\right)^2\left(\frac{\sqrt{10+2\sqrt5}}{4}\right)^2$

$=\frac{(10)^2-(2\sqrt5)^2}{16\times 16}$

$=\frac{100-20}{16}=\frac{80}{16\times16}=\frac{5}{16}=$ RHS $\longrightarrow$ (2m)

S_{14} (11)(i). If A is not integral multiple of $\frac{\pi}{2}$, prove that

(i) $\tan A + \cot A = 2\operatorname{Cosec} 2A$

Sol: $LHS = \tan A + \cot A$

$$= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}$$

$$= \frac{\sin^2 A + \cos^2 A}{\cos A \cdot \sin A}$$

$$= \frac{1}{\cos A \sin A}$$

Multiply & divide by 2

$$= \frac{2}{2} \times \frac{1}{\cos A \sin A}$$

$$\boxed{\sin 2A = 2\sin A \cos A}$$

$$= \frac{2}{\sin 2A}$$

$$= 2\operatorname{Cosec} 2A$$

$= RHS$. $\longrightarrow$ (2M)

(ii) $\cot A - \tan A = 2\cot 2A$

Sol: $LHS = \cot A - \tan A$

$$= \frac{\cos A}{\sin A} - \frac{\sin A}{\cos A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\sin A \cos A}$$

$$\boxed{\cos 2A = \cos^2 A - \sin^2 A}$$

$$= \frac{\cos 2A}{\sin A \cos A}$$

Multiply & divide by 2

$$= \frac{2\cos 2A}{2\sin A \cos A}$$

$= \frac{2\cos 2A}{\sin 2A} = 2\cot 2A$ $\longrightarrow$ (2M)

S14 (12) If none of the denominators is zero, then prove that

$$\left(\frac{\cos A+\cos B}{\sin A-\sin B}\right)^n + \left(\frac{\sin A+\sin B}{\cos A-\cos B}\right)^n = \begin{cases} 2\cot^n\left(\frac{A-B}{2}\right), & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases}$$

Sol: Given, $\left(\frac{\cos A+\cos B}{\sin A-\sin B}\right)^n + \left(\frac{\sin A+\sin B}{\cos A-\cos B}\right)^n$

$\boxed{\because \cos C+\cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)}$

$$= \left[\frac{2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)}\right]^n + \left[\frac{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{-2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)}\right]^n \longrightarrow (2m)$$

$\boxed{\because \cos C-\cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)}$

$$= \left[\frac{\cos\left(\frac{A-B}{2}\right)}{\sin\left(\frac{A-B}{2}\right)}\right]^n + \left[\frac{\cos\left(\frac{A-B}{2}\right)}{-\sin\left(\frac{A-B}{2}\right)}\right]^n$$

$$= \cot^n\left(\frac{A-B}{2}\right) + (-1)^n\cot^n\left(\frac{A-B}{2}\right) \longrightarrow (2m)$$

$= 0$ if n is odd $(-1)^n = -1$

$= 2\cot^n\left(\frac{A-B}{2}\right)$ if n is even $(-1)^n = 1$ $\longrightarrow$ (1m)

———— × ————

S_{14} 13. If $\sec(\theta+\alpha)+\sec(\theta-\alpha)=2\sec\theta$, prove that $\cos\theta = \pm\sqrt{2}\cos\frac{\alpha}{2}$

Sol: $\sec(\theta+\alpha)+\sec(\theta-\alpha)=2\sec\theta$

$$\frac{1}{\cos(\theta+\alpha)}+\frac{1}{\cos(\theta-\alpha)}=2\sec\theta$$

$$\frac{\cos(\theta-\alpha)+\cos(\theta+\alpha)}{\cos^2\theta-\sin^2\theta}=2\sec\theta \longrightarrow (1M)$$

$$\frac{\cancel{2}\cos\theta\cos\alpha}{\cos^2\theta-\sin^2\alpha}=\frac{\cancel{2}}{\cos\theta}$$

$$\boxed{\because \cos(A-B)+\cos(A+B)=2\cos A\cos B}$$

$$\cos^2\theta\cdot\cos\alpha=\cos^2\theta-\sin^2\alpha$$

$$\sin^2\alpha=\cos^2\theta\,(1-\cos\alpha)$$

$$(1-\cos^2\alpha)=\cos^2\theta\,(1-\cos\alpha)$$

$$(1+\cos\alpha)(1-\cos\alpha)=\cos^2\theta\,(1-\cos\alpha) \longrightarrow (2M)$$

$$\cos^2\theta=2\cos^2\left(\frac{\alpha}{2}\right)$$

$$\boxed{\because 1+\cos A=2\cos^2\frac{A}{2}}$$

$$\cos\theta=\pm\sqrt{2}\cos\left(\frac{\alpha}{2}\right)$$

Hence proved. $\longrightarrow (1M)$

S_{14} 14. If $\cos x+\cos y=\frac{4}{5}$ and $\cos x-\cos y=\frac{2}{7}$. Find the Value of $14\tan\frac{(x-y)}{2}+5\cot\frac{(x+y)}{2}$

Sol: Take LHS $=\cos x+\cos y=\frac{4}{5}$ $\longrightarrow (1M)$

$$\Rightarrow 2\cos\frac{x+y}{2}\cdot\cos\frac{x-y}{2}=\frac{4}{5}\rightarrow (1)$$

$$\boxed{\because \cos C+\cos D=2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)}$$

$$\cos x-\cos y=\frac{2}{7}$$

$$\Rightarrow -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}=\frac{2}{7}\rightarrow (2)$$

$$\boxed{\because \cos C-\cos D=-2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)}$$

$\rightarrow (1M)$

Now, $\frac{①}{②} \Rightarrow \dfrac{\cancel{2}\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)}{-\cancel{2}\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)} = \dfrac{\frac{4}{5}}{\frac{2}{7}}$ $\longrightarrow$ (1m)

$$-\cot\left(\frac{x+y}{2}\right)\cot\left(\frac{x-y}{2}\right) = \frac{14}{5}$$

$$5\cot\left(\frac{x-y}{2}\right) = -14\tan\left(\frac{x+y}{2}\right)$$

$$14\tan\left(\frac{x-y}{2}\right) + 5\cot\left(\frac{x+y}{2}\right) = 0$$ $\longrightarrow$ (1m)

Hence proved

S14 ⑮ Prove that $\sqrt{3}\operatorname{cosec}20^\circ - \sec 20^\circ = 4$

Sol: $LHS = \sqrt{3}\operatorname{cosec}20^\circ - \sec 20^\circ$

$$= \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ}$$

$$= \frac{\sqrt{3}\cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cdot \cos 20^\circ}$$ $\longrightarrow$ (1m)

$$= \frac{2\left(\frac{\sqrt{3}}{2}\cos 20^\circ - \frac{1}{2}\sin 20^\circ\right)}{\frac{1}{2}\left(2\sin 20^\circ\cos 20^\circ\right)} = \frac{2\left(\sin 60^\circ\cos 20^\circ - \cos 60^\circ\sin 20^\circ\right)}{\frac{1}{2}\sin 40^\circ}$$

$$= \frac{4\sin(60-20)}{\sin 40^\circ} = 4\frac{\cancel{\sin 40}}{\cancel{\sin 40}} = 4$$ $\longrightarrow$ (3m)

S14 ⑯ If $0 < A < B < \frac{\pi}{4}$, $\sin(A+B) = \frac{24}{5}$, $\cos(A-B) = \frac{4}{5}$, find the value of $\tan 2A$

Sol: $\sin(A+B) = \frac{24}{25}$, $\cos(A-B) = \frac{4}{5}$

$\tan(A+B) = \frac{24}{7}$, $\tan(A-B) = \frac{-3}{4}$

$\because A+B>0,\ A-B<0$
$A-B \in Q_4,\ A+B \in Q_1$

$\longrightarrow$ (2m)

$$\tan 2A = \tan\left[(A+B) + (A-B)\right]$$

$$= \frac{\tan(A+B) + \tan(A-B)}{1 - \tan(A+B)\tan(A-B)}$$

$$= \frac{\frac{24}{7} - 3/4}{1 + \left(\frac{24}{7}\right)\left(\frac{3}{4}\right)} = \frac{96-21}{28+72} = \frac{75}{100} = \frac{3}{4}$$ $\longrightarrow$ (2m)

PROPERTIES OF TRIANGLES

SAQ'S

$S_{17}(1)$: If $a=(b-c)\sec\theta$, prove that $\tan\theta=\frac{2\sqrt{bc}}{b-c}\sin\frac{A}{2}$

Solution: $a=(b-c)\sec\theta$

$$\frac{a}{(b-c)}=\sec\theta$$

$\boxed{\text{W.K.T} \quad \sec^2\theta=1+\tan^2\theta}$

$$\Rightarrow \tan^2\theta=\sec^2\theta-1$$

$$\tan^2\theta=\left[\frac{a}{b-c}\right]^2-1 \longrightarrow (1m)$$

$\boxed{\begin{array}{l}\sec^2\theta-\tan^2\theta=1\\ a^2-b^2=(a+b)(a-b)\\ a+b+c=2s\\ \sin\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{bc}}\end{array}}$

$$=\frac{a^2}{(b-c)^2}-1$$

$$=\frac{a^2-(b-c)^2}{(b-c)^2}$$

$$=\frac{(a+b-c)(a-b+c)}{(b-c)^2}$$

$$=\frac{(2s-2c)(2s-2b)}{(b-c)^2}$$

$$=\frac{2(s-c)2(s-b)}{(b-c)^2}$$

$$\tan^2\theta=\frac{4(s-c)(s-b)}{(b-c)^2}\times\frac{bc}{bc} \longrightarrow (1m)$$

$$\tan\theta=\sqrt{\frac{4(s-c)(s-b)}{(b-c)^2}\times\frac{bc}{bc}}$$

$$\tan\theta=\frac{2\sqrt{bc}}{b-c}\sqrt{\frac{(s-c)(s-b)}{bc}}$$

$$\tan\theta=\frac{2\sqrt{bc}}{b-c}\sin\frac{A}{2} \longrightarrow (2m)$$

$S_{17}(2)$: If $a=(b+c)\cos\theta$, prove that $\sin\theta=\frac{2\sqrt{bc}}{b+c}\cos\frac{A}{2}$

Sol: Given $a=(b+c)\cos\theta$

$$\cos\theta=\frac{a}{b+c}$$

$\boxed{\begin{array}{l}\sin^2\theta+\cos^2\theta=1\\ a^2-b^2=(a+b)(a-b)\\ a+b+c=2s\\ \cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}}\end{array}}$

$\boxed{\text{W.K.T} \quad \sin^2\theta+\cos^2\theta=1}$

$$\Rightarrow \sin^2\theta=1-\cos^2\theta$$

$$=1-\left[\frac{a}{b+c}\right]^2 \longrightarrow (1m)$$

$$= 1 - \frac{a^2}{(b+c)^2}$$

$$= \frac{(b+c)^2 - a^2}{(b+c)^2}$$

$$= \frac{(b+c-a)(b+c-a)}{(b+c)^2}$$

$$= \frac{(a+b+c)(a+b+c-2a)}{(b+c)^2}$$

$$= \frac{2S(2S-2a)}{(b+c)^2} \longrightarrow (2M)$$

$$\sin^2\theta = \frac{4S(S-a)}{(b+c)^2} \times \frac{bc}{bc}$$

$$\sin\theta = \sqrt{\frac{4S(S-a)}{(b+c)^2} \times \frac{bc}{bc}}$$

$$= \frac{2\sqrt{bc}}{b+c}\sqrt{\frac{S(S-a)}{bc}}$$

$$\sin\theta = \frac{2\sqrt{bc}}{b+c}\cos\frac{A}{2} \longrightarrow (1M)$$

$S_{17}(3)$: If $\sin\theta = \frac{a}{b+c}$, prove that $\cos\theta = \frac{2\sqrt{bc}}{b+c}\cos\frac{A}{2}$

Sol: Given $\sin\theta = \frac{a}{b+c}$

$$\boxed{\text{W.K.T } \sin^2\theta + \cos^2\theta = 1}$$

$$\Rightarrow \cos^2\theta = 1 - \sin^2\theta$$

$$\cos^2\theta = 1 - \left[\frac{a}{b+c}\right]^2$$

$$= 1 - \frac{a^2}{(b+c)^2} \longrightarrow (1M)$$

$$= \frac{(b+c)^2 - a^2}{(b+c)^2}$$

$$= \frac{(b+c+a)(b+c-a)}{(b+c)^2}$$

$$\cos^2\theta = \frac{(a+b+c)(b+c-2a)}{(b+c)^2}$$

$$= \frac{(2S)(2S-2a)}{(b+c)^2}$$

$$\cos^2\theta = \frac{2S(2)(S-a)}{(b+c)^2} \times \frac{bc}{bc} \longrightarrow (2M)$$

$$\cos\theta = \sqrt{\frac{4s(s-a)}{(b+c)^2} \times \frac{bc}{bc}}$$

$$= \frac{2\sqrt{bc}}{b+c}\sqrt{\frac{s(s-a)}{bc}}$$

$$\cos\theta = \frac{2\sqrt{bc}}{b+c}\cos\frac{A}{2} \longrightarrow (1m)$$

$S_{17}(4)$: $\cot A + \cot B + \cot C = \dfrac{a^2+b^2+c^2}{4\Delta}$

Sol: $LHS = \cot A + \cot B + \cot C$

$$= \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C} \longrightarrow (1m)$$

$$= \frac{b^2+c^2-a^2}{2bc\sin A} + \frac{a^2+c^2-b^2}{2ac\sin B} + \frac{a^2+b^2-c^2}{2ab\sin C}$$

$$= \frac{b^2+c^2-a^2}{4\Delta} + \frac{a^2+c^2-b^2}{4\Delta} + \frac{a^2+b^2-c^2}{4\Delta}$$

$$= \frac{b^2+c^2-a^2+a^2+c^2-b^2+a^2+b^2-c^2}{4\Delta}$$

$$= \frac{a^2+b^2+c^2}{4\Delta} = RHS$$

$LHS = RHS \longrightarrow (3m)$

$\cos A = \dfrac{b^2+c^2-a^2}{2bc}$

$\cos B = \dfrac{a^2+c^2-b^2}{2ac}$

$\cos C = \dfrac{a^2+b^2-c^2}{2ab}$

$4\Delta = 2bc\sin A$

$4\Delta = 2ac\sin B$

$4\Delta = 2ab\sin C$

$S_{17}(5)$: Show that $\dfrac{\cos A}{a} + \dfrac{\cos B}{b} + \dfrac{\cos C}{c} = \dfrac{a^2+b^2+c^2}{2abc}$

Sol: $LHS = \dfrac{\cos A}{a} + \dfrac{\cos B}{b} + \dfrac{\cos C}{c}$

$$= \frac{\frac{b^2+c^2-a^2}{2bc}}{a} + \frac{\frac{c^2+a^2-b^2}{2ac}}{b} + \frac{\frac{a^2+b^2-c^2}{2ab}}{c} \longrightarrow (2m)$$

$$= \frac{b^2+c^2-a^2}{2abc} + \frac{c^2+a^2-b^2}{2abc} + \frac{a^2+b^2-c^2}{2abc}$$

$$= \frac{b^2+c^2-a^2+c^2+a^2-b^2+a^2+b^2-c^2}{2abc}$$

$$= \frac{a^2+b^2+c^2}{2abc} = RHS \longrightarrow (2m)$$

$LHS = RHS$

S17(6): In ΔABC, show that $\frac{b^2-c^2}{a^2} = \frac{\sin(B-C)}{\sin(B+C)}$

Sol:-

$$LHS = \frac{b^2-c^2}{a^2}$$

$$= \frac{(2R\sin B)^2-(2R\sin C)^2}{(2R\sin A)^2}$$

$$= \frac{4R^2\sin^2 B-4R^2\sin^2 C}{4R^2\sin^2 A}$$

$$= \frac{4R^2(\sin^2 A-\sin^2 C)}{4R^2\sin^2 A}$$

$$= \frac{\sin^2 B-\sin^2 C}{\sin^2 A}$$

$$= \frac{\sin(B-C)}{\sin^2(B+C)}$$

$$= \frac{\sin(B-C)}{\sin(B+C)} = RHS.$$

$\longrightarrow$ (4M)

∵ From Sine rule:

$b = 2R\sin B$

$c = 2R\sin C$

$a = 2R\sin A$

$A+B+C = 180$

$B+C = 180-A$

$\sin(B+C) = \sin(180-A)$
$= \sin A$

$\sin^2 B-\sin^2 C = \sin(B+C)\sin(B-C)$

S17(7):- In ΔABC, if $\frac{1}{a+c}+\frac{1}{b+c} = \frac{3}{a+b+c}$ show that $C = 60^\circ$

Sol: Given $\frac{1}{a+c}+\frac{1}{b+c} = \frac{3}{a+b+c}$

$$\frac{b+c+a+c}{(b+c)(a+c)} = \frac{3}{a+b+c}$$

$$\frac{a+b+2c}{(a+c)(b+c)} = \frac{3}{a+b+c}$$

$$(a+b+c)(a+b+2c) = 3(ab+ac+bc+c^2)$$

$\longrightarrow$ (2M)

From cosine rule,

$c^2 = a^2+b^2-2ab\cos C$

$2ab\cos C = a^2+b^2-c^2$

$\cos 60^\circ = \frac{1}{2}$

$$ab+b^2+bc+a^2+ab+ac+2ac+2bc+2c^2 = 3ab+3ac+3bc+3c^2$$

$$2ab+b^2+a^2+2c^2-3ab-3c^2 = 0$$

$$b^2+a^2-c^2-ab = 0$$

$$b^2+a^2-c^2 = ab$$

$$\cos C = \frac{ab}{2ab}$$

$$\cos C = \frac{1}{2} = \cos 60^\circ$$

$$\boxed{\angle C = 60^\circ}$$

$\longrightarrow$ (2M)

$S_{17}(8)$: Show that $\frac{1}{r^2}+\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}=\frac{a^2+b^2+c^2}{\Delta^2}$

Sol: $LHS=\frac{1}{r^2}+\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}$

$=\frac{1}{\left[\frac{\Delta}{s}\right]^2}+\frac{1}{\left[\frac{\Delta}{s-a}\right]^2}+\frac{1}{\left[\frac{\Delta}{s-b}\right]^2}+\frac{1}{\left[\frac{\Delta}{s-c}\right]^2}$ $\longrightarrow$ (1M)

$r=\frac{\Delta}{s}$

$r_1=\frac{\Delta}{s-a}$

$r_2=\frac{\Delta}{s-b}$

$r_3=\frac{\Delta}{s-c}$

$a+b+c=2s$

$=\frac{s^2}{\Delta^2}+\frac{(s-a)^2}{\Delta^2}+\frac{(s-b)^2}{\Delta^2}+\frac{(s-c)^2}{\Delta^2}$

$=\frac{s^2+s^2+a^2-2sa+s^2+b^2-2sb+s^2+c^2-2sc}{\Delta^2}$

$=\frac{4s^2-2s(a+b+c)+a^2+b^2+c^2}{\Delta^2}$

$=\frac{4s^2-4s^2+a^2+b^2+c^2}{\Delta^2}$

$=\frac{a^2+b^2+c^2}{\Delta^2}=RHS$ $\longrightarrow$ (3M)

$LHS=RHS$

$S_{17}(9)$: If $C=60^\circ$, then show that

i. $\frac{a}{b+c}+\frac{b}{c+a}=1$ (ii) $\frac{b}{c^2-a^2}+\frac{a}{c^2-b^2}=0$.

i) Given $C=60^\circ$

From cosine rule

$c^2=a^2+b^2-2ab\cos C$

$c^2=a^2+b^2-2ab\cos 60^\circ$

$c^2=a^2+b^2-\cancel{2}ab\left[\frac{1}{\cancel{2}}\right]$

$c^2=a^2+b^2-ab$

$c^2+ab=a^2+b^2$ $\longrightarrow$ (1M)

$LHS=\frac{a}{b+c}+\frac{b}{c+a}$

$=\frac{a(c+a)+b(b+c)}{(b+c)(c+a)}$

$=\frac{ac+a^2+b^2+bc}{bc+ab+c^2+ac}=1=RHS$ $\longrightarrow$ (1M)

iii) $LHS=\frac{b}{c^2-a^2}+\frac{a}{c^2-b^2}$

$= \frac{b}{b(b-a)} + \frac{a}{(a-b)a}$

$= \frac{1}{b-a} + \frac{1}{a-b}$

$= \frac{1}{b-a} - \frac{1}{b-a} = 0 = RHS$ $\longrightarrow$ (2M)

$S_{17}(10)$:- Show that $(b-c)^2\cos^2\frac{A}{2} + (b+c)^2\sin^2\frac{A}{2} = a^2$

Sol: LHS:- $(b-c)^2\cos^2\frac{A}{2} + (b+c)^2\sin^2\frac{A}{2}$

$= (b^2+c^2-2bc)\cos^2\frac{A}{2} + (b^2+c^2+2bc)\sin^2\frac{A}{2}$ $\longrightarrow$ (1M)

$= (b^2+c^2)\cos^2\frac{A}{2} - 2bc\cos^2\frac{A}{2} + (b^2+c^2)\sin^2\frac{A}{2} + 2bc\sin^2\frac{A}{2}$

$= (b^2+c^2)(\cos^2\frac{A}{2} + \sin^2\frac{A}{2}) - 2bc(\cos^2\frac{A}{2} - \sin^2\frac{A}{2})$.

$= (b^2+c^2)(1) - 2bc\cos A$

$= b^2+c^2-2bc\cos A = a^2 = RHS$ $\longrightarrow$ (3M)

$\therefore$ LHS = RHS

$S_{17}(11)$:- Show that $a^2\cot A + b^2\cot B + c^2\cot C = \frac{abc}{R}$

Sol. LHS $\rightarrow a^2\cot A + b^2\cot B + c^2\cot C$

$= 4R^2\sin^2 A\frac{\cos A}{\sin A} + 4R^2\sin^2 B\frac{\cos B}{\sin B} + 4R^2\sin^2 C\cdot\frac{\cos C}{\sin C}$ $\longrightarrow$ (1M)

$= 4R^2(\sin A\cos A + \sin B\cos B + \sin C\cos C)$

$= 4R^2\left[\frac{1}{2}\right][2\sin A\cos A + 2\sin B\cos B + 2\sin C\cos C]$

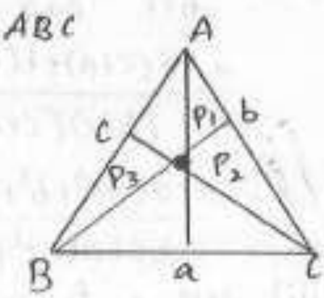
$= 2R^2[\sin 2A + \sin 2B + \sin 2C]$ $\longrightarrow$ (2M)

$= 8R^2\left[\frac{a}{2R}\right]\left[\frac{b}{2R}\right]\left[\frac{c}{2R}\right] = \frac{abc}{R} = RHS$

LHS = RHS $\longrightarrow$ (1M)

$S_{17}(12)$: If $P_1P_2P_3$ are the altitudes of the vertices A, B, C of a triangle respectively. Show that $\frac{1}{P_1^2} + \frac{1}{P_2^2} + \frac{1}{P_3^2} = \frac{\cot A + \cot B + \cot C}{\Delta}$

Sol: Given P_1, P_2, P_3 are altitudes of ΔABC



Area of $\Delta ABC = \Delta = \frac{1}{2}$ base $\times$ height

$\Delta = \frac{1}{2}aP_1 = \frac{1}{2}bP_2 = \frac{1}{2}cP_3$

$2\Delta = aP_1$ $\quad$ $2\Delta = bP_2$ $\quad$ $2\Delta = cP_3$.

$P_1 = \frac{2\Delta}{a}$ $\quad$ $P_2 = \frac{2\Delta}{b}$ $\quad$ $P_3 = \frac{2\Delta}{c}$

$\longrightarrow$ (1M)

$$LHS = \frac{1}{P_1^2} + \frac{1}{P_2^2} + \frac{1}{P_3^2}$$

$$= \frac{1}{\left[\frac{2\Delta}{a}\right]^2} + \frac{1}{\left[\frac{2\Delta}{b}\right]^2} + \frac{1}{\left[\frac{2\Delta}{c}\right]^2} = \frac{a^2}{4\Delta^2} + \frac{b^2}{4\Delta^2} + \frac{c^2}{4\Delta^2}$$

$$= \frac{a^2+b^2+c^2}{4\Delta^2} \longrightarrow (1M)$$

$$R.H.S = \frac{\cot A + \cot B + \cot C}{\Delta}$$

$$\boxed{\cot A = \frac{\cos A}{\sin A}}$$

$$= \frac{1}{\Delta}\left[\frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}\right]$$

$$= \frac{1}{\Delta}\left[\frac{b^2+c^2-a^2}{2bc\sin A} + \frac{a^2+c^2-b^2}{2ac\sin B} + \frac{a^2+b^2-c^2}{2ab\sin C}\right]$$

$$= \frac{1}{\Delta}\left[\frac{b^2+c^2-a^2}{4\Delta} + \frac{a^2+c^2-b^2}{4\Delta} + \frac{a^2+b^2-c^2}{4\Delta}\right]$$

$$= \frac{1}{\Delta}\left[\frac{b^2+c^2-a^2+a^2+c^2-b^2+a^2+b^2-c^2}{4\Delta}\right]$$

$$= \frac{1}{\Delta}\left[\frac{a^2+b^2+c^2}{4\Delta}\right]$$

$$= \frac{a^2+b^2+c^2}{4\Delta^2} \longrightarrow (2M)$$

$\therefore$ LHS = RHS

S_{17} (13):- If $a:b:c = 7:8:9$. Find $\cos A : \cos B : \cos C$

Sol: Given $a:b:c = 7:8:9$ $\longrightarrow (1M)$

$$\frac{a}{7} = \frac{b}{8} = \frac{c}{9}$$

$$a = 7K \quad b = 8K \quad c = 9K$$

$$\boxed{\begin{aligned}\cos A &= \frac{b^2+c^2-a^2}{2bc}\\ \cos B &= \frac{c^2+a^2-b^2}{2ac}\\ \cos C &= \frac{a^2+b^2-c^2}{2ab}\end{aligned}}$$

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{64K^2+81K^2-49K^2}{2(8K)(9K)}$$

$$= \frac{145(K^2)-49K^2}{(16\times 9)K^2} = \frac{96K^2}{(16\times 9)K^2} = \frac{2}{3} \longrightarrow (1M)$$

$$\cos B = \frac{c^2+a^2-b^2}{2ac}$$

$$= \frac{81K^2+49K^2-64K^2}{2(9K)(7K)}$$

$$= \frac{145(k^2) - 49k^2}{(16\times 9)k^2}$$

$$= \frac{130k^2 - 64k^2}{126k^2} = \frac{66k^2}{126k^2} = \frac{11}{21} \longrightarrow (1M)$$

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

$$= \frac{49k^2+64k^2-81k^2}{2(7k)(8k)}$$

$$= \frac{113k^2-81k^2}{112k^2} = \frac{32k^2}{112k^2} = \frac{2}{7}$$

$$\cos A : \cos B : \cos C = \frac{2}{3} : \frac{11}{21} : \frac{1}{7}$$

$$= 14 : 11 : 6. \longrightarrow (1M)$$

——— x ———

Very short answer questions:

V 1) If $A=\{0,\frac{\pi}{6},\frac{\pi}{4},\frac{\pi}{3},\frac{\pi}{2}\}$ and $f:A\to B$ is a surjection defined by $f(x)=\cos x$ then find B.

Sol: Given $A=\{0,\frac{\pi}{6},\frac{\pi}{4},\frac{\pi}{3},\frac{\pi}{2}\}$

B = Range of f and

$f:A\to B$ is a surjection. $\longrightarrow$ (1m)

$f(x)=\cos x$

$B=f(A)$

$B=\{f(0),f(\frac{\pi}{6}),f(\frac{\pi}{4}),f(\frac{\pi}{3}),f(\frac{\pi}{2})\}$

$B=\{\cos 0,\cos\frac{\pi}{6},\cos\frac{\pi}{4},\cos\frac{\pi}{3},\cos\frac{\pi}{2}\}$

$B=\{1,\frac{\sqrt{3}}{2},\frac{1}{\sqrt{2}},\frac{1}{2},0\}$ $\longrightarrow$ (1M)

V 2) If $A=\{-2,-1,0,1,2\}$ & $f:A\to B$ is a surjection defined by $f(x)=x^2+x+1$ then find B.

Sol: $f(x)=x^2+x+1$

$A=\{-2,-1,0,1,2\}$

$f:A\to B$ is a surjection & $B=f(A)$

$f(-2)=(-2)^2+(-2)+1=4-2+1=3$ $\longrightarrow$ (1m)

$f(-1)=(-1)^2+(-1)+1=1-1+1=1$

$f(0)=0+0+1=1$

$f(1)=1+1+1=3$

$f(2)=2^2+2+1=7$

$B=\{3,1,1,3,7\}$

$B=\{1,3,7\}$ $\longrightarrow$ (1M)

V 3) If $A=\{1,2,3,4\}$ & $f:A\to A$ is a function defined by $f(x)=\frac{x^2-x+1}{x+1}$ then find range of f?

Sol: $f(1)=\frac{1^2-1+1}{2}=\frac{1}{2}$

$f(2)=\frac{2^2-2+1}{2+1}=\frac{4-2+1}{3}=\frac{3}{3}$

$f(2)=1$

$\longrightarrow$ (1m)

$f(3)=\frac{3^2-3+1}{3+1}=\frac{9-3+1}{3+1}=\frac{7}{4}$

$f(4)=\frac{4^2-4+1}{4+1}=\frac{16-4+1}{4+1}=\frac{13}{5}$

Range of $f=\{\frac{1}{2},1,\frac{7}{4},\frac{13}{5}\}$ $\longrightarrow$ (1m)

V 4) If $f(x)=\frac{\cos^2x+\sin^4x}{\sin^2x+\cos^4x}$ $\forall x\in R$ then show that $f(2012)=1$

Sol: $f(x)=\frac{\cos^2x+\sin^4x}{\sin^2x+\cos^4x}$

$=\frac{\cos^2x+\sin^2x\cdot\sin^2x}{\sin^2x+\cos^2x\cdot\cos^2x}$

$=\frac{\cos^2x+\sin^2x(1-\cos^2x)}{\sin^2x+\cos^2x(1-\sin^2x)}$ $\longrightarrow$ (1m)

$\boxed{\begin{aligned}1-\cos^2x&=\sin^2x\\1-\sin^2x&=\cos^2x\end{aligned}}$

$=\frac{\cos^2x+\sin^2x-\sin^2x\cos^2x}{\sin^2x+\cos^2x-\cos^2x\sin^2x}$

$=\frac{1-\sin^2x\cos^2x}{1-\sin^2x\cos^2x}$

$=1$

$f(x)=1$

$f(2012)=1$ $\longrightarrow$ (1m)

V 5) (i) a) If $f:R\to R$ is defined by $f(x)=\frac{1-x^2}{1+x^2}$ then show that $f(\tan\theta)=\cos 2\theta$

Put $x=\tan\theta$

$f(\tan\theta)=\frac{1-\tan^2\theta}{1+\tan^2\theta}$ $\longrightarrow$ (1m)

$=\cos 2\theta$

$\longrightarrow$ (1m)

V1

5) b) If $f: R-\{\pm 1\} \to R$ is defined by $f(x)=\log\left|\frac{1+x}{1-x}\right|$ then show that $f\left(\frac{2x}{1+x^2}\right)=2f(x)$

Sol: $f(x)=\log\left|\frac{1+x}{1-x}\right|$ $\quad f\left(\frac{2x}{1+x^2}\right)$

$$=\log\left|\frac{1+\frac{2x}{1+x^2}}{1-\frac{2x}{1+x^2}}\right| \longrightarrow (1M)$$

$$=\log\left|\frac{(1+x)^2}{(1-x)^2}\right|$$

$$=\log\left|\frac{1+x}{1-x}\right|^2$$

$$=2\log\left|\frac{1+x}{1-x}\right|$$

V1

$$=2f(x) \longrightarrow (1M)$$

6) If $f: R\to R$ is defined as $f(x+y)=f(x)+f(y)\ \forall\ x,y\in R$. $f(1)=7$ then find $\sum_{r=1}^{n} f(r)$

Sol: $f(x+y)=f(x)+f(y)$

$f(1)=7$

$x=1$ & $y=1$

$f(1+1)=f(1)+f(1)$

$f(2)=2f(1) \longrightarrow (1M)$

$f(3)=3f(1)$

$f(4)=4f(1)$ ---- up to so on

$$\sum_{r=1}^{n} f(r)=f(1)+f(2)+f(3)+f(4)+\dots+f(n)$$

$$=f(1)+2f(1)+3f(1)+4f(1)+\dots+nf(1)$$

$$=f(1)[1+2+3+\dots+n]$$

$$=f(1)\frac{n(n+1)}{2}$$

$$=\frac{7n(n+1)}{2} \longrightarrow (1M)$$

V1

7) If $2^x+2^y=2$ then domain of x.

Sol: $2^x+2^y=2$

$2^x=2-2^y<2 \longrightarrow (1M)$

$2^x<2^1$

$x<1$

$x\in(-\infty,1)$

Domain of x is $(-\infty,1) \longrightarrow (1M)$

V1

8) To find domain of $f(x)=\log(x^2-4x+3)$

$\boxed{\log f(x) \Leftrightarrow f(x)>0}$

$x^2-4x+3>0$

$x(x-3)-1(x-3)>0$

$(x-1)(x-3)>0 \longrightarrow (1M)$

$\boxed{\text{If } (x-\alpha)(x-\beta)>0\ (\alpha<\beta) \text{ then } x\in(-\infty,\alpha)\cup(\beta,\infty)}$

$x\in(-\infty,1)\cup(3,\infty)$

Domain of 'f' is $R-[1,3] \longrightarrow (1M)$

V2

9) To find domain of $f(x)=\sqrt{4x-x^2}$

$(4x-x^2)\geq 0$

$x^2-4x\leq 0$

$x(x-4)\leq 0 \longrightarrow (1M)$

$\boxed{\text{If } (x-\alpha)(x-\beta)\leq 0\ (\alpha<\beta) \text{ then } x\in[\alpha,\beta]}$

$x\in[0,4]$

Domain of 'f' is $[0,4] \longrightarrow (1M)$

10) To find domain of $f(x)=\sqrt{x^2-25}$

$\boxed{\sqrt{f(x)}=f(x)\geq 0}$

$(x-5)(x+5)\geq 0 \longrightarrow (1M)$

$\boxed{\text{If } (x-\alpha)(x-\beta)\geq 0\ (\alpha<\beta) \text{ then } x\in(-\infty,\alpha]\cup[\beta,\infty)}$

$(-\infty,-5]\cup[5,\infty)$

Domain of $f=(-\infty,-5]\cup[5,\infty) \longrightarrow (1M)$

11) To find domain of

$$f(x) = \frac{\sqrt{3+x} + \sqrt{3-x}}{x}$$

$\boxed{\sqrt{f(x)} \Rightarrow f(x) \geq 0}$

$\boxed{\frac{1}{f(x)} \Leftrightarrow f(x) \neq 0}$ $\longrightarrow$ (1m)

$3+x \geq 0$

$3-x \geq 0$

$x \neq 0$

$x \geq -3,\ x \leq 3,\ x \neq 0$

$x \in [-3,3] - \{0\}$

(or)

$[-3,0) \cup (0,3]$

Domain of f is $[-3,3]-\{0\}$

(or)

$[-3,0) \cup (0,3]$ $\longrightarrow$ (1m)

$\frac{1}{2}$ 12) To find domain of

$$f(x) = \frac{1}{6x - x^2 - 5}$$

Sol:

$$\frac{1}{5x + x - x^2 - 5}$$

$$= \frac{1}{(5x-5) + (x - x^2)}$$

$$= \frac{1}{5(x-1) - x(x-1)}$$

$$= \frac{1}{(5-x)(x-1)}$$

$\boxed{\frac{1}{f(x)} = f(x) \neq 0}$ $\longrightarrow$ (1m)

$f(x)$ define when $(x-1)(5-x) \neq 0$

$x \neq 1, 5$

Domain of f is $R - \{1,5\}$ $\longrightarrow$ (1m)

$\frac{1}{2}$ 13) To find domain $f(x) = \frac{1}{(x^2-1)(x+3)}$

Sol: $f(x) = \frac{1}{(x^2-1)(x+3)}$

$$= \frac{1}{(x-1)(x+1)(x+3)}$$

$\boxed{\frac{1}{f(x)} = f(x) \neq 0}$ $\longrightarrow$ (1m)

$f(x)$ is definite when $f(x) \neq 0$

$(x-1)(x+1)(x+3) \neq 0$

$x \neq (-3, -1, 1)$

Domain of f is $R - \{-3, -1, 1\}$ $\longrightarrow$ (1m)

$\frac{1}{2}$ 14) To find the domain of

$$f(x) = \frac{1}{\sqrt{x^2 - a^2}}$$

Sol:

$\boxed{\frac{1}{\sqrt{f(x)}} = f(x) > 0}$ $\longrightarrow$ (1m)

$x^2 - a^2 > 0$

$(x-a)(x+a) > 0$

$(x-(-a))(x-a) > 0$

Domain of $x \in (-\infty, -a) \cup (a, \infty)$ $\longrightarrow$ (1m)

$\frac{1}{2}$ 15) a) find range of $f(x) = \log|1-x^2|$

Sol: $y = f(x) = \log|4-x^2|$

$\boxed{\text{Range of } \log|f(x)| \text{ is } R}$ $\longrightarrow$ (1m)

$f(x) \in R$

$4 - x^2 \neq 0$

$x \neq \pm 2$

$|4-x^2| = e^y$

$e^y > 0 \quad \forall y \in R$

Range of f is R $\longrightarrow$ (1m)

$\frac{1}{2}$ 15) b) Range of $f(x) = \frac{x^2-4}{x-2}$

Domain of f is $R - \{2\}$ $\longrightarrow$ (1m)

$y = x+2$

$x \neq 2$

$y \neq 4$

The range of f is $R - \{4\}$ $\longrightarrow$ (1m)

16) To find domain & range of

$f(x) = \sqrt{9-x^2}$

Sol: $\sqrt{f(x)} = f(x) \geq 0$

$9-x^2 \geq 0$ → (1M)

$\boxed{(\alpha-x)(x-\beta) \geq 0 \ (\alpha<\beta) \text{ then } x \in [\alpha,\beta]}$

∴ $x \in [-3,3]$ is domain

let $y = f(x)$

$x = \sqrt{9-y^2}$

$-3 \leq x \leq 3$

$0 \leq x^2 \leq 9$

$-9 \leq 9-x^2 \leq 0$

$0 \leq 9-x^2 \leq 9$

$0 \leq \sqrt{9-x^2} \leq 3$

Range of $f = [0,3]$ → (1M)

VSAQ's

V_3 ①: If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 8 \\ 7 & 2 \end{bmatrix}$ and $2x + A = B$ then find x.

Sol:- Given that $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 8 \\ 7 & 2 \end{bmatrix}$

$$2x + A = B \Rightarrow 2x = B - A$$ → (1M)

$$2x = \begin{bmatrix} 3 & 8 \\ 7 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$$ → (1M)

V_3 ②: If $\begin{bmatrix} x-3 & 2y-8 \\ z+2 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ -2 & a-4 \end{bmatrix}$ find x, y, z and a.

Sol:- Given that $\begin{bmatrix} x-3 & 2y-8 \\ z+2 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ -2 & a-4 \end{bmatrix}$

$x-3=5$	$2y-8=2$	$z+2=-2$	$6=a-4$
$x=8$	$2y=10$	$z=-4$	$a=10$
	$y=5$		

→ (2M)

$\therefore x=8, y=5, z=-4, a=10$

V_3 ③: Define trace of a matrix and find the trace of A. If $A = \begin{bmatrix} 1 & 2 & -1/2 \\ 0 & -1 & 2 \\ -1/2 & 2 & 1 \end{bmatrix}$

Sol:- Trace :- The sum of the principal diagonal elements of a square matrix is called a "Trace" of a matrix. → (1M)

Given $A = \begin{bmatrix} 1 & 2 & -1/2 \\ 0 & -1 & 2 \\ -1/2 & 2 & 1 \end{bmatrix}$

$Tr(A) = 1-1+1$
$= 1$ $\longrightarrow (1m)$

V_3 ④: Define symmetric and skew-symmetric matrix.

Symmetric Matrix :- A square matrix 'A' is said to be symmetric if $\boxed{A^T = A}$. $\longrightarrow (1m)$

Skew-Symmetric Matrix :- A square matrix 'A' is said to be skew-symmetric if $\boxed{A^T = -A}$. $\longrightarrow (1m)$

V_3 ⑤: If $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & -6 & x \end{bmatrix}$ and det $A = 45$ then find x.

Sol:- Given $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & -6 & x \end{bmatrix}$ det $A = 45$

$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & -6 & x \end{vmatrix} = 45$ $\longrightarrow (1m)$

$\Rightarrow 1(3x+24) = 45$

$\Rightarrow 3x = 45 - 24 = 21$

$\Rightarrow 3x = 21$

$\Rightarrow x = \frac{21}{3}$

$\Rightarrow \boxed{x = 7}$ $\longrightarrow (1m)$

V_3 ⑥: If ω is a complex cube root of units then show that $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix} = 0$

Sol:- L.H.S. $= \begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$

$R_1 \to R_1 + R_2 + R_3$ $\longrightarrow (1m)$

$= \begin{vmatrix} 1+\omega+\omega^2 & 1+\omega+\omega^2 & 1+\omega+\omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$

$= \begin{vmatrix} 0 & 0 & 0 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$ $\because \boxed{1+\omega+\omega^2 = 0}$

L.H.S = R.H.S. $\longrightarrow (1m)$

V_3 7: If $A = \begin{bmatrix} 2 & 4 \\ -1 & k \end{bmatrix}$ and $A^2 = 0$. Find the value of k.

Sol:- Given $A = \begin{bmatrix} 2 & 4 \\ -1 & k \end{bmatrix}$

$A^2 = 0 \Rightarrow A \cdot A = 0$

$\Rightarrow \begin{bmatrix} 2 & 4 \\ -1 & K \end{bmatrix} \begin{bmatrix} 2 & 4 \\ -1 & K \end{bmatrix} = 0$ $\longrightarrow (1m)$

$= \begin{bmatrix} 4-4 & 8+4K \\ -2-K & -4+K^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$-2 - K = 0$

$\boxed{K = -2}$ $\longrightarrow (1m)$

V_3 (8): Construct a 3×2 matrix whose elements are defined by $a_{ij} = \frac{1}{2}|i-3j|$.

Sol:- Consider 3×2 matrix is

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \longrightarrow (1m)$$

Given $a_{ij} = \frac{1}{2}|i-3j|$

$a_{11} = \frac{1}{2}|1-3| = \frac{1}{2}|-2| = \frac{1}{2}(2) = 1$

$a_{12} = \frac{1}{2}|1-6| = \frac{1}{2}|-5| = \frac{1}{2}(5) = \frac{5}{2}$

$a_{21} = \frac{1}{2}|2-3| = \frac{1}{2}|-1| = \frac{1}{2}(1) = \frac{1}{2}$

$a_{22} = \frac{1}{2}|2-6| = \frac{1}{2}|-4| = \frac{1}{2}(4) = 2$

$a_{31} = \frac{1}{2}|3-3| = \frac{1}{2}(0) = 0.$

$a_{32} = \frac{1}{2}|3-6| = \frac{1}{2}|-3| = \frac{1}{2}(3) = \frac{3}{2}$

$$\therefore A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix} \longrightarrow (1m)$$

V_3 (9): If $A = \begin{bmatrix} -1 & 2 & 3 \\ 2 & 5 & 6 \\ 2 & x & 7 \end{bmatrix}$ is a symmetric matrix find x.

Sol:- Given $A = \begin{bmatrix} -1 & 2 & 3 \\ 2 & 5 & 6 \\ 2 & x & 7 \end{bmatrix}$ is a symmetric matrix $\therefore \boxed{A^T = A} \longrightarrow (1m)$

$$\begin{bmatrix} -1 & 2 & 3 \\ 2 & 5 & \boxed{6} \\ 3 & \boxed{x} & 7 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 3 \\ 2 & 5 & \boxed{x} \\ 3 & \boxed{6} & 7 \end{bmatrix}$$

$\therefore \boxed{x = 6} \longrightarrow (1m)$

V_3 (10): If $A = \begin{bmatrix} 0 & 2 & 1 \\ -2 & 0 & -2 \\ -1 & x & 0 \end{bmatrix}$ is a skew-symmetric matrix. Find x.

Sol:- Given $A = \begin{bmatrix} 0 & 2 & 1 \\ -2 & 0 & -2 \\ -1 & x & 0 \end{bmatrix}$ is a skew-symmetric matrix.

$\boxed{A^T = -A} \longrightarrow (1m)$

$$\begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & x \\ 1 & -2 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 2 & 1 \\ -2 & 0 & -2 \\ -1 & x & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & x \\ 1 & \boxed{-2} & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & 2 \\ 1 & \boxed{-x} & 0 \end{bmatrix}$$

$\Rightarrow -x = -2 \therefore \boxed{x = 2} \longrightarrow (1m)$

V_3 (11): Find the determinent of $\begin{bmatrix} 1^2 & 2^2 & 3^2 \\ 2^2 & 3^2 & 4^2 \\ 3^2 & 4^2 & 5^2 \end{bmatrix}$

Sol:- Given $A = \begin{bmatrix} 1^2 & 2^2 & 3^2 \\ 2^2 & 3^2 & 4^2 \\ 3^2 & 4^2 & 5^2 \end{bmatrix}$

$$\text{Det } A = \begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{vmatrix} \longrightarrow (1m)$$

$= 1(225-256) - 4(100-144) + 9(64-81)$

$= 1(-31) - 4(-44) + 9(-17)$

$= -31 + 176 - 153$

$= -184 + 176 = -8$

$\boxed{|A| = -8} \longrightarrow (1m)$

V4 (12): If $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ then show that $A \cdot A^T = A^T A = I$

Sol:- Given $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$

$A^T = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

$A \cdot A^T = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$

$= \begin{bmatrix} \cos^2\alpha + \sin^2\alpha & -\cos\alpha\sin\alpha + \sin\alpha\cos\alpha \\ -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha & \sin^2\alpha + \cos^2\alpha \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\boxed{\therefore \sin^2\theta + \cos^2\theta = 1}$ $\longrightarrow (1m)$

$A \cdot A^T = I$, Similarly $A^T \cdot A = I$

$A \cdot A^T = A^T \cdot A = I \longrightarrow (1m)$

V4 (13): Find the Rank of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

Sol:- Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

det $A = 1(1-1) - 1(1-1) + 1(1-1)$

det $A = 0 \longrightarrow (1m)$

All 2×2 submatrices det is zero.

$\therefore$ Rank of $A = 1$. $\longrightarrow (1m)$

V4 (14): Find the Rank of $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$

Sol:- Given $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$

det $A = 1(0-2) - 2(1-0) + 1(-1-0)$

$= 1(-2) - 2(1) + 1(-1)$

$= -2-2-1 = -5 \neq 0 \longrightarrow (1m)$

$\therefore$ Rank of $A = 3 \longrightarrow (1m)$

V4 (15) Find the Rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$

Sol:- Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$

det $A = 1(6-4) - 2(4-0) + 3(2-0)$

$\Rightarrow 1(2) - 2(4) + 3(2) \Rightarrow 2 - 8 + 6$

$\Rightarrow 8 - 8 = 0 \longrightarrow (1m)$

Let $B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ be a submatrix

det $B = 3 - 4 = -1 \neq 0$

$\therefore$ Rank of $A = 2$. $\longrightarrow (1m)$

ADDITION OF VECTORS VSAQ

(V5)

(1) (i) Find the unit vector in the direction of vector $\bar{a} = 2\bar{i} + 3\bar{j} + \bar{k}$

Sol: Given $\bar{a} = 2\bar{i} + 3\bar{j} + \bar{k}$

$$|\bar{a}| = \sqrt{2^2 + 3^2 + 1^2}$$

$$= \sqrt{4+9+1} \longrightarrow (1M)$$

$$= \sqrt{14}$$

$\therefore$ The unit vector in the direction of $\bar{a} = \dfrac{\bar{a}}{|\bar{a}|} = \dfrac{2\bar{i} + 3\bar{j} + \bar{k}}{\sqrt{14}}$ $\longrightarrow (1M)$

V5

(ii) Let $a = 2\bar{i} + 4\bar{j} - 5\bar{k}$, $b = \bar{i} + \bar{j} + \bar{k}$ & $c = \bar{j} + 2\bar{k}$. Find the unit vector in the opposite direction of $a+b+c$.

Sol: Given $\bar{a} = 2\bar{i} + 4\bar{j} - 5\bar{k}$; $\bar{b} = \bar{i} + \bar{j} + \bar{k}$; $\bar{c} = \bar{j} + 2\bar{k}$

$$(\bar{a} + \bar{b} + \bar{c}) = 2\bar{i} + 4\bar{j} - 5\bar{k} + \bar{i} + \bar{j} + \bar{k} + \bar{j} + 2\bar{k}$$

$$= 3\bar{i} + 6\bar{j} - 2\bar{k} \longrightarrow (1M)$$

Now $|\bar{a} + \bar{b} + \bar{c}| = \sqrt{3^2 + 6^2 + (-2)^2} = \sqrt{9+36+4} = \sqrt{49} = 7$

$$\boxed{\therefore \text{The unit vector in the opposite direction of } \bar{a} + \bar{b} + \bar{c} = \frac{-(\bar{a} + \bar{b} + \bar{c})}{|\bar{a} + \bar{b} + \bar{c}|}}$$

$$= \frac{-(3\bar{i} + 6\bar{j} - 2\bar{k})}{7} \longrightarrow (1M)$$

V5 (2) If the position vectors of the points A, B & C are $-2\bar{i} + \bar{j} - \bar{k}$, $-4\bar{i} + 2\bar{j} + 2\bar{k}$, & $6\bar{i} - 3\bar{j} - 13\bar{k}$ respectively and $AB = \lambda AC$ then find the value of λ.

Sol: Let $\bar{OA} = -2\bar{i} + \bar{j} - \bar{k}$; $\bar{OB} = -4\bar{i} + 2\bar{j} + 2\bar{k}$; $\bar{OC} = 6\bar{i} - 3\bar{j} - 13\bar{k}$

$$\bar{AB} \Rightarrow \bar{OB} - \bar{OA} = -4\bar{i} + 2\bar{j} + 2\bar{k} + 2\bar{i} - \bar{j} + \bar{k}$$

$$\bar{AB} = -2\bar{i} + \bar{j} + 3\bar{k}$$

$$\bar{AC} \Rightarrow \bar{OC} - \bar{OA} = 6\bar{i} - 3\bar{j} - 13\bar{k} + 2\bar{i} - \bar{j} + \bar{k}$$

$$\bar{AC} = 8\bar{i} - 4\bar{j} - 12\bar{k} \Rightarrow -4(-2\bar{i} + \bar{j} + 3\bar{k}) \longrightarrow (1M)$$

Now $\bar{AB} = (\lambda \bar{AC})$

$$(-2\bar{i} + \bar{j} + 3\bar{k}) = -4\lambda(-2\bar{i} + \bar{j} + 3\bar{k})$$

$$1 = -4\lambda$$

$$\lambda = \frac{-1}{4} \longrightarrow (1M)$$

(V5)

(3) If the vectors $-3\bar{i}+4\bar{j}+\lambda\bar{k}$ & $\mu\bar{i}+8\bar{j}+6\bar{k}$ are collinear vectors, then find λ and μ

Sol:- Let $\bar{a} = -3\bar{i}+4\bar{j}+\lambda\bar{k}$; $\bar{b} = \mu\bar{i}+8\bar{j}+6\bar{k}$; are collinear.

$$\boxed{\text{If } \bar{a} = a_1\bar{i}+b_1\bar{j}+c_1\bar{k} \text{ \& } \bar{b} = a_2\bar{i}+b_2\bar{j}+c_2\bar{k} \text{ are collinear then } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

$\longrightarrow$ (1M)

$$\frac{-3}{\mu} = \frac{4}{8} = \frac{\lambda}{6}$$

$$\frac{-3}{\mu} = \frac{1}{2} \quad \Big| \quad \frac{1}{2} = \frac{\lambda}{6}$$

$$\boxed{\mu = -6} \quad \Big| \quad \boxed{\lambda = 3}$$

$\therefore$ The values of μ and λ are $-6, 3$ $\longrightarrow$ (1M)

(V6)

(4) If $\bar{a} = 2\bar{i}+5\bar{j}+\bar{k}$ and $\bar{b} = 4\bar{i}+m\bar{j}+n\bar{k}$ are collinear vectors then find the values of m and n.

Sol:- Given, $\bar{a} = 2\bar{i}+5\bar{j}+\bar{k}$; $\bar{b} = 4\bar{i}+m\bar{j}+n\bar{k}$

$$\boxed{\text{If } \bar{a} = a_1\bar{i}+b_1\bar{j}+c_1\bar{k} \text{ and } \bar{b} = a_2\bar{i}+b_2\bar{j}+c_2\bar{k} \text{ are collinear then } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

$\longrightarrow$ (1M)

$$\frac{2}{4} = \frac{5}{m} = \frac{1}{n}$$

$$\frac{1}{2} = \frac{5}{m} \quad \Big| \quad \frac{1}{2} = \frac{1}{n}$$

$$m = 10 \quad \Big| \quad n = 2$$

$\therefore$ The values of m, n are 10, 2. $\longrightarrow$ (1M)

(5) If $\overline{OA} = \bar{i}+\bar{j}+\bar{k}$ and $\overline{AB} = 3\bar{i}-2\bar{j}+\bar{k}$, $\overline{BC} = \bar{i}+2\bar{j}-2\bar{k}$ and $\overline{CD} = 2\bar{i}+\bar{j}+3\bar{k}$ then find the vector $\overline{OD}$.

Sol:- Given $\overline{OA} = \bar{i}+\bar{j}+\bar{k}$; $\overline{AB} = 3\bar{i}-2\bar{j}+\bar{k}$

$\overline{BC} = \bar{i}+2\bar{j}-2\bar{k}$; $\overline{CD} = 2\bar{i}+\bar{j}+3\bar{k}$

$$\overline{OD} = \overline{OA}+\overline{AB}+\overline{BC}+\overline{CD}$$

$\longrightarrow$ (1M)

$$\overline{OD} = \bar{i}+\bar{j}+\bar{k}+3\bar{i}-2\bar{j}+\bar{k}+2\bar{i}+\bar{j}+3\bar{k}+\bar{i}+2\bar{j}-2\bar{k}$$

$$\overline{OD} = 7\bar{i}+2\bar{j}+3\bar{k}$$

$\longrightarrow$ (1M)

(V5)

(6) OABC is a parallelogram. If OA = a and OC = c then find the vector equation of the side BC.

Sol:- Let 'O' be the origin

$\overline{OA} = a$, $\overline{OC} = c$

$\overline{CB} = \overline{OA} = \bar{a}$ the vector equation of $\overline{BC}$ is vector equation of the line passing through $\bar{c}$ and parallel to $\bar{a}$

$$\Rightarrow \boxed{\bar{r} = \bar{c}+t\bar{a}}$$

$\longrightarrow$ 1M

$\therefore$ The vector equation of the line passing through $\bar{a}$ and parallel to the $\bar{b}$ is $\bar{r} = \bar{a} + t\bar{b}$ $\longrightarrow$ (1m)

V_6 (7) Find the equation of the plane which passes through the points $2\bar{i}+4\bar{j}+2\bar{k}$, $2\bar{i}+3\bar{j}+5\bar{k}$ and parallel to the vectors $3\bar{i}-2\bar{j}+\bar{k}$

Sol: Let $\bar{a} = 2\bar{i}+4\bar{j}+2\bar{k}$; $\bar{b} = 2\bar{i}+3\bar{j}+5\bar{k}$; $\bar{c} = 3\bar{i}-2\bar{j}+\bar{k}$

The vector equation of the plane passing through the points $\bar{a},\bar{b}$ and parallel to vector $\bar{c}$ is
$\bar{r} = (1-t)\bar{a} + t\bar{b} + s\bar{c}$; $t, s \in R$ $\longrightarrow$ (1m)

$\bar{r} = (1-t)(2\bar{i}+4\bar{j}+2\bar{k}) + t(2\bar{i}+3\bar{j}+5\bar{k}) + s(3\bar{i}-2\bar{j}+\bar{k})$; $t, s \in R$ $\longrightarrow$ (1m)

V_6 (8) Find the vector equation of the line joining the points $2\bar{i}+\bar{j}+3\bar{k}$ and $-4\bar{i}+3\bar{j}-\bar{k}$

Sol: Let $\bar{a} = 2\bar{i}+\bar{j}+3\bar{k}$ $\quad \bar{b} = -4\bar{i}+3\bar{j}-\bar{k}$

The vector equation of the line passing through the points $\bar{a},\bar{b}$ is $\bar{r} = (1-t)\bar{a} + t\bar{b}$; $t \in R$

$\bar{r} = (1-t)(2\bar{i}+\bar{j}+3\bar{k}) + t(-4\bar{i}+3\bar{j}-\bar{k})$; $t \in R$ $\longrightarrow$ (2m)

V_6 (9) Find the vector equation of the line passing through the points $2\bar{i}+3\bar{j}+\bar{k}$ and parallel to the vector $4\bar{i}-2\bar{j}+3\bar{k}$

Sol: Let $\bar{a} = 2\bar{i}+3\bar{j}+\bar{k}$ $\quad \bar{b} = 4\bar{i}-2\bar{j}+3\bar{k}$

The vector equation of the line passing through the point $\bar{a}$ and parallel to vector $\bar{b}$ is $\bar{r} = \bar{a} + t\bar{b}$; $t \in R$

$\bar{r} = (2\bar{i}+3\bar{j}+\bar{k}) + t(4\bar{i}-2\bar{j}+3\bar{k})$ $\longrightarrow$ (2m)

V_6 (10) Find the vector equation of the plane passing through the ~~pts~~ points $\bar{i}-2\bar{j}+5\bar{k}$, $-5\bar{j}-\bar{k}$ and $-3\bar{i}+5\bar{j}$

Sol: Let $\bar{a} = \bar{i}-2\bar{j}+5\bar{k}$ $\quad \bar{b} = -5\bar{j}-\bar{k}$ $\quad \bar{c} = -3\bar{i}+5\bar{j}$

The vector equation of the plane passing through the points $\bar{a},\bar{b}$ and $\bar{c}$ is $\bar{r} = (1-t-s)\bar{a} + t\bar{b} + s\bar{c}$; $t, s \in R$

$\bar{r} = (1-t-s)(\bar{i}-2\bar{j}+5\bar{k}) + t(-5\bar{j}-\bar{k}) + s(-3\bar{i}+5\bar{j})$ $\longrightarrow$ 2M

V_6 (11) If $\bar{a},\bar{b},\bar{c}$ are the position vectors of the vertices A, B, C respectively of ΔABC then the vector equations of the median through vertex A.

Sol: Let $\overline{OA} = \bar{a}$ $\quad \overline{OB} = \bar{b}$ $\quad \overline{OC} = \bar{c}$

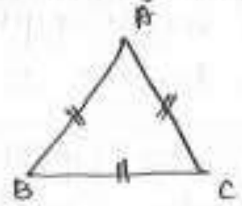
D is midpoint of BC $\Rightarrow$ $\overline{OD} = \frac{\bar{b}+\bar{c}}{2}$

$\therefore$ The vector equation of the lines passing through the points $\bar{a},\bar{b}$ is $\bar{r} = (1-t)\bar{a} + t\bar{b}$; $(t \in R)$

$\bar{r} = (1-t)\overline{OA} + t(\overline{OD})$

$\bar{r} = (1-t)\bar{a} + t\left(\frac{\bar{b}+\bar{c}}{2}\right)$, $t \in R$ $\longrightarrow$ (2m)

V_6 (12) Is the triangle formed by the vectors $3\bar{i}+5\bar{j}+2\bar{k}$, $2\bar{i}-3\bar{j}-5\bar{k}$ and $-5\bar{i}-2\bar{j}+3\bar{k}$ equilateral.



Sol:- $\bar{AB} = 3\bar{i}+5\bar{j}+2\bar{k}$

$|\bar{AB}| = \sqrt{3^2+5^2+2^2} = \sqrt{9+25+4} = \sqrt{38}$

$\bar{BC} = 2\bar{i}-3\bar{j}-5\bar{k}$

$|\bar{BC}| = \sqrt{2^2+(-3)^2+(-5)^2} = \sqrt{4+9+25} = \sqrt{38}$ $\longrightarrow$ (1M)

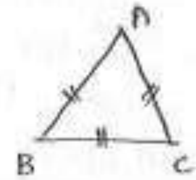
$\bar{CA} = -5\bar{i}-2\bar{j}+3\bar{k}$

$|\bar{CA}| = \sqrt{25+4+9} = \sqrt{38}$

$\therefore |\bar{AB}| = |\bar{BC}| = |\bar{CA}|$

$\therefore$ the given vectors forms an equilateral triangle. $\longrightarrow$ (1M)

V_6 (13) Is the triangle formed by the vectors $3\bar{i}+5\bar{j}+2\bar{k}$, $2\bar{i}-3\bar{j}-5\bar{k}$ & $-5\bar{i}-2\bar{j}+3\bar{k}$ equilateral?



Sol:- Let $\bar{AB} = 3\bar{i}+5\bar{j}+2\bar{k}$

$|\bar{AB}| = \sqrt{3^2+5^2+2^2} = \sqrt{9+25+4} = \sqrt{38}$

$\bar{BC} = 2\bar{i}-3\bar{j}-5\bar{k}$

$|\bar{BC}| = \sqrt{2^2+(-3)^2+(-5)^2} = \sqrt{4+9+25} = \sqrt{38}$ $\longrightarrow$ (1M)

$\bar{CA} = -5\bar{i}-2\bar{j}+3\bar{k}$

$|\bar{CA}| = \sqrt{25+4+9} = \sqrt{38}$

$\therefore |\bar{AB}| = |\bar{BC}| = |\bar{AC}|$

$\therefore$ The given vectors are forms equilateral triangle $\longrightarrow$ (1M)

V_6 (14) Find the vector equation of the plane passing through the points $(0,0,0)$, $(0,5,0)$, $(2,0,1)$

Sol:- Let $\bar{a} = (0,0,0) = \bar{0}$ $\quad \bar{b} = (0,5,0) = 5\bar{j}$ $\quad \bar{c} = (2,0,1) = 2\bar{i}+\bar{k}$

$\boxed{\text{The vector eq of the plane passing through the points } \bar{a}, \bar{b}, \bar{c} \text{ is } \bar{r} = (1-t-s)\bar{a} + t\bar{b} + s\bar{c}\ ;\ t, s \in R}$

$\bar{r} = (1-t-s)\bar{0} + t(5\bar{j}) + s(2\bar{i}+\bar{k})$

$\bar{r} = 5t\bar{j} + 2s\bar{i} + s\bar{k}\ ;\ t, s \in R$ $\longrightarrow$ (2M)

V_6 (15) ABCDE is a pentagon. If the sum of the vectors $\bar{AB}$, $\bar{BC}$, $\bar{AE}$, $\bar{DC}$, $\bar{ED}$ and $\bar{AC}$ is $\lambda\bar{AC}$ then find the value of λ.

Sol:- Given, ABCDE is a pentagon.

$\bar{AB} + \bar{AE} + \bar{BC} + \bar{DE} + \bar{ED} + \bar{AC} = \lambda\bar{AC}$ $\longrightarrow$ (1M)

$(\bar{AB}+\bar{BC}) + (\bar{AE}+\bar{ED}+\bar{DC}) + \bar{AC} = \lambda\bar{AC}$

$\bar{AC} + \bar{AC} + \bar{AC} = \lambda\bar{AC}$

$3\bar{AC} = \lambda\bar{AC}$

$\lambda = 3$ $\longrightarrow$ (1M)

PRODUCT OF VECTORS

V.S.AQ's

$V_7(1)$:- If $\bar{a}=\bar{i}+2\bar{j}-3\bar{k}$ and $\bar{b}=3\bar{i}-\bar{j}+2\bar{k}$ then show that $\bar{a}+\bar{b}$ and $\bar{a}-\bar{b}$ are perpendicular to each other

Sol: Given $\bar{a}=\bar{i}+2\bar{j}-3\bar{k}$ $\quad \bar{b}=3\bar{i}-\bar{j}+2\bar{k}$

$$\bar{a}+\bar{b}=\bar{i}+2\bar{j}-3\bar{k}+3\bar{i}-\bar{j}+2\bar{k}$$

$$=4\bar{i}+\bar{j}-\bar{k}$$

$$\bar{a}-\bar{b}=\bar{i}+2\bar{j}-3\bar{k}-(3\bar{i}-\bar{j}+2\bar{k})$$

$$=\bar{i}+2\bar{j}-3\bar{k}-3\bar{i}+\bar{j}-2\bar{k}$$

$$=-2\bar{i}+3\bar{j}-5\bar{k} \longrightarrow (1M)$$

If $\bar{a},\bar{b}$ are perpendicular then $\bar{a}\cdot\bar{b}=0$

$$(\bar{a}+\bar{b})(\bar{a}-\bar{b})=(4\bar{i}+\bar{j}-\bar{k})(-2\bar{i}+3\bar{j}-5\bar{k})$$

$$=(-8+3+5)$$

$$=(-8+8)$$

$$=0$$

$\therefore \bar{a}+\bar{b}$ and $\bar{a}-\bar{b}$ are perpendicular to each other $\longrightarrow (1M)$

$V_7(2)$:- If the vectors $\lambda\bar{i}-3\bar{j}+5\bar{k}$ and $2\lambda\bar{i}-\lambda\bar{j}-\bar{k}$ are perpendicular to each other. Find λ

Sol: Let $\bar{a}=\lambda\bar{i}-3\bar{j}+5\bar{k}$, $\bar{b}=2\lambda\bar{i}-\lambda\bar{j}-\bar{k}$

$\boxed{\because \bar{a} \text{ is perpendicular to } \bar{b} \Rightarrow \bar{a}\cdot\bar{b}=0}$ $\longrightarrow (1M)$

$$(\lambda\bar{i}-3\bar{j}+5\bar{k})(2\lambda\bar{i}-\lambda\bar{j}-\bar{k})=0$$

$$2\lambda^2+3\lambda-5=0 \Rightarrow (2\lambda+5)(\lambda-1)=0$$

$$\boxed{\lambda=-\frac{5}{2}},\ \boxed{\lambda=1} \longrightarrow (1M)$$

$V_7(3)$:- If $4\bar{i}+\frac{2}{3}p\bar{j}+p\bar{k}$ is parallel to the vector $\bar{i}+2\bar{j}+3\bar{k}$ find p.

Sol: Let $\bar{a}=4\bar{i}+\frac{2p}{3}\bar{j}+p\bar{k}$, $\bar{b}=\bar{i}+2\bar{j}+3\bar{k}$ are parallel

$\boxed{\text{If } \bar{a}=a_1\bar{i}+b_1\bar{j}+c_1\bar{k} \ \& \ \bar{b}=a_2\bar{i}+b_2\bar{j}+c_2\bar{k} \text{ are parallel then } \frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}}$ $\longrightarrow (1M)$

$$\frac{4}{1}=\frac{\frac{2p}{3}}{2}=\frac{p}{3}$$

$$\frac{4}{1}=\frac{p}{3}$$

$$\boxed{P=12} \longrightarrow (1M)$$

$V_7(4)$: Find the angle between the vector $\bar{i}+2\bar{j}+3\bar{k}$ and $3\bar{i}-\bar{j}+2\bar{k}$

Sol: Given $\bar{a}=\bar{i}+2\bar{j}+3\bar{k}$ $\quad \bar{b}=3\bar{i}-\bar{j}+2\bar{k}$

$$|\bar{a}|=\sqrt{1^2+2^2+3^2}=\sqrt{1+4+9}=\sqrt{14}$$

$$|\bar{b}|=\sqrt{3^2+(-1)^2+2^2}=\sqrt{9+1+4}=\sqrt{14}$$

$$\bar{a}\cdot\bar{b}=(\bar{i}+2\bar{j}+3\bar{k})(3\bar{i}-\bar{j}+2\bar{k})$$

$$=3-2+6$$

$$=9-2=7 \longrightarrow (1M)$$

Angle between two vectors $\bar{a}$ and $\bar{b}$ is

$$\cos\theta=\frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$$

$$\cos\theta=\frac{7}{\sqrt{14}\sqrt{14}}=\frac{7}{14}=\frac{1}{2}$$

$$\cos\theta=\cos 60^\circ$$

$$\boxed{\theta=60^\circ} \longrightarrow (1M)$$

$V_7(5)$: Find the angle between the planes $\bar{r}\cdot(2\bar{i}-\bar{j}+2\bar{k})=3$ $\bar{r}\cdot(3\bar{i}+6\bar{j}+\bar{k})=4$

Sol: Given $\bar{r}\cdot(2\bar{i}-\bar{j}-2\bar{k})=3$

$$\bar{a}=2\bar{i}-\bar{j}-2\bar{k}$$

$$\bar{r}\cdot(3\bar{i}+6\bar{j}+\bar{k})=4$$

$$\bar{b}=3\bar{i}+6\bar{j}+\bar{k}$$

$$|\bar{a}|=\sqrt{2^2+(-1)^2+(-2)^2}=\sqrt{4+1+4}=\sqrt{9}=3$$

$$|\bar{b}|=\sqrt{3^2+6^2+1^2}=\sqrt{9+36+1}=\sqrt{46}$$

$$\bar{a}\cdot\bar{b}=(2\bar{i}-\bar{j}-2\bar{k})\cdot(3\bar{i}+6\bar{j}+\bar{k})$$

$$=6-6+2=2 \longrightarrow (1M)$$

$$\boxed{\cos\theta=\frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}}$$

$$\cos\theta=\frac{2}{3\sqrt{46}}$$

$$\theta=\cos^{-1}\left(\frac{2}{3\sqrt{46}}\right) \longrightarrow (1M)$$

$V_7(6)$:- Find the area of Parallelogram having $2\bar{i}-3\bar{j}$ and $3\bar{i}-\bar{k}$ as adjacent sides

Sol: Let $\bar{a}=2\bar{i}-3\bar{j}$; $\bar{b}=3\bar{i}-\bar{k}$

Area of parallelogram with adjacent sides $\bar{a}$ and $\bar{b}$ is $|\bar{a}\times\bar{b}|$

$$\bar{a}\times\bar{b}=\begin{vmatrix}\bar{i} & \bar{j} & \bar{k}\\ 2 & -3 & 0\\ 3 & 0 & -1\end{vmatrix} \longrightarrow (1M)$$

$$=\bar{i}(3-0)+\bar{j}(-2-0)+\bar{k}(0+9)$$

$$=3\bar{i}+2\bar{j}+9\bar{k}$$

Required area is $|\bar{a}\times\bar{b}|=\sqrt{3^2+2^2+9^2}$

$$=\sqrt{9+4+81}=\sqrt{94} \longrightarrow (1M)$$

$V_7(7)$:- If $\bar{a}=2\bar{i}-\bar{j}+\bar{k}$ and $\bar{b}=\bar{i}-3\bar{j}-5\bar{k}$ then find $|\bar{a}\times\bar{b}|$

Sol: Given $\bar{a}=2\bar{i}-\bar{j}+\bar{k}$ $\quad \bar{b}=\bar{i}-3\bar{j}-5\bar{k}$

$$[\bar{a}\times\bar{b}]=\begin{vmatrix} i & j & k\\ 2 & -1 & 1\\ 1 & -3 & -5\end{vmatrix}$$

$$=\bar{i}(5+3)-\bar{j}(-10-1)+\bar{k}(-6+1)$$

$$=8\bar{i}+11\bar{j}-5\bar{k} \longrightarrow (1M)$$

$$|\bar{a}\times\bar{b}|=\sqrt{8^2+11^2+5^2}=\sqrt{64+121+25}=\sqrt{210} \longrightarrow (1M)$$

$V_7(8)$:- Find the area of Parallelogram whose diagonals are $3\bar{i}+\bar{j}-2\bar{k}$ and $\bar{i}-3\bar{j}+4\bar{k}$

Sol: Let $\bar{a}=3\bar{i}+\bar{j}-2\bar{k}$ $\quad \bar{b}=\bar{i}-3\bar{j}+4\bar{k}$

Area of parallelogram with diagonals $\bar{a}$ & $\bar{b}=\frac{1}{2}|\bar{a}\times\bar{b}|$

$$\bar{a}\times\bar{b}=\begin{vmatrix}\bar{i} & \bar{j} & \bar{k}\\ 3 & 1 & -2\\ 1 & -3 & 4\end{vmatrix}$$

$$=i(4-6)-\bar{j}(12+2)+\bar{k}(-9-1)$$

$$=-2\bar{i}-14\bar{j}-10\bar{k} \longrightarrow (2M)$$

$$|\bar{a}\times\bar{b}|=\sqrt{(-2)^2+(-14)^2+(-10)^2}=\sqrt{4+196+100}=\sqrt{300}$$

$$=\sqrt{4\times75}=2\sqrt{75}$$

Required area $=\frac{1}{2}|a\times b| \Rightarrow \frac{1}{\cancel{2}}|\cancel{2}\sqrt{75}|=\sqrt{75} \longrightarrow (1M)$

V_7(9): If $\frac{1}{2}|\bar{e}_1-\bar{e}_2| = \sin\lambda\theta$, where $\bar{e}_1$ and $\bar{e}_2$ are unit vectors including an angle θ, show that $\lambda = 1/2$

Sol:- Given $|\bar{e}_1| = |\bar{e}_2| = 1$ and $\frac{1}{2}|\bar{e}_1-\bar{e}_2| = \sin\lambda\theta$

Squaring

$$\left[\frac{1}{2}|\bar{e}_1-\bar{e}_2|\right]^2 = (\sin\lambda\theta)^2$$

$$= \frac{1}{4}\left[|\bar{e}_1|^2+|\bar{e}_2|^2-2(\bar{e}_1\cdot\bar{e}_2)\right] = \sin^2\lambda\theta \longrightarrow (1M)$$

$$= \frac{1}{4}\left[1+1-2(|\bar{e}_1||\bar{e}_2|\cos\theta)\right] = \sin^2\lambda\theta$$

$$= \frac{1}{4}\left[2-2(1\cdot1\cos\theta)\right] = \sin^2\lambda\theta$$

$$= \frac{1}{\cancel{4}_2}\times\cancel{2}(1-\cos\theta) = \sin^2\lambda\theta$$

$$= \frac{1}{\cancel{2}}\times\cancel{2}\sin^2\theta/_2 = \sin^2\lambda\theta$$

$$= \sin^2\theta/_2 = \sin^2\lambda\theta$$

$$\cancel{\theta}/_2 = \lambda\cancel{\theta}$$

$$\boxed{\lambda = \frac{1}{2}} \longrightarrow (1M)$$

V_7(10):- $\bar{a} = \bar{i}-\bar{j}-\bar{k}$ and $\bar{b} = 2\bar{i}-3\bar{j}+\bar{k}$ then find the projection vector of $\bar{b}$ and $\bar{a}$.

Sol: $\bar{a} = \bar{i}-\bar{j}-\bar{k}$ $\quad \bar{b} = 2\bar{i}-3\bar{j}+\bar{k}$

Projection vector of $\bar{b}$ on $\bar{a} = \dfrac{(\bar{b}\cdot\bar{a})\bar{a}}{|\bar{a}|^2} \longrightarrow (1M)$

$$= \frac{\left[(2\bar{i}-3\bar{j}+\bar{k})(\bar{i}-\bar{j}-\bar{k})\right](\bar{i}-\bar{j}-\bar{k})}{(\sqrt{1+1+1})^2}$$

$$= \frac{(2+3-1)(\bar{i}-\bar{j}-\bar{k})}{3} = \frac{4}{3}(\bar{i}-\bar{j}-\bar{k}) \longrightarrow (1M)$$

V_7(11): Find the unit vectors perpendicular to plane determined by vectors $\bar{a} = 4\bar{i}+3\bar{j}-\bar{k}$, $\bar{b} = 2\bar{i}-6\bar{j}-3\bar{k}$

Sol:- $\bar{a} = 4\bar{i}+3\bar{j}-\bar{k}$ $\quad \bar{b} = 2\bar{i}-6\bar{j}-3\bar{k}$

$$\bar{a}\times\bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 4 & 3 & -1 \\ 2 & -6 & -3 \end{vmatrix}$$

$$= \bar{i}(-9-6)-\bar{j}(-12+2)+\bar{k}(-24-6)$$

$$= -15\bar{i}+10\bar{j}-30\bar{k}$$

$$= 5(-3\bar{i}+2\bar{j}-6\bar{k}) \longrightarrow (1M)$$

$$|\bar{a}\times\bar{b}| = 5\sqrt{9+4+36} = 5\sqrt{49} = 5(7) = 35 \longrightarrow (1M)$$

$V_7(12)$: If $|\bar{p}|=2$, $|\bar{q}|=3$ and $(\bar{p},q)=\pi/6$ then find $|\bar{p}\times\bar{q}|^2$

Sol: $|\bar{p}\times\bar{q}|^2 = |\bar{p}|^2|\bar{q}|^2\sin^2\theta$ $\longrightarrow$ (1M)

$= 2^2\, 3^2 \sin^2 30$

$= 4\cdot 9 \frac{1}{4} = 9$ $\longrightarrow$ (1M)

$V_7(13)$: If $a=2\bar{i}+2\bar{j}-3\bar{k}$, $b=3i-j+2\bar{k}$, the find the angle between $2a+b$ and $a+2b$.

Sol: $2\bar{a}+\bar{b}=2(2\bar{i}+2\bar{j}-3\bar{k})+(3\bar{i}-\bar{j}+2\bar{k})=7\bar{i}+3\bar{j}-4\bar{k}$

$\bar{a}+2\bar{b}=(2\bar{i}+2\bar{j}-3\bar{k})+2(3i-\bar{j}+2\bar{k})=8\bar{i}+\bar{k}$

Suppose θ be angle b/w $2\bar{a}+\bar{b}$ & $\bar{a}+2\bar{b}$

$\therefore \cos\theta = \dfrac{(2\bar{a}+\bar{b})(\bar{a}+2\bar{b})}{|2\bar{a}+\bar{b}||\bar{a}+2\bar{b}|}$ $\longrightarrow$ (1M)

$\cos\theta = \dfrac{(7\bar{i}+3\bar{j}-4\bar{k})(8\bar{i}+\bar{k})}{\sqrt{49+9+16}\sqrt{64+1}}$

$= \dfrac{56-4}{\sqrt{74}\sqrt{65}} = \dfrac{52}{\sqrt{74}\sqrt{65}}$

$\theta = \cos^{-1}\left(\dfrac{52}{\sqrt{74}\sqrt{65}}\right)$ $\longrightarrow$ (1M)

$V_7(14)$:- If $|a|=2$, $|b|=3$ & $|c|=4$ and each of a,b,c is perpendicular to the sum of the other two vectors. then find the magnitude of $a+b+c$.

Sol: $|\bar{a}|=2$ $|\bar{b}|=3$ $|\bar{c}|=4$

$\bar{a}\cdot(\bar{b}+\bar{c})+\bar{b}\cdot(\bar{c}+\bar{a})+\bar{c}(\bar{a}+\bar{b})=0$ $\longrightarrow$ (1M)

$2[\bar{a}\bar{b}+\bar{b}\bar{c}+\bar{c}\bar{a}]=0$

$\boxed{\because (\bar{a}+\bar{b}+\bar{c})^2=|\bar{a}|^2+|\bar{b}|^2+|\bar{c}|^2+2[\bar{a}\bar{b}+\bar{b}\bar{c}+\bar{c}\bar{a}]}$

$|\bar{a}|^2+|\bar{b}|^2+|\bar{c}|^2=4+9+16+0=29$

$\therefore |\bar{a}+\bar{b}+\bar{c}|=\sqrt{29}$. $\longrightarrow$ (1M)

$V_7(15)$:- If $a=2\bar{i}-3\bar{j}+5k$, $b=-\bar{i}+4\bar{j}+2\bar{k}$ then find $\bar{a}\times\bar{b}$ and unit vector perpendicular to both a and b.

Sol: $\bar{a}\times\bar{b}=\begin{vmatrix} i & j & k \\ 2 & -3 & 5 \\ -1 & 4 & 2 \end{vmatrix} = \bar{i}(-6-20)-\bar{j}(4+5)+\bar{k}(8-3)$

$= -26\bar{i}-9\bar{j}+5\bar{k}$

$|\bar{a}\times\bar{b}|=\sqrt{(-26)^2+(-9)^2+5^2}=\sqrt{782}$ $\longrightarrow$ (1M)

$\therefore$ The unit vector $\perp$ as to both $\bar{a}$ & $\bar{b}$ is $= \boxed{\pm \dfrac{\bar{a}\times\bar{b}}{|\bar{a}\times\bar{b}|}}$

$$= \pm \frac{1}{\sqrt{782}}(-26\bar{i}-9\bar{j}+5\bar{k}) \longrightarrow (1M)$$

V_7(16): If $a = 2\bar{i}-\bar{j}+k$ and $b = 3\bar{i}+4\bar{j}-k$. If θ is the angle between a and b, then find $\sin\theta$.

Sol: $\bar{a}\times\bar{b} = \begin{vmatrix} i & j & k \\ 2 & -1 & 1 \\ 3 & 4 & -1 \end{vmatrix}$

$= \bar{i}(1-4)-\bar{j}(-2-3)+\bar{k}(8+3)$

$= -3\bar{i}+5\bar{j}+11\bar{k}$

$|\bar{a}| = \sqrt{4+1+1} = \sqrt{6}$ $|b| = \sqrt{9+16+1} = \sqrt{26}$ $\longrightarrow (1M)$

$|\bar{a}\times\bar{b}| = \sqrt{9+25+121} = \sqrt{155}$

$\boxed{\because \sin\theta = \dfrac{|\bar{a}\times\bar{b}|}{|\bar{a}||\bar{b}|}} = \dfrac{\sqrt{155}}{\sqrt{6}\sqrt{26}} = \dfrac{\sqrt{155}}{\sqrt{156}}$ $\longrightarrow (1M)$

V_7(17): for any vector a, show that $|\bar{a}\times\bar{i}|^2+|\bar{a}\times\bar{j}|^2+|\bar{a}\times\bar{k}|^2 = 2|\bar{a}|^2$

Sol: Let $\bar{a} = a_1\bar{i}+a_2\bar{j}+a_3\bar{k}$

$\bar{a}\times\bar{i} = (a_1\bar{i}+a_2\bar{j}+a_3\bar{k})\times\bar{i} = a_1(\bar{i}\times\bar{i})+a_2(\bar{j}\times\bar{i})+a_3(\bar{k}\times\bar{i}) = -a_2k+a_3\bar{j}$

$|\bar{a}\times\bar{i}| = \sqrt{a_2^2+a_3^2}$ $\longrightarrow (1m)$

Similarly $|\bar{a}\times\bar{j}| = \sqrt{a_1^2+a_3^2}$, $|\bar{a}\times\bar{k}| = \sqrt{a_1^2+a_2^2}$

$\therefore |a\times\bar{i}|^2+|a\times\bar{j}|^2+|a\times\bar{k}|^2 = (a_2^2+a_3^2)+(a_1^2+a_3^2)+(a_1^2+a_2^2)$

$= 2[a_1^2+a_2^2+a_3^2] = 2|a|^2 = RHS$ $\longrightarrow (1m)$

V_7(18): If $|\bar{p}| = 2$, $|\bar{q}| = 3$ and $(\bar{p},\bar{q}) = \frac{\pi}{6}$ then find $|\bar{p}\times\bar{q}|^2$

Sol: $|\bar{p}\times\bar{q}|^2 = |p|^2|q|^2\sin^2\theta$

$= 2^2\cdot 3^2\cdot\sin^2 30^\circ = 4\cdot 9\cdot\frac{1}{4} = 9.$ $\longrightarrow (2m)$

V_7(19): If the vectors $a = 2i-j+k$, $b = i+2j-3k$ and $c = 3\bar{i}+p\bar{j}+5k$ are coplanar the find p.

Sol: $[\bar{a}\ \bar{b}\ \bar{c}] = \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & p & 5 \end{vmatrix}$ $[i\ j\ k] = 0$ $\longrightarrow (1M)$

$= 2(10)+3p)+1(5+9)+1(p-6) = 0$

$= 20 + 6P + 14 + P - 6 = 0$

$= 7P = -28$

$P = -4$ $\longrightarrow$ (1m)

V7(20): If $\bar{a} = 2\bar{i} - 3\bar{j} + \bar{k}$ and $\bar{b} = \bar{i} + 4\bar{j} - 2\bar{k}$ the find $(\bar{a}+\bar{b})\times(\bar{a}-\bar{b})$

Sol: Given

$$\bar{a} = 2\bar{i} - 3\bar{j} + \bar{k}$$

$$\bar{b} = \bar{i} - 4\bar{j} - 2\bar{k}$$

$$\bar{a} + \bar{b} = (2\bar{i} - 3\bar{j} + \bar{k}) + (\bar{i} + 4\bar{j} - 2\bar{k})$$

$$\boxed{\bar{a} + \bar{b} = 3\bar{i} + \bar{j} - \bar{k}}$$

$$\bar{a} - \bar{b} = (2\bar{i} - 3\bar{j} + \bar{k}) - (\bar{i} + 4\bar{j} - 2\bar{k})$$

$$\bar{a} - \bar{b} = 2\bar{i} - 3\bar{j} + \bar{k} - \bar{i} - 4\bar{j} + 2\bar{k}$$

$$\boxed{\bar{a} - \bar{b} = \bar{i} - 7\bar{j} + 3\bar{k}} \longrightarrow (1m)$$

$$(\bar{a}+\bar{b})\times(\bar{a}-\bar{b}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 3 & 1 & -1 \\ 1 & -7 & 3 \end{vmatrix}$$

$$= \bar{i}\begin{vmatrix} 1 & -1 \\ -7 & 3 \end{vmatrix} - \bar{j}\begin{vmatrix} 3 & -1 \\ 1 & 3 \end{vmatrix} + \bar{k}\begin{vmatrix} 3 & 1 \\ 1 & -7 \end{vmatrix}$$

$$= \bar{i}(3-7) - \bar{j}(9+1) + \bar{k}(-21-1)$$

$$= -4\bar{i} - 10\bar{j} - 22\bar{k} \longrightarrow (1m)$$

V7(21): Find the equation of the plane passing through the point $(3, -2, 1)$ and perpendicular to the vector $(4, 7, -4)$

Sol: Let $\bar{a} = 3\bar{i} - 2\bar{j} + \bar{k}$

$$\bar{n} = 4\bar{i} + 7\bar{j} - 4\bar{k}$$

The equation of the plane passing through $\bar{a}$ and perpendicular $\bar{n}$ is $\boxed{(\bar{r} - \bar{a})\cdot\bar{n} = 0}$ $\longrightarrow$ (1m)

$$= \bar{r}\cdot\bar{n} - \bar{a}\cdot\bar{n} = 0$$

$$= \bar{r}(4\bar{i} + 7\bar{j} - 4\bar{k}) - (12 - 14 - 4) = 0$$

$$= \bar{r}(4\bar{i} + 7\bar{j} - 4\bar{k}) = -6$$

$\longrightarrow$ (1m)

TRIGNOMETRY UPTO TRANSFORMATIONS

V8

① Find the period of $f(x) = \tan 5x$

Sol: Given,

$$f(x) = \tan 5x$$
$$a = 5$$

$\boxed{\text{period of } \tan x \text{ is } \frac{\pi}{|a|}}$ → (1m)

∴ period of $\tan 5x$ is

$$\frac{\pi}{|5|} = \frac{\pi}{5} \rightarrow (1m)$$

V8 ② Find the period of $f(x) = \cos\left(\frac{4x+9}{5}\right)$

Sol: Given,

$$f(x) = \cos\left(\frac{4x+9}{5}\right)$$
$$\cos\left(\frac{4x}{5} + \frac{9}{5}\right)$$
$$a = \frac{4}{5}$$

$\boxed{\text{period of } \cos ax = \frac{2\pi}{|a|}}$ (1m)

∴ period of $\cos\left(\frac{4x+9}{5}\right) = \frac{2\pi}{\left|\frac{4}{5}\right|}$

$$\frac{10\pi}{4} = \frac{5\pi}{2} \rightarrow (1m)$$

V8 ③ Find the period of $f(x) = \tan(x + 4x + 9x + \cdots n^2x)$

Sol: Given, $f(x) = \tan(x + 4x + \cdots n^2x)$

$$= \tan[x(1+4+9+\cdots+n^2)]$$
$$= \tan[x(1^2+2^2+3^2+\cdots+n^2)]$$
$$= \tan\left[x\left(\frac{n(n+1)(2n+1)}{6}\right)\right]$$
$$a = \frac{n(n+1)(2n+1)}{6},$$

$\boxed{\text{period of } \tan x \text{ is } \frac{\pi}{|a|}}$

∴ period of $f(x) =$ → (1m)

$$= \frac{\pi}{\frac{n(n+1)(2n+1)}{6}}$$
$$= \frac{6\pi}{n(n+1)(2n+1)} \rightarrow (1m)$$

V8

④ Find the maximum and minimum value of $f(x) = 7\cos x - 24\sin x + 5$

Sol: Given,

$$f(x) = 7\cos x - 24\sin x + 5$$
$$a = 7;\ b = -24;\ c = 5$$

∴ Maximum Value $= c + \sqrt{a^2+b^2}$

$$= 5 + \sqrt{7^2+(-24)^2}$$
$$= 5 + \sqrt{49+576}$$
$$= 5 + \sqrt{625}$$
$$= 5 + 25$$
$$= 30$$

→ (1m)

Minimum value $= c - \sqrt{a^2+b^2}$

$= 5 - \sqrt{7^2+(-24)^2}$

$= 5 - \sqrt{49+576}$

$= 5 - \sqrt{625}$

$= 5 - 25$

$= -20$ → (1m)

V8 ⑤ Find the maximum and minimum value of

$f(x) = \cos\left(x+\frac{\pi}{3}\right) + 2\sqrt{2}\left(x+\frac{\pi}{3}\right) - 3$

Sol:

$a = 1$; $b = 2\sqrt{2}$; $c = -3$

$\therefore$ maximum value $= c + \sqrt{a^2+b^2}$

$= -3 + \sqrt{1^2+(2\sqrt{2})^2}$

$= -3 + \sqrt{1-18}$

$\Rightarrow -3+3 = 0$

$\therefore$ Minimum value $= c - \sqrt{a^2+b^2}$

$\Rightarrow -3 - \sqrt{1^2+(2\sqrt{2})^2}$

$\Rightarrow -3 - \sqrt{1-18}$

$\Rightarrow -3 - \sqrt{9}$

$\Rightarrow -3-3 = -6$

V8 ⑥ Find the maximum and minimum values of $f(x) = 3\sin x - 4\cos x$

Sol: Given

$f(x) = 3\sin x - 4\cos x$

$a = 3$; $b = -4$; $c = 0$

Maximum value $= c + \sqrt{a^2+b^2}$

$= 0 + \sqrt{(3)^2+(-4)^2}$

$= 0 + \sqrt{9+16}$

$= \sqrt{25} = 5$

Minimum value $= c - \sqrt{a^2+b^2}$

$0 - \sqrt{3^2+(-4)^2}$

$-\sqrt{9-16}$

$-\sqrt{25} = -5$

V8 ⑦ Find the maximum and minimum values of

$f(x) = 13\cos x + 3\sqrt{3}\sin x - 4$

Sol: Given

$f(x) = 13\cos x + 3\sqrt{3}\sin x - 4$

$a = 13$; $b = 3\sqrt{3}$; $c = -4$

Maximum value $= c + \sqrt{a^2+b^2}$

$= -4 + \sqrt{(13)^2+(3\sqrt{3})^2}$

$= -4 + \sqrt{169+27}$

$= -4 + \sqrt{196}$

$\Rightarrow -4+14 = 10$

minimum value $= c - \sqrt{a^2+b^2}$

$= -4 - \sqrt{(13)^2+(3\sqrt{3})^2}$

$= -4 - \sqrt{169+27}$

$= -4 - \sqrt{196}$

$= -4 - 14$

$= -18$ → (2m)

V8 ⑧ Find the value of

$\sin^2 82\frac{1}{2} - \sin^2 22\frac{1}{2}$

Sol: Given

$\sin\left(82\frac{1}{2}+22\frac{1}{2}\right)\sin\left(82\frac{1}{2}-22\frac{1}{2}\right)$

$\sin 105 \sin 60$

$\sin(90+15)\sin 60$

$\cos 15 \cdot \sin 60$

$\frac{\sqrt{3}+1}{2\sqrt{2}} \cdot \frac{\sqrt{3}}{2}$

$\frac{\sqrt{3}+3}{4\sqrt{2}} \longrightarrow (2m)$

$\boxed{\because \sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)}$

V8 (9) Find the value of

$\cos^2 112\frac{1}{2} - \sin^2 52\frac{1}{2}$

Sol: $\boxed{\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)}$

$\cos\left(112\frac{1}{2}+52\frac{1}{2}\right)\cos\left(112\frac{1}{2}-52\frac{1}{2}\right)$

$= \cos 165 \cdot \cos 60$

$= -\cos 15 \cdot \cos 60$

$= \frac{-\sqrt{3}+1}{2\sqrt{2}}\left(\frac{1}{2}\right)$

$= -\frac{(\sqrt{3}+1)}{4\sqrt{2}} \longrightarrow (2m)$

V8 (10) Find the value of

$\sin^2 52\frac{1}{2} - \sin^2 22 \cdot \frac{1}{2}$

Sol: $\boxed{\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)}$

Now,

$\sin\left(52\cdot\frac{1}{2}+22\cdot\frac{1}{2}\right)\sin\left(52\frac{1}{2}-22\frac{1}{2}\right)$

$\sin 75 \cdot \sin 30$

$= \frac{\sqrt{3}+1}{2\sqrt{2}}\left(\frac{1}{2}\right)$

$= \frac{\sqrt{3}+1}{4\sqrt{2}}$

V8 (11) prove that

$\cot\frac{\pi}{20}\cdot\cot\frac{3\pi}{20}\cdot\cot\frac{5\pi}{20}\cdot\cot\frac{7\pi}{20}\cdot\cot\frac{9\pi}{20} = 1$

Sol: L.H.S $= \cot 9 \cdot \cot 27 \cdot \cot 45 \cdot \cot 63 \cdot \cot 81$

$$\boxed{\begin{array}{c} A+B=90 \\ \frac{\pi}{20}=9 \\ \cot A \cdot \tan A = 1 \end{array}}$$

$= (\cot 9 \cdot \cot 81)(\cot 27 \cdot \cot 63)(\cot 45)$

$= (\cot 9 \cdot \tan 9)(\cot 27 \cdot \tan 27) \cdot \cot 45$

$= 1 \times 1 \times 1 = 1 \longrightarrow (2m)$

V8 (12) If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$ prove that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$

Sol: $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$

$\sin\theta = \sqrt{2}\cos\theta - \cos\theta$

$\sin\theta = (\sqrt{2}-1)\cos\theta$

Now multiply $(\sqrt{2}+1)$ on B.S

$(\sqrt{2}+1)\sin\theta = (\sqrt{2}+1)(\sqrt{2}-1)\cos\theta$

$\sqrt{2}\sin\theta + \sin\theta = (2-1)\cos\theta$

$\therefore \sqrt{2}\sin\theta = \cos\theta - \sin\theta$

V8 (13). If $3\sin\theta + 4\cos\theta = 5$ then find the value of $4\sin\theta - 3\cos\theta$

Sol: Given,

$$3\sin\theta + 4\cos\theta = 5 \longrightarrow (1)$$

Let,

$$4\sin\theta - 3\cos\theta = a \longrightarrow (2)$$

$(1)^2 + (2)^2$

$$5^2 + a^2 = (3\sin\theta + 4\cos\theta)^2 + (4\sin\theta - 3\cos\theta)^2$$

$$25 + a^2 = 9\sin^2\theta + 16\cos^2\theta + 24\sin\theta\cos\theta + 16\sin^2\theta + 9\cos^2\theta - 24\sin\theta\cos\theta$$

$$25 + a^2 = 9(\sin^2\theta + \cos^2\theta) + 16(\sin^2\theta + \cos^2\theta)$$

$$25 + a^2 = 9 + 16$$

$$a^2 = 0$$

$\therefore 4\sin\theta - 3\cos\theta = 0 \longrightarrow (2m)$

V9 (14) prove that

$$\frac{1}{\sin 10} - \frac{\sqrt{3}}{\cos 10} = 4$$

LHS: $\frac{1}{\sin 10} - \frac{\sqrt{3}}{\cos 10}$

$$= \frac{\cos 10 - \sqrt{3}\sin 10}{\sin 10 \cdot \cos 10}$$

$$= \frac{2\left(\frac{1}{2}\cos 10 - \frac{\sqrt{3}}{2}\sin 10\right)}{\frac{1}{2}(2\sin 10\cos 10)}$$

$$= \frac{2(\sin 30\cos 10 - \cos 30\sin 10)}{\frac{1}{2}\sin 2(10)}$$

$$= \frac{4\sin 20}{\sin 20} = 4 \longrightarrow (2m)$$

V9 (15) If $\tan 20 = p$ then prove that $\frac{\tan 610 + \tan 700}{\tan 560 - \tan 470} = \frac{1-p^2}{1+p^2}$

Sol: Given

$$\tan 20 = p; \cot 20 = \frac{1}{p}$$

L.H.S =

$$\frac{\tan(360+250) + \tan(360+340)}{\tan(360+200) - \tan(360+110)}$$

$$= \frac{\tan 250 + \tan 340}{\tan 200 - \tan 110}$$

$$= \frac{\tan(270-20) + \tan(360-20)}{\tan(180+20) - \tan(90+20)}$$

$$= \frac{\cot 20 - \tan 20}{\tan 20 + \cot 20} = \frac{\frac{1}{p} - p}{p + \frac{1}{p}}$$

$$= \frac{1-p^2/p}{p^2+1/p} = \frac{1-p^2}{1+p^2} = R.H.S \longrightarrow (2m)$$

V9 (16) prove that $\frac{\cos 9 + \sin 9}{\cos 9 - \sin 9} = \cot 36$

Sol:

$$L.H.S = \frac{\cos 9 + \sin 9}{\cos 9 - \sin 9}$$

$$= \cot(45-9)$$

$\boxed{\because \dfrac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta} = \cot(45-\theta)}$

$= \cot 36 \longrightarrow (2m)$

(17) Find the value of
V_9 $\cos 42 + \cos 78 + \cos 162$

Sol: $\cos 42 + \cos 78 + \cos 162$

$\cos(60-18) + \cos(60+18) + \cos(180-18)$

$= 2\cos 60 \cos 18 - \cos 18$

$2\left(\frac{1}{2}\right)\cos 18 - \cos 18 = 0$

$\boxed{\because \cos(A-B)+\cos(A+B) = 2\cos A \cos B}$

(18) Find the value of $\sin 34 + \cos 64 - \cos 4$
V_9

Sol: $\sin 34 + \cos 64 - \cos 4$

$\sin(34) + 2\sin\left(\frac{64-14}{2}\right)\sin\left(\frac{4-64}{2}\right)$

$\sin 34 - 12\left(\frac{-1}{2}\right)\sin 34$

$\sin 34 - \sin 34 = 0 \longrightarrow (2m)$

V_9 (19) If $\sin\theta = \frac{-1}{3}$ and θ does not lie on 3rd Quadrant. find the value of $\cos\theta$.

Sol: Given that.

$\sin\theta = \frac{-1}{3} < 0$

$\theta \in Q_3 \,\&\, Q_4$

$\theta \notin Q_3 \Rightarrow \theta \in Q_4$

$\cos\theta > 0$

$\cos^2\theta = 1 - \sin^2\theta$

$= 1 - \left(\frac{-1}{3}\right)^2 \Rightarrow 1 - \frac{1}{9}$

$\cos^2\theta = \frac{8}{9}$

$\cos\theta = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3} \longrightarrow (2m)$

V_9

(20) Find the cosine function whose period is 7

Sol: Let the required function

$f(x) = \cos ax$

Given period $= 7$

$\frac{2\pi}{|a|} = 7 \longrightarrow (1m)$

$|a| = \frac{2\pi}{7}$

$\therefore f(x) = \cos \frac{2\pi}{7} x \longrightarrow (1m)$

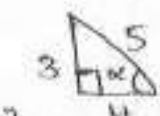
V_9

(21) If $\sin\alpha = \frac{3}{5}$ where $\frac{\pi}{2} < \alpha < \pi$
Evaluate $\cos 3\alpha$ and $\tan 2\alpha$.

Sol: Given

$\sin\alpha = \frac{3}{5}, \frac{\pi}{2} < \alpha < \pi$

$\alpha \in Q_2$



$\cos\alpha = \frac{-4}{5} \quad \tan\alpha = \frac{-3}{4}$

$\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha \longrightarrow (1m)$

$= 4\left[\frac{-4}{5}\right]^3 - 3\left[\frac{-4}{5}\right]$

$= 4\left[\frac{-64}{125}\right] + \frac{12}{5}$

$$= \frac{-256+300}{125} = \frac{44}{125}$$

$$\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$$

$$= \frac{2\left(-3/4\right)}{1-\left(-3/4\right)^2} = \frac{-3/2}{1-9/16}$$

$$= \frac{\frac{-3}{2}}{\frac{16-9}{16}} = \frac{\frac{-3}{2}}{\frac{7}{16}} = \frac{-24}{7}$$

$\longrightarrow$ (1m)

V9 (22) If $\cos\theta = -1$ $(0 < t < 1)$ and θ doesnt lie in the Q_1. find value of $\sin\theta$ and $\tan\theta$.

Sol: $\cos\theta = \frac{t}{1}$ $(0 < t < 1)$ and

$\theta \in Q_4$ $(\because \theta \notin Q_1)$

$\therefore \sin\theta = -\sqrt{1-t^2}$

$\tan\theta = \frac{\sqrt{1-t^2}}{t}$

(23) prove that $\cos 48 \cdot \cos 12 = \frac{\sqrt{5}+3}{8}$ V9

Sol: $\cos 48 \cdot \cos 12$

$$= \frac{1}{2}\left(2\cos 48 \cdot \cos 12\right)$$

$$= \frac{1}{2}\left(\cos(48+12) + \cos(48-12)\right)$$

$$= \frac{1}{2}\left(\cos 60 + \cos 36\right)$$

$$= \frac{1}{2}\left(\frac{1}{2} + \frac{\sqrt{5}+1}{4}\right)$$

$$= \frac{1}{2}\left(\frac{2+\sqrt{5}+1}{4}\right)$$

$$\frac{\sqrt{5}+3}{8} = R.H.S$$

$\longrightarrow$ (2m)

V9 (24) Eliminate θ from $x = a\cos^3\theta$ $y = b\sin^3\theta$

Sol: Given

$x = a\cos^3\theta$ $\quad y = b\sin^3\theta$

$\frac{x}{a} = \cos^3\theta$ $\quad \frac{y}{b} = \sin^3\theta$

$\left(\frac{x}{a}\right)^{1/3} = \cos\theta$ $\quad \left(\frac{y}{b}\right)^{2/3} = \sin\theta$

$\cos^2\theta = \left(\frac{x}{a}\right)^{2/3}$, $\sin^2\theta = \left(\frac{y}{b}\right)^{2/3}$

$$\boxed{\cos^2\theta + \sin^2\theta = 1}$$

$$\therefore \left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$$

$\longrightarrow$ (2m)

V9 (25) Find the extreme values of $\cos 2x + \cos^2 x$

Sol: Let $f(x) = \cos 2x + \cos^2 x$

$= 2\cos^2 x - 1 + \cos^2 x$

$= 3\cos^2 x - 1$

$\longrightarrow$ (1m)

W.K.T

$0 \le \cos^2 x \le 1$

$0 \le 3\cos^2 x \le 3$

$-1+0 \le 3\cos^2 x - 1 \le 3-1$

$-1 \le 3\cos^2 x - 1 \le 2$

$-1 \le f(x) \le 2$

$\therefore$ The extreme points are $-1, 2$

$\longrightarrow$ (1m)

26 (V9) If $\tan 20 = \lambda$ then show that $\dfrac{\tan 160 - \tan 110}{1 + \tan 160 \tan 110} = \dfrac{1-\lambda^2}{2\lambda}$

$\underline{\underline{L.H.S}} = \dfrac{\tan 160 - \tan 110}{1 + \tan 160 \tan 110}$

$= \dfrac{\tan(180-20) - \tan(90+20)}{1 + \tan(180-20)\tan(90+20)}$

$= \dfrac{-\tan 20 + \cot 20}{1 + (-\tan 20)(-\cot 20)}$ $\longrightarrow$ (1m)

$= \dfrac{-\lambda + \frac{1}{\lambda}}{1 + \lambda\left(\frac{1}{\lambda}\right)}$

$= \dfrac{\frac{-\lambda^2+1}{\lambda}}{1 + \lambda\left(\frac{1}{\lambda}\right)}$

$= \dfrac{\frac{-\lambda^2+1}{\lambda}}{1+1} = \dfrac{1-\lambda^2}{2\lambda} = R.H.S.$

$\longrightarrow$ (1m)

——— × ———

V10

HYPERBOLIC FUNCTIONS

1) If $\cosh x = 5/2$, find the values of i) $\cosh 2x$ ii) $\sinh 2x$

sol: Given $\cosh x = 5/2$

$$\boxed{\cosh 2x = 2\cosh^2 x - 1}$$

$$\cosh(2x) = 2\times\frac{25}{4} - 1 = \frac{25-2}{2} = 23/2.$$

$\longrightarrow$ (1M)

ii) $\sinh^2(2x) = \cosh^2(2x) - 1$

$$= \frac{(23)^2}{2^2} - 1 = \frac{23^2 - 2^2}{4} = \frac{(23+2)(23-2)}{4} = \frac{25\times21}{4}$$

$$\sinh(2x) = \pm\sqrt{\frac{25\times21}{4}}$$

$\longrightarrow$ (1M)

V10

2) If $\sinh x = 3/4$, find $\cosh(2x)$ & $\sinh(2x)$

sol: $\sinh x = 3/4$

$$\boxed{\cosh(2x) = 1 + 2\sinh^2 x}$$

i) $\cosh(2x) = 1 + 2\times\frac{9}{16}$

$$\cosh(2x) = 1 + \frac{9}{8} = \frac{8+9}{8} = 17/8$$

$\longrightarrow$ (1M)

ii) $\boxed{\sinh^2(2x) = \cosh^2(2x) - 1}$

$$= \frac{17^2}{8^2} - 1 = \frac{17^2 - 8^2}{8^2} = \frac{(17+8)(17-8)}{64}$$

$$\sinh^2(2x) = \frac{25\times9}{4}$$

$$\sinh(2x) = \pm\sqrt{\frac{25\times9}{64}} = \pm\frac{5\times3}{8} = \pm 15/8$$

$\longrightarrow$ (1M)

V10

3) If $\cosh x = \sec\theta$ then prove that $\tanh^2\left(\frac{x}{2}\right) = \tan^2\theta/2$.

sol: $\cosh x = \sec\theta$

$$LHS = \boxed{\tanh^2\left(\frac{x}{2}\right) = \frac{\cosh x - 1}{\cosh x + 1}}$$

$\longrightarrow$ (1M)

$$= \frac{\sec\theta - 1}{\sec\theta + 1} = \frac{\frac{1}{\cos\theta} - 1}{\frac{1}{\cos\theta} + 1}$$

$$= \frac{1-\cos\theta}{1+\cos\theta} = \frac{2\sin^2\theta/2}{2\cos^2\theta/2} = \boxed{\tan^2\theta/2 = \tanh^2 x/2}$$

$\longrightarrow$ (1M)

V10

4) Prove that $(\cosh x-\sinh x)^n = \cosh(nx)-\sinh(nx)$ $n\in R$

Sol: L.H.S $=(\cosh x-\sinh x)^n$

$= \frac{e^x+e^{-x}}{2}, -\frac{e^x-e^{-x}}{2}$

$=\left(\frac{e^x+e^{-x}+e^{-x}-e^{x}}{2}\right)^n = \left(\frac{2e^{-x}}{2}\right)^n = (e^{-x})^n = e^{-nx}$ $\longrightarrow$ (1m)

RHS $= \cosh(nx)-\sinh(nx)$

$=\frac{e^{nx}+e^{-nx}}{2}-\frac{e^{nx}-e^{-nx}}{2}$

$=\frac{e^{nx}+e^{-nx}-e^{nx}+e^{-nx}}{2}$

$=\frac{2e^{-nx}}{2}=e^{-nx}$ $\longrightarrow$ (1m)

$(\cosh x-\sinh x)^n = \cosh(nx)-\sinh(nx)$

V10

5) Prove that $(\cosh x+\sinh x)^n = \cosh(nx) - \sinh(nx)$

LHS $= (\cosh x+\sinh x)^n$

$=\left(\frac{e^x+e^{-x}}{2}+\frac{e^x-e^{-x}}{2}\right)^n$

$=\frac{2e^x}{2} = e^{nx}$ $\longrightarrow$ (1m)

RHS $= \cosh(nx)+\sinh(nx)$

$=\frac{e^{nx}+e^{-nx}}{2}+\frac{e^{nx}-e^{-nx}}{2} = \frac{e^{nx}+e^{-nx}+e^{nx}-e^{-nx}}{2} = \frac{2e^{nx}}{2} = e^{nx}$

$\therefore (\cosh x+\sinh x)^n = \cosh(nx)+\sinh(nx)$ $\longrightarrow$ (1m)

V10

6) For any $x\in R$ prove that $\cosh^4 x-\sinh^4 x = \cosh(2x)$

LHS $= \cosh^4 x-\sinh^4 x$

$=(\cosh^2 x)^2-(\sinh^2 x)^2 = (\cosh^2 x+\sinh^2 x)(\cosh^2 x-\sinh^2 x)$

$\therefore \boxed{\cosh^2 x+\sinh^2 x = \cosh(2x)}$ $\therefore \boxed{\cosh^2 x-\sinh^2 x=1}$ $\longrightarrow$ (2m)

$=\cosh(2x) =$ RHS.

V10

7) Prove that $\tanh^{-1}(\frac{1}{2}) = \frac{1}{2}\log_e 3$. LHS $= \tanh^{-1}(\frac{1}{2}) = \frac{1}{2}\log\left|\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right|$

$=\frac{1}{2}\log\left|\frac{2+1}{2-1}\right| = \frac{1}{2}\log 3$ $\therefore \boxed{\tanh^{-1}x=\frac{1}{2}\log\left|\frac{1+x}{1-x}\right|}$

8) $\longrightarrow$ (2m)

8) If $\sinh x = 5$ show that $x = \log(5+\sqrt{26})$

Given $\sinh x = 5$

$x = \sinh^{-1}(5)$

$$\boxed{\sinh^{-1}(x) = \log_e(x+\sqrt{x^2+1})} \longrightarrow (1m)$$

$x = \log_e(5+\sqrt{5^2+1})$

$x = \log_e(5+\sqrt{25+1})$

$x = \log_e(5+\sqrt{26}) \longrightarrow (1m)$

9) If $u = \log_e\left(\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right)$ & if $(\cos\theta > 0)$ then prove that $\cosh u = \sec\theta$.

sol: $u = \log_e\left(\tan\left(\frac{\pi}{4}+\theta/2\right)\right)$

$e^{u} = \tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)$

$e^{-u} = \cot\left(\frac{\pi}{4}+\theta/2\right)$

$\boxed{e^{-u} = \frac{1}{e^{u}}}$ & $\boxed{\frac{1}{\tan\theta} = \cot\theta}$

$\cosh u = \frac{e^{u}+e^{-u}}{2} \longrightarrow (1m)$

$\cosh x = \frac{e^{x}+e^{-x}}{2}$

$= \frac{1}{2}\left[\tan\left(\frac{\pi}{4}+\theta/2\right)+\cot\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]$

$= \frac{1}{2}\left[2\operatorname{cosec} 2\left(\frac{\pi}{4}+\theta/2\right)\right]$

$\boxed{\cot A + \tan A = 2\operatorname{cosec} 2A}$

$= \operatorname{cosec}\left(\frac{\pi}{2}+\theta\right)$

$= \sec\theta$.

$\boxed{\operatorname{cosec}\left(\frac{\pi}{2}+\theta\right) = \sec\theta}$

$\longrightarrow (1m)$

10) Prove that $\tanh^{-1}x = \frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right)$

sol: $x \in (-1,1)$ $y = \tanh^{-1}x$.

$x = \tanh y$.

$\frac{x}{1} = \frac{e^{y}-e^{-y}}{e^{y}+e^{-y}}$.

using componendo & dividendo

$\frac{x+1}{x-1} = \frac{(e^{y}-e^{-y})+(e^{y}+e^{-y})}{(e^{y}-e^{-y})-(e^{y}+e^{-y})}$

$$\boxed{\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}} \longrightarrow (1m)$$

componendo & dividendo

$\frac{x+1}{x-1} = \frac{2e^{y}}{-2e^{-y}} = -e^{2y}$.

$e^{2y} = \frac{1+x}{1-x} = 2y = \log_e\left(\frac{1+x}{1-x}\right)$

$= y = \frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right)$

$= \tanh^{-1}x = \frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right) \longrightarrow (1m)$

11) For any $x \in R$ Show that $\cosh 2x = 2\cosh^2 x - 1$

sol: $\cosh^2 x + \sinh^2 x = \cosh 2x$

$\cosh^2 x - \sinh^2 x = 1$

$\sinh^2 x = \cosh^2 x - 1 \longrightarrow (1m)$

LHS $= \cosh 2x = \cosh^2 x + \sinh^2 x$

$= \cosh^2 x + \cosh^2 x - 1$

$= 2\cosh^2 x - 1$

RHS.

$\longrightarrow (1m)$